

HUGO Midterm Report

DSE Group 28

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1. Executive Overview

In order to achieve a High-speed Unmanned General-purpose Observatory (HUGO) mission of testing aeroservoelastic phenomena in different kinds of high aspect ratio wings, five possible configurations were proposed in the Baseline Report ⁽¹⁾. Upon further revision, four of the configurations were upheld and are shown in [Table 7.1](#). The goal of the Midterm Report is to provide a final assessment on which configuration will be chosen for further development. To facilitate this, a manufacturing, assembly, and integration plan will be established, followed by the internal and external layouts of the chosen design configured, and the sustainable development strategy. In addition, the final project objectives will be established, the operations and logistics of HUGO's mission, and the certification and risk assessment in accordance with European Union Aviation Safety Agency (EASA) regulations.

Table 7.1: Final integrated candidate architectures for detailed trade-off⁽¹⁾

ID	Wing Configuration	Wing Modularity	Canard Modularity
HUG-CFG-301	Low Wing	Static Port	Static Port
HUG-CFG-302	High Wing	Static Port	No Port
HUG-CFG-303	High Wing	Static Port	Static Port
HUG-CFG-304	High Wing	Variable Port (X)	Static Port

First, the project objectives were formulated according to the SMART methodology to help guide the trade-off between the different configurations selected in the baseline report. They cover the financial, quality, performance, business, and compliance objectives of HUGO and its mission.

As previously stated, the goal of the midterm report is to select the configuration best suited to the mission requirements, which relies on structured planning. This begins with the midterm Workflow Diagram (WFD), followed by its translation into a Work Breakdown Structure defining responsibilities, methods, and durations. Dependencies are then captured in an N2 diagram, the full lifecycle is outlined in the Project Design & Development Logic diagram, and the planning is completed through a team organogram and Gantt chart that integrates roles and scheduling into a coherent project timeline.

Operations and Logistics are a crucial part in the development of a successful mission. Together they cover all aspects of HUGO's day-to-day usage. For a mission to be successful, HUGO must operate in a controlled airspace, fly over 100km in a straight line, and to perform a test at Mach 0.75 for at least 5 minutes.

To that end, Den Helder airport was chosen as HUGO's base of operations in the Netherlands. It is a joint military and civil airfield located next to the sea, with a controlled airspace around it. This is ideal from a safety standpoint for other aircraft in the shared airspace, and greatly reduces the transit time from take-off to the testing location over the North Sea where the population density is low enough to allow operations at the target Mach 0.75 flight speed as seen in [Figure 4.1](#). A test day timeline from Den Helder Airport was made, emphasizing the pre-, in- and post-flight operations for each of the four daily tests, for 100 days per year. A preliminary maintenance and continuance plan was then drafted in accordance with EU No. 1321/2014 and a stakeholder responsibility analysis.

To facilitate the operations and reduce transport costs, HUGO is to be stored in a hangar in the area adjacent to Den Helder airport. Although not a commercial airport, Den Helder is the main gateway to off-shore platforms in the North Sea, thus daily traffic is expected. As such, the test flight plans will be coordinated with De Kooy Air Traffic Control. HUGO's propulsion system uses Sustainable Aviation Fuel (SAF), which will need to be stored in the hangar and brought to the airport for each flight day as Den Helder airport does not offer SAF refuelling services. Lastly, the handling of HUGO; transport, flight, and ground support will be performed by certified personnel.

¹ [Baseline Report](#) (accessed on 21.05.2026)

The technical risk assessment was also improved. The technical risk matrix was updated after new risks were identified, and the preventative measures and mitigation strategies refined. Some of the identified risks include: certification duration in the range of years, rejection of type certificate and/or DOA application, operating risks specific to Den Helder airport like fuel shortage, lack of Air Traffic Control (ATC) clearance, ground station equipment failure, or corrupted measurement data. The chosen preventive measures and contingency scenarios reduce all the risks to acceptable levels of likelihood and severity.

EASA Certification was also a major concern in the Baseline Report, as HUGO's risk class and operational environment were highly uncertain. However, in consultation with, and with guidance from EASA, HUGO was classified under CS-23 for aerobatic aircraft, with supplemental special conditions for unmanned aerial vehicles considered high risk. Successful certification of an Unmanned Aerial Vehicle (UAV) with such a high degree of modularity would require constant and expensive certification. However, through the simultaneous application for a type certificate and a Design Organisation Approval (DOA), HUGO will be allowed to undergo major changes without requiring full recertification. Though more versatile, it is a process that will require an initial investment of €10,170 for the certification, and then a yearly fee of €11,050 for the maintenance of the Type Certificate. For compliance, a team of engineers will therefore have to apply for the DOA, and be responsible for approving any major changes. A flight envelope combining both manoeuvring and gust loads borne by the structure was then calculated to ensure compliance with the CS-23 aerobatic requirements. Furthermore, it was necessary to verify the present environmental requirements issued by the European Union. The relevant legalisation for UAVs of HUGO's specifications emphasise a focus on the reduction of noise and emissions.

To determine if each of the four configurations complied with the requirements, a cursory analysis and estimation of design parameters had to be performed. Several trade-off criteria were established for parts of the design. First, the airfoil and analogous wing estimation was performed. To do that, performance and geometric parameters were obtained from reference aircraft. These aircraft were selected based on the mission profile, size and cruise conditions. Drag buildup methods described by Torenbeek (1982) [1] were used, namely: stall speed, Aspect Ratio (AR), wingspan, Mean Aerodynamic Chord (MAC), sweep, wing area, wing loading, and taper ratio. Then, design aerodynamic parameters were estimated using the previously obtained data and drag build-up methods described by Raymer (design lift coefficient, cruise Reynolds number, maximum lift coefficient). Finally, based on the reference aircraft and HUGO cruise condition requirements, a NASA symmetrical supercritical airfoil was selected to limit compressibility effects and prevent shock wave formation on the lifting surfaces.

To generate numerical data to serve as basis for the trade-off, the four design options were conceptually sized, taking into account the design aspects where differences between the configurations were expected. To provide initial values for the design iterations, reference aircraft data was gathered. No reference aircraft of similar scale were developed by the time of writing this report, however, several aircraft approach it. One of the aircraft close to HUGO's mission is the NASA DAST UAS with the ARW-2 wing used for aeroservoelastic effects testing. Although the mission objective was very similar, the aircraft maximum take-off weight was above 1000 kg, placing it far from the requirements for HUGO. Next, the NASA X-56 and FLEXOP experimental drones were used. These aircraft approach more closely the weight and scale of HUGO but operate in a highly subsonic low-speed regime. Nevertheless, the data obtained from all the considered aircraft helped to guide the first stage of the design process.

To model the aerodynamic performance of the aircraft an inviscid parametric lifting line model of the aircraft was constructed using Aerosandbox. This was combined with a component build-up method to also account for parasitic drag.

The fuel mass fraction of the aircraft was obtained by breaking down its mission profile into multiple climb and cruise segments, for which standard equivalent range formulae are available, and evaluating the fuel mass fractions for each segment with performance parameters for that segment. Then, the mass fractions were combined

into the total fuel mass fraction, which was around 27 % for all configurations.

Due to HUGO's varying modularity across most configurations, its effect on structural loading had to be investigated. The variable longitudinal location of the wing has to be accounted for structurally at all possible wing positions. Regarding configurations with a canard port, loads from a canard change the load on the aircraft as well. Since multiple configurations were modular, it was necessary to account for the worst-case scenario with the highest load. To investigate this, the fuselage was initially modelled as a simple cylindrical shape with applied point loads from the canard, wing and tail. The quasi-isotropic mechanical properties of carbon fibre reinforced polymer (CFRP) were used for this analysis. Code was developed to calculate the required skin thickness along the length of the fuselage and compare the weight penalties caused by the multiple configurations considered.

The design wing loading and thrust-weight ratio were obtained by means of a matching diagram, with constraints tailored to account for the high-speed operations of HUGO, compliance with CS-23, and its compatibility with den Helder airport.

As a next step, a preliminary empennage size estimation was performed. Analogously to the wing preliminary design, statistical data from reference aircraft were collected. Based on that an initial size of the empennage is obtained and iterated upon using standard stability and controllability equations. The sizing point for the empennage is neutral stability, to allow or the aircraft to be easily reconfigurable into stable and unstable configurations.

The results of the sizing were assessed in a sensitivity study, whose main finding include a highly constrained design space for the landing gear placement, and significant dependency of the overall aircraft mass on fuselage dimensions. Other assumptions, such as the design C.G. Excursion and the geometry of particular landing gear struts, drive the results to a lesser extent.

To evaluate the candidate vehicle configurations, a weighted trade-off matrix was made using criteria including cost, versatility, reliability, and sustainability. This qualitative assessment was integrated with quantitative performance metrics derived from the parametric sizing code. The results of the code showed that the performance of the different configurations were largely the same, even after accounting for the structural weight penalties of the added modularity. By comparing the configurations against these combined criteria, HUG-CFG-303 was identified as the optimal architecture. This configuration, featuring a high-wing layout, T-tail, fuselage-mounted propulsion, and static modular ports, achieved the highest overall score by maximizing reliability and aerodynamic efficiency while maintaining the essential modularity required for a flying laboratory.

Subsequently, a material selection trade-off was performed to determine the optimal structural composition of the aircraft. Several material candidates were evaluated against weighted criteria, including structural properties, durability, manufacturing cost, and design freedom. Carbon-fiber-reinforced polymer (CFRP) was selected as the primary material for the fuselage and main aerodynamic surfaces. Its superior specific modulus and plate buckling efficiency are critical for supporting a high-aspect-ratio airframe operating at Mach 0.75, while simultaneously offering the design freedom necessary to manufacture complex, aerodynamically optimized profiles. For localized, high-stress functional components such as the landing gear and pylons, Aluminum 7075-T76 was selected due to its high specific fracture toughness and ease of manufacturability.

Once the final configuration and material was established, a Manufacturing, Assembly and Integration (MAI) plan was created. The manufacturing techniques were chosen based on labour costs, part size, production time, and material efficiency. The chosen materials, carbon fibre composite and Al 7075 T76, require different manufacturing processes. The carbon fibre parts will be manufactured using a pre-preg lay up and autoclave method. Due to the primary use of the aluminium alloy for the landing gear, and the limited number of HUGOs set to be produced, CNC machining paired with deep-hole drilling, heat treatment, and corrosion protection was chosen as the manufacturing method. The assembly will involve the joining of different parts. As such,

different joining techniques such as adhesive bonding, riveting, bolting, and welding were analysed, and the pros and cons of each method described for each planned application. Additionally, it was determined that the specialised modular ports will join the primary structure subsystems. The integration of the remaining systems (i. e. avionics, power supply, engine, and interior) was subsequently performed. The main considerations when planning the integration of subsystems were: accessibility of the components for maintenance, and avoiding time delays and safety risks due to improper subsystem integration.

The external configuration of the chosen design was illustrated through a sketch, elucidating the positions of all outside mounted ports and holes, such as antennas. The internal configuration is not yet defined at this design stage. Instead, all relevant components that must be contained within the fuselage are listed. A detailed explanation of the electronics system, major components, as well as specific components like the autopilot, and a comprehensive list of sensors used to quantify aeroservoelastic phenomena is provided further. It was determined that the battery will likely be the limiting factor dictating the dimensions of the fuselage, as it is the largest component. The fuel and propulsion system was also analysed and visualised graphically. The components related to the propulsion system were identified. Additionally, the components were sized to ensure compliance with the 5 litre payload requirement. The required positioning of the landing gear and the associated retraction mechanisms were also estimated and accounted for in the internal volume of the fuselage. Subsequently, a cost analysis of all components will be performed.

To ensure minimal environmental impact, HUGO's sustainable development strategy was devised. A preliminary Life Cycle Analysis (LCA) was performed, resulting in an estimated total Global Warming Potential (GWP) of $1.92794 \cdot 10^6$ [kg CO₂ eq] for the production, operation and end-of-life solutions, with fuel-burn and the production of SAF being the biggest contributors. The social sustainability strategy from the Baseline Report was expanded on with the recommendations set out by the European Commission in regulation (EU) 2018/1139 Annex IX.

To conclude, HUG-CFG-303, a high-wing configuration with static wing and canard ports, and retractable landing gear, was chosen as the final configuration for its balance of experimental versatility, cost efficiency, and reliability.

HUG-CFG-303 will undergo further development, as there are several factors limiting the precision of the models at this time. In particular, the aerodynamic model has not been validated against experimental data specific to HUGO. Additionally, known discrepancies with Data Compendium (DATCOM) predictions at higher angles of attack suggest that lift estimates may carry non-negligible uncertainty, particularly for the swept wing configuration. Lastly, the structural weight penalty analysis relied on simplified beam models with point loads, which will require refinement once the final configuration geometry is fixed.

Further recommendations are proposed to refine the final configuration. Namely, it is recommended to proceed with converging the weight and centre-of-gravity iteration loop, refining the aerodynamic model through higher-fidelity analysis at high subsonic conditions, completing the detailed structural design of the wing-fuselage and tail-fuselage static ports, and finalising the internal layout and fuel system sizing for HUG-CFG-303. In addition, to improve the organisational structure, it is also recommended to initiate preliminary contact with EASA regarding the DOA application and confirming hangar availability at Den Helder, which can be done in parallel with the development of the final design.

2. Project Objectives

The project objectives are formulated according to the SMART method: they are specific, measurable, actionable, relevant, and time-bound. They serve as a roadmap for planning and execution, defining the tangible deliverables and outcomes required within a project's timeline, budget, and scope.

2.1. Financial Objectives

The financial objectives focus on budget management.

1. The total manufacturing cost per aircraft configuration shall not exceed €45000 ⁽¹⁾.
2. The aircraft shall have a maximum operational cost of €2000 per flight day ⁽¹⁾.

2.2. Quality Objectives

The quality objectives define standards for the deliverables.

1. A HUGO configuration shall undergo 50 days of maintenance per year ⁽¹⁾.
2. The configuration trade-off shall evaluate the candidate architectures against weighted criteria, with a consistency ratio below 0.1 for the Analytic Hierarchy Process (AHP) weighting ([section 7.2](#)).

2.3. Performance Objectives

The performance objectives focus on the results and effectiveness of the project.

1. The initial HUGO fleet shall comprise of five aircraft to ensure continuous operational availability across 100 operational days per year with 4 flight tests per day ⁽¹⁾.
2. The selected configuration shall withstand manoeuvre load factors ranging from -3 to +6 ([section 5.4](#)).
3. The aircraft shall enable continuous flight and collection of flight test data at 0.75 Mach for at least 5 minutes ⁽¹⁾.
4. The selected configuration shall carry sufficient fuel for the nominal mission of 30 minutes and four landing attempts ⁽¹⁾.

2.4. Business Objectives

The business requirements align with the strategic goals of market entry and client acquisition.

1. The Netherlands shall serve as the primary market for initial operations, with Den Helder Airport as the designated base of operations ⁽¹⁾.
2. HUGO shall be capable of serving customers from at least four market segments: universities and academic laboratories, research centres, aerospace companies, and defence organisations ⁽¹⁾.

2.5. Compliance Objectives

The compliance objectives ensure adherence to regulations and standards.

1. Each HUGO configuration shall comply with the environmental performance requirements set out in Annex III of Regulation (EU) 2018/1139 of the European Parliament and of the Council ([section 5.5](#)).
2. Each HUGO configuration shall comply with the CS 23 Aerobatic Airplanes requirements and Special Condition for Light Unmanned Aircraft Systems - High Risk ([subsection 5.2.1](#)).

3. Project Planning

This chapter outlines the project planning approach and tools used to define and monitor the workflow. [section 3.1](#) describes the approach towards constructing the workflow diagram, from which the work breakdown structure is derived in [section 3.2](#). Mutual task dependencies are defined and evaluated using an N2 diagram in [section 3.3](#). The future steps for project design and development are outlined in [section 3.4](#). Organisational aspects such as team responsibility allocation and work timeline are described in [section 3.5](#) and [section 3.6](#), respectively.

3.1. Midterm Workflow Diagram (WFD)

The workflow for converging towards the midterm review is shown in [Figure A.1](#) in [Appendix A](#). The main input is the short list of possible design configurations to be considered further.

In the first work phase, the preliminary weight of each configuration needs to be set before proceeding with future sizing. This, in turn, contributes to the initial definition of the geometry parameters. At the same time, the V-n diagram is created, combining the stall speed, the maximum positive and negative load limits, and gust load requirements. In parallel to defining the initial mass, the airfoil is selected based on the constraints, such as the critical Mach number. Finally, the trade-off criteria are substantiated and developed to the extent at which they can all be unambiguously quantified.

The second work phase begins with an low-fidelity aerodynamic analyses to determine the parameters such as: the maximum lift coefficient, the lift curve slope, and the lift-to-drag-ratio. Then, the parameters such as the thrust-to-weight ratio and wing loading have to be estimated. At the same time, the static stability characteristics can be analyzed, such as the c.g. range required for stability. Endurance performance can be quantified, followed directly by the Class I weight estimation. All the tasks from this phase contribute to the preliminary sizing.

In the final stage, differences between candidate HUGO configurations can be quantified, a preliminary life-cycle assessment can be performed. Consequently, a trade-off is performed in a feedback loop with the sensitivity study, and a final design configuration is established.

3.2. Midterm Work Breakdown Structure (WBS)

Following from the WFD, the WBS is constructed to provide a more hierarchical overview of all project activities. It ensures that progress can be tracked and responsibilities clearly assigned at every level. HUGO's WBS was divided into four work phases reflecting the natural progression of the engineering design process:

1. **Low-Fidelity Design Analysis:** initial aerodynamic characterisation of candidate configurations, incl. airfoil selection, V-n diagram, endurance estimation, and static stability determination.
2. **Preliminary Sizing:** constraint analysis and weight estimation activities that yield the first quantitative geometric and performance parameters of the aircraft.
3. **Trade-off Framework & Parameter Computation:** establishing evaluation criteria, computing trade-off parameters for each configuration, preliminary Life Cycle Assessment.
4. **Trade-off & Configuration Selection,** where the weighted scoring matrix is applied, a sensitivity study is conducted on the trade-off results, and the best configuration is formally selected.

Each task has a responsibility defined based on one of the disciplinary roles (Aerodynamics, Structures, Systems Engineering, Operations, and Sustainability). An estimated duration was also attributed to each task, allowing the WBS to feed directly into the project Gantt chart (see: [section 3.6](#)).

Identified methods used for specific tasks (e. g. Roskam statistical fractions for weight estimation, or XFOIL for airfoil analysis) are noted explicitly at the subtask level. In cases where a method has not yet been determined, a TBD flag is placed. This highlights areas that require further investigation before work can begin. This approach ensures the WBS serves not only as a planning tool but also as a live record of methodological decisions

The colour scheme of the WBS diagram follows the work phases. Lighter coloured cards correspond to tasks belonging to earlier phases of the project, while progressively darker colours indicate tasks that are executed later in the workflow. The WBS diagram is shown in [Figure A](#) in [Appendix A](#).

3.3. N2 diagram

The purpose of an N2 diagram is to break down a complete system into functional units or subsystems, and define the dependencies between them ⁽²⁾. The N2 diagram for the HUGO, describing the relationships between its subsystems is shown in [Figure A.2](#) in [Appendix A](#).

3.4. Project Design & Development Logic

It is important to keep an outlook for the future of the project and the subsequent development of the aircraft after the design has been completed. To create a more comprehensible diagram, the project design and development were broken down into five different phases [2]:

1. **Development:** Takes the preliminary design as input and evaluate the viability of the design. If it is deemed inviable, it is checked whether or not the issues can be solved. If not the project is ended, otherwise the preliminary design is altered until it is viable.
2. **Prototyping and testing:** Each subsystem is designed, manufactured, verified, and validated in isolation. Then, all the subsystems are integrated into a fully functional UAV (Unmanned Aerial Vehicle). The UAV is tested on the ground and in the air to validate the design as a whole.
3. **Manufacturing and Certification:** The finalized, validated design is taken into production. In the beginning, the aircraft design is certified by EASA. If it passes certification, the production can begin. If it does not, the design is reiterated. This involves the procurement of off-the-shelf components, and working with manufacturers and suppliers.
4. **Operations:** The bulk of the life of an aircraft. Technicians and pilots must be hired and trained. HUGO will perform tests required by the client, and undergo necessary design modifications and certification steps requested by the competent authority.
5. **End-of-life:** The airframe must be decommissioned responsibly, recycled, or scrapped. The useful components, such as the avionics and actuators, can be repurposed. All other parts must be recycled or disposed of.

The Project Design & Development Logic diagram can be seen in [Figure A.3](#) in [Appendix A](#).

3.5. Team Organogram & HR Allocation

To optimise technical development, the team identified and distributed critical organisational responsibilities, as shown in [Figure A.4](#) in [Appendix A](#). These roles structure the development process, mitigating internal risks and enhancing the overall quality of the final product. The designated roles and their core functions are as follows:

- **Chairman:** Leads meetings and discussions, prepares agendas, and ensures alignment with team goals.
- **Secretary:** Documents meetings, acts as the primary point of contact for all external stakeholders.
- **Project Manager:** Maintains the project schedule, ensures timely progression and keeps track of deadlines.
- **Chief Systems Engineer:** Oversees technical architecture, ensures integration of the final system.

²[Project Guide Design Synthesis Exercise 2025-2026](#) (accessed on 20.04.2026)

- **Documentation:** Documentation Manager, Git Manager, and CAD Manager are responsible for managing all text, code, and CAD files. They enforce relevant standards and implement efficient organisation for their respective file types.
- **Sustainability Manager:** Guides the project towards social, economic, and environmental sustainability.
- **Quality Assurance Manager:** Ensures all work products meet the standards required by external stakeholders and internal protocols.

Additional technical roles were defined to create focused engineering design departments. Each team member was assigned one organisational role and one technical role, and backup members were appointed for each department (names written in bold and italics). Additional members expected to contribute to a specific department were highlighted as "Contributors". The complete Team Organogram is shown in [Figure A.4](#) in [Appendix A](#).

3.6. Gantt Chart

[Figure A.5](#) in [Appendix A](#) shows the project Gantt chart outlining the project tasks, along with relevant dependencies and a preliminary timeline.

3.7. Statement on AI usage

During the project, AI tools such as ChatGPT, or Claude Code were used for $\text{\LaTeX}$ formatting and troubleshooting code. Grammarly was used for grammar corrections. Those tools were not used to develop new code from scratch or write sections of this report.

4. Operations and Logistics

This chapter presents the operational and logistical framework required to bring HUGO into service following completion of the design and certification process. In [section 4.1](#), the intended use of the system is described, covering the concept of operations, mission objectives, mission phases, and maintenance provisions. Then, [section 4.2](#) addresses the supporting resources needed for sustained operation, including: ground personnel, deployment logistics, facility and location requirements, consumables, spare parts, and supply-chain considerations.

4.1. Operations

This section contains a detailed description of HUGO's intended operations, an overview of the system users, testing location and conditions, mission phases, and approximate timeline. A brief discussion is dedicated to operational constraints and maintenance procedures to ensure HUGO's successful, and continuous operation.

4.1.1. High-level Mission Description

HUGO's main objective aeroservoelastic phenomena in high aspect ratio wings. The mission is considered successful if a flight speed of Mach 0.75 is achieved for 5 consecutive minutes of straight flight, all control systems required by the user were activated/actuated without error, and the collected data is successfully recorded, downloaded, and delivered to the user.

Based on the identified beachhead marker in the baseline report, a suitable airport was found in the north-east of The Netherlands for HUGO's operations, Den Helder ⁽⁴⁾. Several characteristics make Den Helder Airport the best choice as HUGO's base of operations. It is located 5 km away from the North sea, reducing the transit time from take-off to the test area over populated areas, and has controlled airspace as required by EASA [\[3\]](#). Den Helder is a joint civilian and military airport, consequently, it is not subject to the same congestion as regular airports, and permits special operations. It also has a 1.27 km runway with on call emergency services, which are beneficial for a test aircraft with potentially high take-off speeds.

At Den Helder Airfield, HUGO will take off from runway 21 and depart via airway named FOXTROT according to Air Traffic Control (ATC) instructions ⁽⁵⁾. It will follow the departure regulations until it reaches a distance of approximately 6 km from the airfield over the North Sea. Once it is clear of ground, it will start climbing to test altitude and route towards the testing area at ATC discretion. The test will take place over the North Sea, and HUGO will follow the preplanned path.

Once the take-off area is free, HUGO can proceed to the test site shown in [Figure 4.1](#). The site was chosen to be close to Den Helder airport to comply with the fuel and endurance requirements. The maximum distance in a straight line in the highlighted area is estimated to span 110 km, which allows for the 5 minute test flight to be successfully performed.

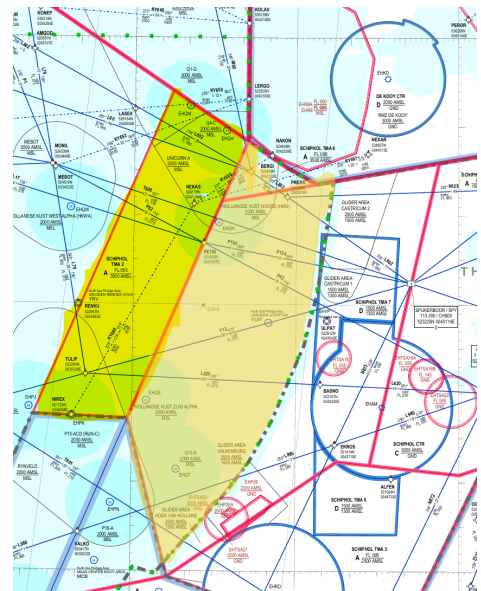


Figure 4.1: Flight area highlighted on Netherlands AIP chart - flight area highlighted for clarity, taken from the Netherlands AIP ⁽³⁾

⁴[Den Helder Airport](#) (accessed on 22.05.2026)

⁵[Den Helder Departure Map](#) (accessed on 11.05.2026)

4.1.2. Mission Phase and Timeline

A reference test day for HUGO is assumed to include four test flights, with 100 operational days per year ⁽¹⁾. Four main phases were identified for a test day: transport to test site, followed by pre-flight, in-flight, and post-flight operations. These shall now be evaluated in more detail.

Transport to Test Site

The day before a test flight is scheduled, the Terminal Aerodrome Forecast (TAF) shall be consulted to ensure that departure will be possible. In case of adverse weather prediction, the test day (or part of it) shall be cancelled and the customer shall be informed.

In case the weather prediction allows for a successful take-off, the test day shall proceed according to schedule. If the test requires parts provided by the client to be mounted onto HUGO, these shall be integrated in the hangar, before proceeding onto the airfield. The ground station and fuel required for the flight shall also be transported onto the airfield.

Pre-Flight Operations

The pre-flight phase begins once HUGO is unloaded from the transportation unit and ends when the aircraft is powered on, all checks are performed, and fuel is loaded on board. The objective of this phase is to verify that the mission plan, operating environment, ground support equipment, aircraft configuration, and safety provisions are suitable for the intended mission.

An overview of the pre-flight steps is provided in [Figure 4.2](#). Once HUGO is unloaded, the mission plan is reviewed, and Notice To Air Missions (NOTAMs), are reviewed for any update lists containing alerts of potential hazards, are checked. If the initial inspection is completed and no problems are found, HUGO can be configured for the mission profile, and the crew briefing and preparation checklists can be performed. Otherwise, the test shall be aborted or delayed. Once all preparation steps are checked and recorded, HUGO can be considered as ready to fly, and the test personnel can proceed to the in-flight phase of the mission.

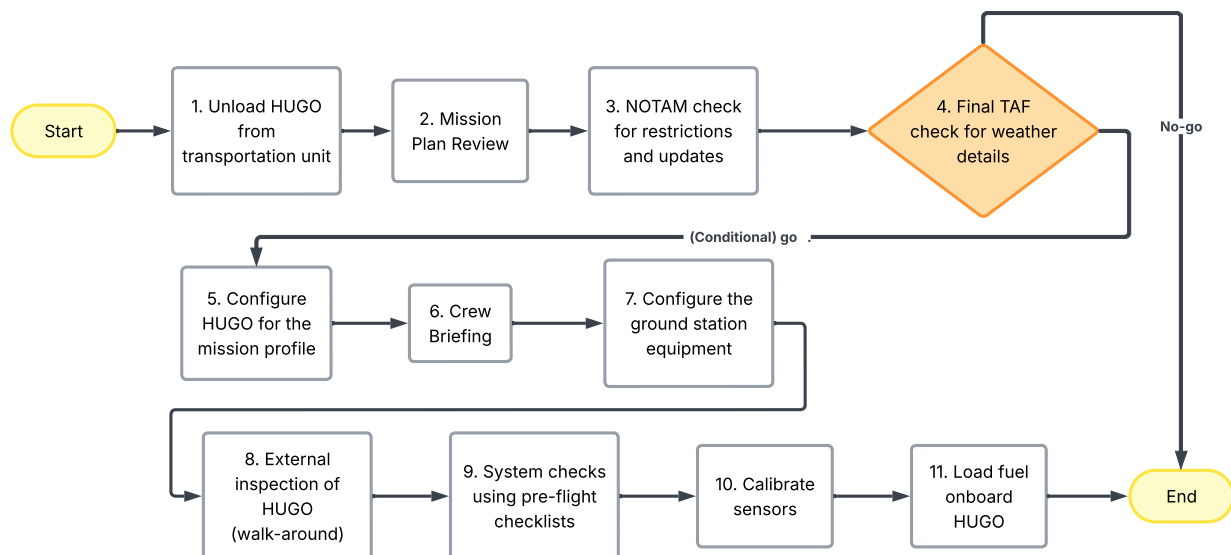


Figure 4.2: *HUGO pre-flight operations*

In-Flight Operations

Once all necessary pre-flight checklists are done and ATC clearance for take-off is obtained, HUGO will be able to enter the next mission phase. This stage of operations begins when the engines are turned on prior to

take-off and ends when they are turned off post-landing. The objective of this mission phase is to perform the test and obtain measurements requested by the customer.

An overview of the expected in-flight operations is provided in [Figure 4.3](#). After engine turn-on, HUGO will taxi onto the runway and take-off. After climbing to test altitude and reaching test area, it shall accelerate to Mach 0.75 and perform the 5 min test, while transmitting measurement data back to the ground station. Subsequently, it shall be ordered to decelerate, return to the airfield, and perform a landing. It is expected that holding or go-around manoeuvres may be necessary prior to landing. These scenarios will be included in the fuel budget.

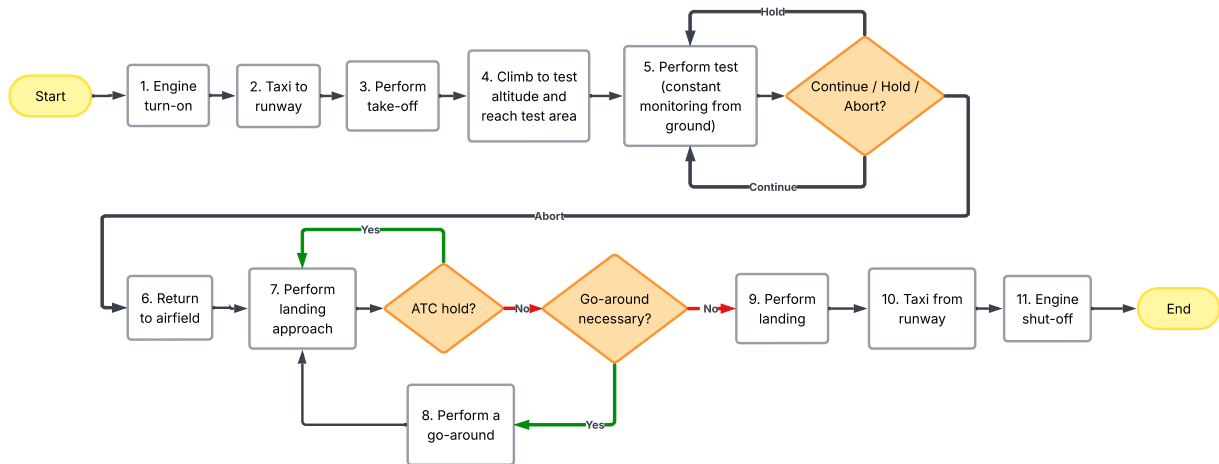


Figure 4.3: *HUGO in-flight operations*

Post-Flight Operations

Once HUGO has performed a successful landing, the post-flight data extraction and operations can commence. In this mission phase, there are two possible scenarios, depending on the mission profile of the day: HUGO can undergo preparations for the next test flight or undergo disassembly and reclaim procedures. An overview of operations for both scenarios are shown in [Figure 4.4](#).

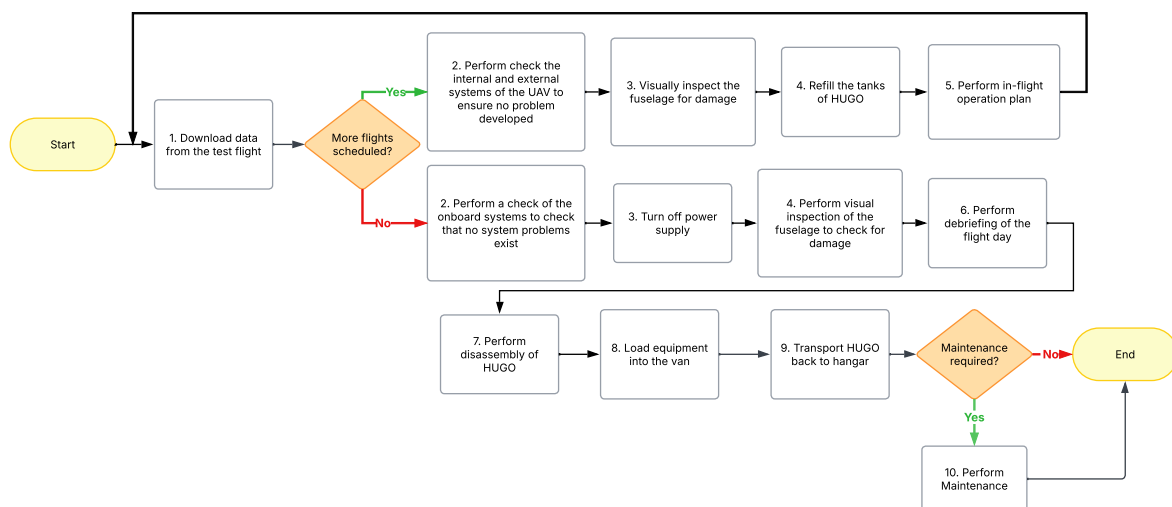


Figure 4.4: *HUGO post-flight operations*

Once HUGO has finished all tests scheduled for a specific flight day, the onboard system checks shall be performed. If no errors are detected, the power supply will be turned off, and an inspection of the external structure shall be performed. Next, the official debrief shall take place, and HUGO shall be disassembled and transported back to the hangar.

The completion of the post-flight operations checklist marks the end of the flight day. If any unforeseen events, anomalies, or deviations occurred during the mission, the flight day shall be followed by an additional debriefing and the completion of the required documentation.

4.1.3. Maintenance

To ensure HUGO's safe and optimal operation, regular maintenance shall be performed, along with regular repairs in case irregularities are found within the workday. Based on considerations made in previous design phases ⁽¹⁾, one HUGO is expected to undergo maintenance for 20% of the working days, which comes down to 50 working days within a year when accounting for free days. Continuing airworthiness is the responsibility of the operator or, in case the operator is not able to comply with the regulations and standards imposed, an outside certified provider ⁽⁶⁾. All maintenance procedures and legislations for HUGO were obtained from EASA documentation on Continuing Airworthiness ⁽⁶⁾, which is applicable for both manned and unmanned aircraft.

General Overview

At this stage of the design, according to the Financial Plan for project HUGO ⁽⁷⁾, the fleet is expected to feature five models in the initial development stage, allowing for the possibility of expansion. Having such a fleet size requires a dedicated location where maintenance can be provided. This problem is ultimately discussed in [section 4.2](#), where the storage of models is expanded upon.

The maintenance operations include: planning of operation, assessment of process, execution of repair/verification, inspection of aircraft, ensure airworthiness, updating the records and release of the model. It is worth stating that modifications brought to the structure for the purpose of fitting a new configuration to be tested will be counted as maintenance. A general maintenance schedule may include the following actions based on the number of flight hours and operating conditions, as defined in [Table 4.1](#).

Table 4.1: Maintenance operations, according to EASA Regulation (EU) No 1321/2014 ⁽⁶⁾

Maintenance Activity	Description
Scheduled maintenance	Planned inspections and maintenance tasks performed at defined intervals, based on flight hours, flight cycles, calendar time, landings, or the aircraft maintenance programme.
Unscheduled or corrective maintenance	Maintenance carried out to correct defects discovered during operation, pre-flight inspections, post-flight inspections, or scheduled maintenance checks.
Airworthiness Directive compliance	Mandatory maintenance or inspection actions required by the competent aviation authority to address unsafe conditions affecting the aircraft, components, or systems.
Structural repair or modification	Repair or modification of the aircraft structure using approved maintenance or repair data, ensuring that the aircraft remains compliant with its certified configuration.

Continued on next page.

⁶ [Easy Access Rules for Continuing Airworthiness](#) (accessed on 11.05.2026)

⁷ [Financial Plan for HUGO](#)

Maintenance Activity	Description
Servicing tasks	Routine servicing activities such as refuelling or battery charging, lubrication, oil replacement, hydraulic fluid checks, tyre pressure checks, filter replacement, and cleaning.
Functional tests and inspections, including test flights	Verification that aircraft systems operate correctly after maintenance, including flight controls, propulsion systems, avionics, sensors, software configuration, safety-critical systems and a flight to assess the aircraft behaviour.

Continuing Airworthiness

As stated previously, it is the operator's responsibility to ensure continuing airworthiness of an aircraft. For operators based in the EU, this typically entails appointing an approved Continuing Airworthiness Management Organisation (CAMO) [4]. The responsibilities of CAMO include: development and control of the maintenance program, monitoring reliability and tracking compliance with Airworthiness Directives (ADs), as well as maintaining records and coordinating with authorities.

The steps towards establishing a cycle for maintaining airworthiness typically begins before the UAV is certified for operations. The competent national authority for operations based in the Netherlands is the Inspectie Leefomgeving en Transport (ILT) (8). The steps towards developing a maintenance cycle include developing safety and system design assurance, and then defining a continuing airworthiness concept. At this stage, the first draft of the Aircraft Maintenance Programme (AMP) is developed [4]. The AMP shall be approved by the appointed CAMO organisation and renewed annually (9).

The AMP is part of Instructions for Continuing Airworthiness (ICA), which define the scope of limitations and maintenance task required to demonstrate that a design is compliant with certification requirements (10). ICA cover both scheduled and unscheduled maintenance procedures, and are necessary for obtaining a Type Certificate (TC). The relevance of a TC shall be explained in chapter 5.

Once the UAV is certified, CAMO takes the responsibility for scheduled maintenance enforcement, AD compliance, reliability tracking and Airworthiness Review Certificates [4]. In case maintenance or repair procedures are required, a Maintenance, Repair, and Overhaul Organization (MRO) approved under EASA Part-145 shall be contracted to perform inspections, repairs, and modifications (11). The MRO is additionally responsible for issuing certificates of release to service.

4.2. Logistics

Logistics represent the backbone of each project, operation, and mission, encompassing the entirety of the unseen processes and actions that bring together the required resources to result in a functional product. For the proper operation of HUGO, a few logistical aspects have to be addressed before it is ready to perform operations. This section addresses the key factors influencing the logistics of operations: required ground crew, storage facility, airfield and ground support, and the part supply chain.

4.2.1. Ground Crew Required for Operation

According to the International Civil Aviation Organization (ICAO)[5], a remote crew comprises licensed crew members whose duties are crucial for the in-flight operations of a remotely-piloted aircraft. This includes the remote pilot in charge of manipulating flight controls, and visual observers (VO) who assist the operations by

⁸ [Inspectie Leefomgeving en Transport - Over de ILT-Luchtvaartautoriteit](#) (accessed on 11.05.2026)

⁹ [EASA - Aircraft Maintenance Programme \(AMP\)](#) (accessed on 11.05.2026)

¹⁰ [Instructions for Continuing Airworthiness \(ICA\): From Certification to Operation](#) (accessed on 11.05.2026)

¹¹ [EASA Part-145 - Maintenance organisation approvals](#) (accessed on 11.05.2026)

visually observing the UAV while it remains within line-of-sight. It is also expected that an ATC controller and personnel responsible for coordinating runway manoeuvres will have to be involved. Since there are no hard requirements regarding the minimum composition of the crew, at this stage of the design it is assumed that HUGO will require one pilot, one VO, and additional personnel as deemed appropriate by the airport authorities.

4.2.2. Storage Facility

On the ground, the aircraft will require an enclosed location to be stored in for maintenance and preparations. As mentioned in [subsection 4.1.3](#), the initial fleet is projected to comprise five HUGOs. This necessitates the use of a large hangar to house parts and machines needed for repairs.

Den Helder Airport has hangar facilities that can accommodate Project HUGO, although the conditions for using these facilities are not specified on the airport website ⁽¹²⁾, Multiple external operators use the airport as a primary base of operations ⁽¹²⁾. Hangar availability is stated to be on request in the EHKD aerodrome information ⁽¹³⁾. The hangar is reported to be 51 m long, 31 m wide, and 11 m high, giving a floor area of approximately 1581 m² [6]. With a maximum wingspan of 15 m for one Project HUGO aircraft, the hangar dimensions are sufficient to place up to three aircraft in line. Once the hangar is secured, it will function as the base of operation for the storage, maintenance and verification of HUGO before and after each flight.

4.2.3. Airfield and Ground Support

Usage of an airfield introduces additional airfield operating costs, ATC coordination requirements, and the mapping of ground routes, taxi areas, and recovery zones. Each drone flight must be approved by De Kooy Air Traffic Control ⁽¹⁴⁾. The departure and landing windows for HUGO will be influenced by the adjacent traffic, necessitating the assumption of multiple missed approaches or holding in the fuel budget. Additionally, Den Helder Airport charges a fee of € 23.00 per landing for a fixed wing aircraft with Maximum Take-Off Mass (MTOM) below 1500kg ⁽¹⁵⁾. Den Helder hosts off-shore flights above the North Sea ⁽¹²⁾, which fits HUGO's mission profile as detailed in [section 4.1](#).

¹²[Den Helder Airport facilities](#) (accessed on 12.05.2026)

¹³[EHKD - Den Helder/De Kooy](#) (accessed on 12.05.2026)

¹⁴[Flight information Den Helder Airport](#) (accessed on 12.05.2026)

¹⁵[Landing fees for fixed wing aircraft at Den Helder](#) (accessed on 12.05.2026)

5. Risk Assessment and Certification

Due to HUGO's unconventional configuration and mission plan, it is important to consider if such a UAV is safe and certifiable by the competent authorities. This chapter focuses on the analysis of technical risks associated with the operation of HUGO, and the steps required to obtain the necessary certification for operations testing the aeroelastic effects of high aspect ratio wings. A qualitative assessment of technical risks is performed in [section 5.1](#). Then, in [section 5.2](#), the classification process is described that places HUGO into a type class category in consultation with EASA. Following the assessment of flight class, the certification for a Design Organisation Approval (DOA) is covered in [section 5.3](#). Establishing a certification basis for HUGO allows for the construction of a preliminary flight envelope [section 5.4](#). Finally, the environmental protection aspect of the certification is expanded in [section 5.5](#).

5.1. Technical Risk Assessment

A preliminary risk assessment for both the development and operation of HUGO has been discussed in the previous design phase ⁽¹⁾, the knowledge of which is assumed here.

Preliminary risk assessment performed in the previous design phase identified five development risks and ten operational risks for HUGO, with likelihood L and severity S scores ranging from 1 to 5, where 1 relates to the least probable/severe outcomes ⁽¹⁾. A more detailed analysis of the design and operations allows for identification of additional risks, their likelihood and expected severity, which can be seen in [Table 5.1](#).

Table 5.1: Continuing development and operation risks

Risk ID	Risk Description	Likelihood	Severity
HUG-RSK-DEV-06	Certification duration exceeding 3 years	2	3
HUG-RSK-DEV-07	Rejection of Certification/DOA application	2	5
HUG-RSK-OP-11	Unavailability of ground crew	3	5
HUG-RSK-OP-12	Ground station equipment failure	2	5
HUG-RSK-OP-13	Fuel shortage at Den Helder airport for HUGO	2	4
HUG-RSK-OP-14	Fuel shortage while holding	2	5
HUG-RSK-OP-15	Runway congestion/lack of ATC clearance	2	3
HUG-RSK-OP-16	Corrupted measurement files	2	4
HUG-RSK-OP-17	Insufficient power to continue tests for the day	2	5

Preventive measures and mitigation strategies developed for the newly identified risks, along with their expected effect on the likelihood or severity, are described in [Table 5.2](#) and [Table 5.3](#), respectively. Preventive measures can be applied before operations take place to reduce the likelihood of the risks occurring. Mitigation refer to the actions that can be taken once the risks occur.

It shall be noted that mitigation strategies cannot be identified for all the risks. This is the case for the development risks associated with certification (HUG-RSK-DEV-06, HUG-RSK-DEV-07), as well as risks dependent on the airport authorities (HUG-RSK-OP-13). In these cases, special emphasis was put on preventive measures.

Table 5.2: Risk preventive measures

Risk ID	Preventive Measure(s)	ΔL
HUG-RSK-DEV-06	Ensuring that the documentation is complete, and all design details comply with relevant regulations and means of compliance.	-1
HUG-RSK-DEV-07	Maintaining contact with EASA and other authorities to resolve any potential issues regarding certification.	-1

Continued on next page.

Risk ID	Preventive Measure(s)	ΔL
HUG-RSK-OP-11	Regular check-ups on crew, establishing reliable communication channels, pre-defining absence regulations, scheduling holidays and foreseen leaves in advance.	-2
HUG-RSK-OP-12	Regular maintenance and equipment checks. Preparing backup equipment.	-1
HUG-RSK-OP-13	Having personnel on standby to supply sustainable aviation fuel (SAF) as soon as a fuel shortage is assessed.	-1
HUG-RSK-OP-14	Implementing a alerting system and constant monitoring of the fuel system.	-1
HUG-RSK-OP-15	Pre-scheduling flights with De Kooy ATC and maintaining communications according to airport regulations.	-1
HUG-RSK-OP-16	Implement continuous data transmission, prepare a backup storage plan for in-flight measurements.	-1
HUG-RSK-OP-17	Regularly checking the condition of the battery	-1

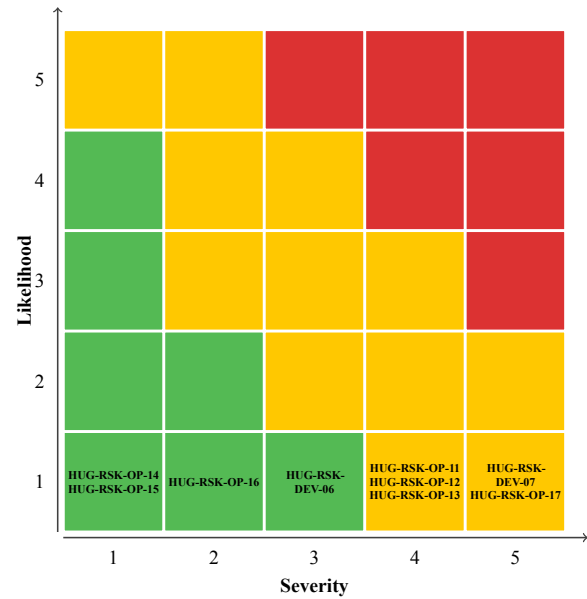
Table 5.3: Risk mitigation strategies

Risk ID	Mitigation Strategy	ΔS
HUG-RSK-OP-11	Appointing pilots/crew members who can be called upon in case of emergency.	-1
HUG-RSK-OP-12	Using the backup equipment, requesting assistance from available airport facilities.	-1
HUG-RSK-OP-14	Sending a "MAYDAY FUEL" emergency signal to the ATC to ensure priority landing.	-3
HUG-RSK-OP-15	Requesting take-off with ATC, asking for clarification on when a take-off window will become available. Including a buffer in the test day planning to account for delay take-off and landings.	-2
HUG-RSK-OP-16	Accessing the backup files, saving the continuously transmitted data during operations.	-2

Applying the preventive measures and mitigation strategies to the newly identified risks results in a mitigated risk list shown in [Figure 5.4](#). The corresponding mitigated risk matrix is shown in [Figure 5.1](#). As can be seen, all of the new risks fall within the green or yellow zone, and therefore are deemed acceptable, provided that all of the identified preventive measures and mitigation strategies are implemented.

Table 5.4: Final development and operational risk assessment

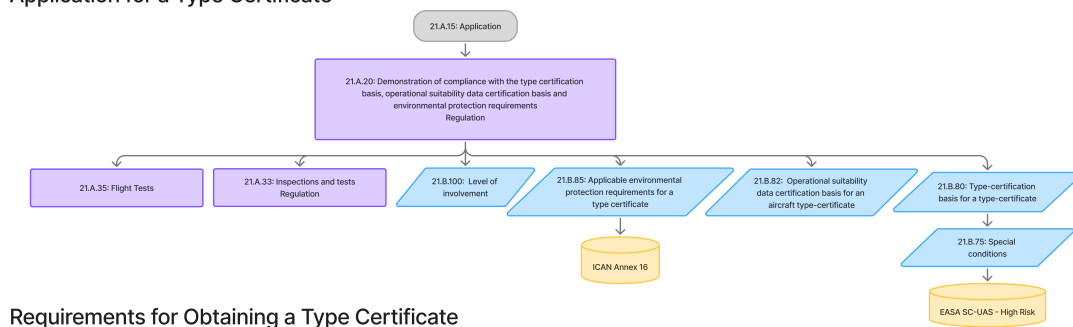
Risk-ID	Likelihood	Severity
HUG-RSK-DEV-06	1	3
HUG-RSK-DEV-07	1	5
HUG-RSK-OP-11	1	4
HUG-RSK-OP-12	1	4
HUG-RSK-OP-13	1	4
HUG-RSK-OP-14	1	1
HUG-RSK-OP-15	1	1
HUG-RSK-OP-16	1	2
HUG-RSK-OP-17	1	5

**Figure 5.1:** Final risk matrix

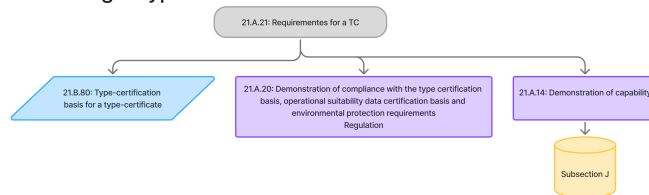
5.2. Type Certificate

According to the EASA eRules for Initial Airworthiness and Environmental Protection (eRules), the Type Certificate (TC) is a document issued by the competent authority declaring an aircraft meets all required airworthiness standards [3]. The eRules describes the process and all necessary documents to obtain a type certificate inside Annex I. Section A details all the obligations of the **applicant**, and section B details those of the **competent authority (EASA)**. The eRules manual is difficult to navigate, as such, Figure 5.2 will serve as a visual guide for the reader. The grey blocks indicate the starting point, purple indicates the applicant's obligations, blue indicates the authority's obligations, and yellow indicates any additional sources of information.

Application for a Type Certificate



Requirements for Obtaining a Type Certificate

**Figure 5.2:** eRules steps for the requirements, and procedure for obtaining a Type Certificate

The requirements for a TC are given in section (21.A.21). They state that the following must be met:

1. demonstration of capabilities (21.A.14),
2. demonstration of compliance with the type certification basis, operational suitability data certification basis and environmental protection requirements (21.A.20),
3. demonstration of compliance with UAS (Unmanned Aircraft System) Type Certification basis (21.B.80).

Section (21.A.14) will be further elaborated in [section 5.3](#), as it pertains to the process for obtaining a Design Organisation Approval (DOA) under 21.A.14(a), with further guidance in Subsection J. Both (21.A.20) and (21.B.80) are covered under the section (21.A.15): Application for a TC, and will be expanded on below.

As previously stated, (21.A.15): Application for a TC, requires compliance with section (21.A.20): Demonstration of compliance with the type certification basis, operational suitability data certification basis and environmental protection requirements Regulation.

5.2.1. Compliance with certification basis, operational suitability, environmental protection

For compliance with (21.A.20) after the certification programme has been accepted by EASA, HUGO's design will have to demonstrate compliance in the following areas: type-certification basis for a TC or restricted TC (21.B.80), operational suitability (21.B.82), and environmental protection (21.B.85).

According to (21.B.80) EASA has determined HUGO falls partially under CS-23 aerobatic, but not fully. As HUGO falls partially outside of the scope of the eRules, special conditions covered under (21.B.75) also apply. For HUGO, EASA instructed the use of the Special Condition for Light Unmanned Aircraft Systems - High Risk [7]. Additionally, HUGO falls outside the scope of (21.B.82) due to its unconventional design and operations.

Section (21.B.85), detailing the necessary compliance with environmental protection requirements as dictated in Regulation (EU) 2018/1139 Annex III, further expanded on in [section 5.5](#).

Sections (21.A.33) and (21.A.35) contains requirements for the applicant regarding inspections and tests regulation, and flight tests respectively. Lastly, (21.B.100) details the authority's obligation to provide guidance on the "Level of involvement", and the likelihood and impact of non-compliance.

With a certification basis established, the remainder of the requirements for the application of a TC achievable, it can be determined that HUGO's intended mission is not regulatorily inviable.

5.2.2. Special Conditions for (Unmanned Aircraft System)

The special conditions detailed under the Special Condition for Light Unmanned Aircraft Systems - High Risk define additional requirements for: Design and Construction, and Systems and Equipment. The scope of these additional requirements covers risk of failure, environmental hazards, robustness in the event of a lightning strike, and additional High Intensity Radiated Fields (HIRF) protection [7].

5.2.3. Application for a Type Certificate

As mentioned in [section 5.2](#), HUGO's application for the TC is done in accordance with (21.A.15) [3]. The detailed procedure after the project is in compliance with (21.A.21) is detailed below:

According to the guidelines, the TC application should be made in a format stipulated by the authority, including as a minimum: a preliminary description of the UAS or the Control and Monitoring Unit (CMU), and the kind of operations for which certification is requested. In addition it shall comply with 21.A.20, which outlines the following:

1. a detailed description of the design, including all the configurations to be certified,

2. the proposed operating characteristics and limitations,
3. the intended use of the UAS or the CMU, and the kind of operations for which certification is requested,
4. a proposal for the initial TC basis, operational sustainability data certification basis, and environmental protection requirements in accordance with (21.B.80), (21.B.82), and (21.B.85),
5. a proposal for the breakdown of the certification program into meaningful groups of compliance demonstration activities and data, including a proposal for the means of compliance and related compliance documents,
6. a proposal for the assessment of meaningful groups of compliance demonstration activities and data addressing the likelihood of an unidentified non-compliance with the TC basis operational sustainability data certification basis, or environmental protection requirements and the potential impact on the non-compliance on the UAS safety or environmental compatibility or on the safety of the CMU. The proposed assessment shall take into account at least the elements set out in points 1 to 4 of (21.B.100(a)),
 - (a) novel or unusual features of the certification project, including operational, organisational and knowledge management aspects;
 - (b) complexity of the design and/or demonstration of compliance;
 - (c) criticality of the design or technology and the related safety and environmental risks, including those identified on similar designs; and
 - (d) performance and experience of the design organisation of the applicant in the domain concerned.

5.3. Design Organisation Approval (DOA)

This section discusses the approach towards certifying modifications to the original configuration of HUGO, following a successful application for a Type Certificate. Special attention is dedicated towards the DOA, its implications for the feasibility of HUGO as a flying testbed, and the necessary steps towards obtaining necessary approval from EASA.

5.3.1. Certification of Design Modifications

For the certification procedure of an aircraft, the competent authority in Europe, EASA, requires extensive documentation and validation procedures of the design. For this stage several EASA-imposed operational restrictions must be considered, including the applicable airworthiness class and the permitted operational environment [3].

DOA application is a step in the process of obtaining a Type Certificate, according to 21.A.14 (a), capability demonstration. Along with 21.A.20 (compliance documentation) and 21.B.80 (configuration demonstration), these three documents will facilitate the application for a Type Certificate.

The application for a DOA involves cooperation with EASA from early stages of the design. The steps necessary for successful certification are ⁽¹⁶⁾: technical familiarisation and certification basis, establishment of the certification programme, compliance demonstration, and technical closure and issue of approval.

Costs involved in this process and the time required for analysis of the design are not well defined, but can range from months to years, depending on the complexity of the concept ⁽¹⁷⁾. This is direct with conflict with HUGO's mission, which requires frequent structural modifications and configuration changes. Applying for a Supplemental Type Certificate or completely re-certifying HUGO for each test case due to design changes

¹⁶[EASA Aircraft Certification](#) (accessed on 04.05.2026)

¹⁷[EASA - From idea to the sky: how EASA ensures that aircraft is safe to fly](#) (accessed on 12.05.2026)

would result in an excess of downtime of the UAV, loss in revenue, and ultimately loss of purpose as a research test-bed [8].

According to EASA classification, re-certification is required each time a change is implemented into the design of an aircraft, with a clear distinction between "minor changes" and "major changes" (21.A.91) [3]. A "minor change" is classified as any type of modification that does not show any discernable effect on weight, structural strength, reliability and other specifications. A change that is not a "minor change" is considered as a "major change". This distinction determines the steps that needs to be taken in order to certify modifications to the design of an aircraft.

5.3.2. The Importance of DOA for HUGO's Mission

The expected structural changes in the wing and tail platforms between experiments are considered as "major changes" according to the EASA classification discussed previously. Typically, this would require direct supervision from EASA, significantly increasing the time and resources required to ensure that the new configuration can be classified as airworthy. To facilitate this process, a DOA application can be made (18).

A DOA is a certification that permits the holder to perform the certification process internally, reducing the workload of EASA to documentation approval as the certification authority. In the context of HUGO, this would allow the design team to internally implement all "minor changes" without unnecessary delay and with minimal dedicated resources. Additionally, "major changes" to the platforms can be classified as research procedures, excluding the need for full re-certification and reducing EASA's involvement (19).

In conversation with Hugo Manuel Rodrigues Lima da Silva, an EASA expert, it was established that in order for HUGO's design team to obtain a DOA: a specific department shall be formed within EASA that will obtain the permission to supervise the changes brought to HUGO, perform required validation steps, and ensure compliance with the applicable certification basis. It was additionally established that personnel comprising of three EASA employees shall be sufficient given the complexity of HUGO's design.

The pricing of a DOA is set at € 10,170 for the initial certification, after which a 12-month billing cycle fee of € 11,050 is charged by EASA to continue oversight of the approved organisation (20).

The average time required to obtain a DOA was estimated as one year, provided that all procedures have been properly developed. The certification process itself is more dependent on the schedule of the design team and EASA employees, with an estimate duration of three years. Notably, both the DOA and the certification processes can run at the same time.

5.4. Flight envelope

The flight envelope of the HUGO aircraft in symmetric flight is based on EASA CS-23 airworthiness requirements for aerobatic aircraft [9]. According to the specifications, except when limited by the stall speeds, the positive limit manoeuvre load factor is 6.0. The negative limit manoeuvre load factor is allowed to vary linearly from -3.0 at the cruise speed up to -1.0 at the dive speed. Additionally, symmetrical vertical gust loads are superimposed on the manoeuvre flight envelope, with gust velocities specified in [9].

Due to the specific flight profile of this aircraft, with essentially two cruise stages (since the testing phase is performed at a separate flight condition from the rest of the cruise), two diagrams have been created, as shown in Figure 5.3, when assuming parameters from Table 5.5. Both diagrams have to be superimposed when designing the structure to ensure that all the critical conditions are covered. Clearly, the gust loads are not constraining for

¹⁸ [EASA - Design Organisations Approvals](#) (accessed on 05.06.2026)

¹⁹ In conversation with EASA employee

²⁰ [EASA Prices](#) (accessed on 13.05.2026)

this aircraft design, due to the high manoeuvre load requirements associated with aerobatic aircraft. Nonetheless, the required structural performance is expected to significantly influence the material selection, as well as the final configuration selection. These will be addressed in [chapter 6](#) and [chapter 7](#), respectively.

Table 5.5: Parameters used for generating the flight envelope diagrams (DATCOM⁽²¹⁾ used for lift curve slope)

(a) Aircraft parameters

Parameter	Value	Unit
Mass	50.0	kg
Aspect ratio	25.0	—
Wing span	2.0	m
Quarter-chord sweep angle	15.0	°
Taper ratio	1.0	—
Positive $C_{L_{\max}}$	1.087	—
Negative $C_{L_{\max}}$	-1.087	—
Wing lift curve slope ([10] and DATCOM)	7.15	rad ⁻¹

(b) Flight conditions

Parameter	Value (condition 1)	Value (condition 2)	Unit
Cruise altitude	5500.0	8230.0	m
Cruise Mach number	0.45	0.75	—
Dive Mach number	0.50	0.80	—

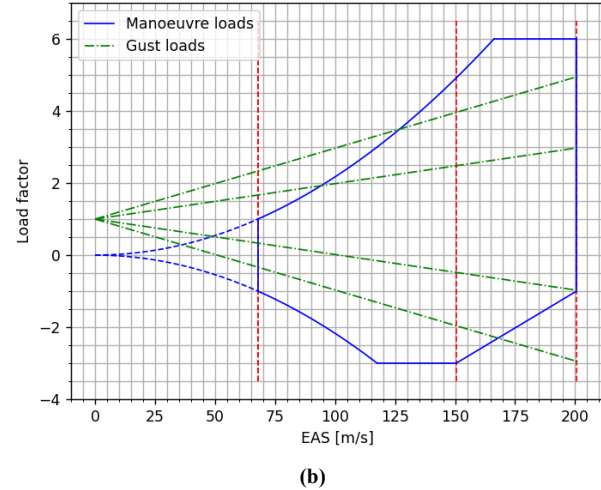
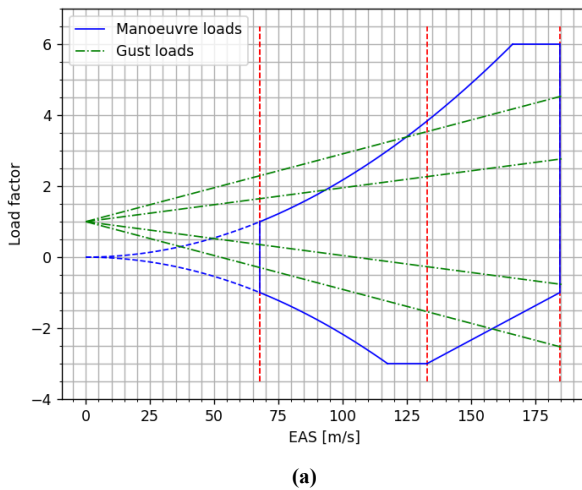


Figure 5.3: Flight envelope diagrams for parameters from [Table 5.5](#), compliant with CS-23 aerobatic category requirements [9].

5.5. Environmental protection

Environmental compatibility was introduced in the European legislation Regulation (EU) 2018/1139⁽²²⁾. The intention was to create a common framework between member states for the environmental certification of aircraft. HUGO is exempt from those rules as stated in recital 1 in Regulation (EU) 2025/870⁽²³⁾: “Aircraft, other than unmanned aircraft, and their engines, propellers, parts and non-installed equipment should comply with the environmental protection requirements. Regulation (EU) 2018/1139” A specific rule-set has been established for unmanned aircraft: Regulation (EU) 2018/1139⁽²²⁾ of the European Parliament and of the Council in Annex IX: “Essential Environmental Requirements for Unmanned Aircraft” states that unmanned aircraft shall comply with the environmental performance requirements set out in Annex III. Looking at Annex III, the main point is a general approach for the design and trade-off between minimising noise and emissions, which has to be taken into account in the area of operation.

²¹ [Scholz](#) (accessed 18.05.2026)

²² [EU Document 32018R1139](#) (accessed on 13.05.2026)

²³ [EU Document 32025R0870](#) (accessed on 13.05.2026).

6. Engineering Design and Analysis

To provide the necessary quantitative background for the trade-off, a preliminary sizing of the design options is performed. [section 6.1](#) describes how statistics from the reference aircraft are used to specify the initial values for design iteration. [section 6.2](#) describes the aerodynamic model used to obtain the lift and drag performance of the aircraft. [section 6.3](#) describe determining the fuel mass fraction, [section 6.4](#) - the empty mass fraction, and [section 6.5](#) - the design point (wing loading and thrust-weight ratio). [section 6.6](#) and [section 6.7](#) cover the design of the empennage and landing gear subsystems. The robustness of the sizing is then assessed in [section 6.8](#) by means of a sensitivity study.

6.1. Airfoil and wing analogous estimation

According to HUG-OPS-005 ⁽¹⁾, HUGO is required to operate at speeds up to Mach 0.75. This is the flight condition at which tests are to be performed. In this section, an estimation of the lift characteristics of HUGO is performed based on statistics from suitable reference aircraft, along with a preliminary airfoil selection suitable for the operations.

The airfoil selection begins with an analysis of reference aircraft and an estimate of the airfoil design lift coefficient $C_{l_{des}}$. Subsequently, the maximum lift coefficient $C_{l_{max}}$ is derived along with relevant aerospace parameters to facilitate the selection. For the purpose of this analysis, it is assumed that HUGO will operate at a speed of Mach 0.75, at an altitude of FL270 (8230 m).

6.1.1. Reference aircraft parameters

For initial estimates, the NASA DAST project ⁽²⁴⁾, NASA X-56A UAS ⁽²⁵⁾ and the FLEXOP UAS [11] are used as reference aircraft. DAST was initially developed with the purpose of remotely testing new aerodynamic designs in the 1980s, featured replaceable wings, and to this day is the only UAV that has approached HUGO's flight speed ⁽²⁴⁾. The X-56A is a NASA testbed flying-wing UAS used to test aeroelastic phenomena on high aspect ratio wings in a low-speed flight regime, and was similar to HUGO in mass and size [12]. Similarly, the FLEXOP UAS was used as a modular testbed with interchangeable wings for researching aeroelastic phenomena and design of flutter-suppressing control systems [13]. Although significantly slower than HUGO, FLEXOP's scale and mission description make it a suitable reference.

The wing design lift coefficient $C_{L_{des}}$ is determined based on an initial wing loading W/S estimate and requirements from previous design phases ⁽¹⁾ at the specified cruise flight condition. [Table 6.1](#) lists the relevant geometric parameters of reference aircraft, as well as their resulting statistical estimation for HUGO.

According to the operations and logistics description provided in [chapter 4](#), HUGO has to perform high subsonic flight, and take-off or land on a paved runway. An initial value of $W/S = 1000 \text{ N} \cdot \text{m}^{-2}$, which lies within the range of the reference aircraft, allows for a sufficiently low stall speed to ensure operations safety. The mass m and aspect ratio AR are determined based on the mass requirement **HUG-GEN-001** and the aspect ratio requirement **HUG-AER-002** ⁽¹⁾. From this requirement, the AR of 23 was chosen as an average of the range specified by the NLR (between 19 and 27) ⁽²⁶⁾. Consequently, the wing area S was obtained from the wing loading. From these parameters, the wing span b and mean chord length MAC can be derived.

The taper ratio $\lambda = 0.5$ for HUGO was selected based on guidelines by Torenbeek [1] and lies within the upper bound considered for jet transport aircraft in a similar flight regime. A slightly larger taper ratio is introduced due to the high aspect ratio of the wing. Torenbeek suggests that the sweep angle is determined primarily by compressibility effects and should increase progressively above a Mach number of 0.8 [1]. An initial leading edge sweep of $\Lambda_{LE} = 15^\circ$ was selected, similar to that of the X-56A and FLEXOP aircraft. This is considered

²⁴[NASA - Firebee Drone: DAST](#) (accessed on 07.05.2026)

²⁵[NASA X-56 UAS testbed](#) (accessed on 08.05.2026)

²⁶In conversation with NLR engineer

a good balance between obtaining reasonable low-speed lift characteristics and delaying drag divergence at the test velocity.

Several additional wing parameters, such as the MAC and wing area, were obtained using the AeroToolbox website (27) and Torenbeek [1].

Table 6.1: Reference aircraft and HUGO initial parameters [11] [13] [14] [15]

Parameter	Symbol	DAST ARW-2	X-56A	FLEXOP	HUGO
Wing span [m]	b	5.79	8.53	7	3.36
MAC [m]	$\bar{c}$	0.56	0.61	0.35	0.15
Aspect ratio [–]	AR	10.3	14	20	23
Taper ratio [–]	λ	0.4	0.46	0.5	0.5
Leading edge sweep [°]	Λ_{LE}	28.8	22	20	15
Wing area [m ²]	S	3.25	5.20	2.45	0.49
Mass (maximum take-off) [kg]	m	1060	238	65	50
Wing loading [N·m ⁻²]	W/S	3216	449	260	1000

The fuselage dimensions are taken from that of FLEXOP drawings [16]. The nose cone length is measured to 0.55 m, the tail cone to 1.12 m, and the total fuselage length was 3.42 m. The fuselage diameter was established as 0.315 m. From the drawing, the tail arm could also be deduced as 2 m.

6.1.2. Design Aerodynamic Parameters

The initial geometric parameters from Table 6.1 can be used to derive HUGO's preliminary aerodynamic characteristics. Those parameters are also used as an initial condition for the iterative sizing, with exception of the aspect ratio, which is set to 27, the most constraining case.

Design Lift Coefficient

To estimate $C_{L_{des}}$, Equation 6.1 [1] can be applied for a cruise air density ρ_{cr} of $0.5112 \text{ kg} \cdot \text{m}^{-3}$ at FL270, and a cruise speed V_{cr} of $230.2945 \text{ m} \cdot \text{s}^{-1}$ corresponding to the test condition. Using the initial value for W/S from Table 6.1, an initial $C_{L_{des}}$ of 0.074 is obtained.

$$C_{L_{des}} = \frac{\left(\frac{W}{S}\right)}{\frac{1}{2}\rho_{cr}V_{cr}^2} \approx 0.074 \quad (6.1)$$

6.1.3. Airfoil Selection

Since the aircraft is expected to operate at Mach 0.75, it is necessary to take transonic phenomena into account. When flying in this region with a traditional airfoil, the airflow might locally be supersonic, which creates strong shock waves on the wing [17]. This leads to a large increase in drag, which is detrimental for the performance of the aircraft and introduces boundary layer separation. Such effects can be mitigated by the use of a supercritical airfoil.

A supercritical airfoil is capable to operate at transonic conditions with only the onset of weak shock waves [18]. An example of such airfoils used in the preliminary design for HUGO is a supercritical SC(2)-XXYY family developed by NASA, where XX is the airfoil design lift coefficient, and YY is the maximum thickness in percent [18]. The advantage of using this system is the inclusion of the design lift coefficient, which makes it possible to narrow down the choices for HUGO using $C_{L_{des}}$ values found in subsection 6.1.2.

²⁷ AeroToolbox wing plotting tool (accessed on 12.05.2026)

Based on the data obtained in the previous subsections, NASA SC(2)-0012 was chosen as the preliminary airfoil for HUGO. It is a symmetrical, supercritical airfoil, with properties matching the requirements for HUGO's mission. The choice follows from the NASA report on supercritical airfoils [18], as well as $C_{l_{des}}$ of 0.074

A symmetrical supercritical airfoil is beneficial for low lift coefficient flight conditions. A symmetrical supercritical airfoil provides the desired lift coefficient at low angles of attack. Moreover, it does not create significant negative pitching moments that necessitate the use of a larger empennage, as opposed to their non-symmetrical counterparts (28). A high thickness-to-chord ratio t/c is expected to alleviate bending loads on the slender wing of HUGO. Although high t/c ratio airfoils are at a relatively high risk of flow separation and weak shock waves forming further from the trailing edge, the reported Mach divergence numbers M_{dd} at which those occur are above the operational regime of HUGO [18].

Cruise Reynolds number

Based on the initial estimates from Table 6.1, as well the cruise conditions ρ_{cr} , V_{cr} and dynamic viscosity μ_{cr} corresponding to Torenbeek's ISA (International Standard Atmosphere) model [19], a cruise Reynold's number Re_{cr} can be calculated using Equation 6.2 [20].

$$Re_{cr} = \frac{\rho_{cr} \bar{c} V_{cr}}{\mu_{cr}} \approx 1.15 \cdot 10^6 \quad (6.2)$$

Maximum Lift Coefficient

The maximum lift coefficient, $C_{L_{max}}$ has to be determined to obtain the stall speed of HUGO, V_{stall} . As stated previously, the use of a supercritical wing profile is highly recommended for the considered test conditions. A preliminary estimate of $C_{L_{max}}$ is made by considering a NASA SC(2)-0010 airfoil and the DATCOM method for estimating the maximum lift coefficient for finite wings.

An estimation of $C_{L_{max}}$ can be made using the software Xfoil (29), and an expression for estimating the wing $C_{L_{max}}$ from the $C_{l_{max}}$ of the wing profile. For a $Re \approx 1.15 \cdot 10^6$ and a critical amplification factor N of 5, a $C_{l_{max}} \approx 1.4$ was found (30). The N value is deemed suitable for this use case due to likely disturbed airflow from the wing surface roughness [21]. Then, Equation 6.3 is used to obtain $C_{L_{max}}$ [1, p. 153].

$$C_{L_{max}} = 0.9 \cdot C_{l_{max}} \cdot \cos \Lambda_{0.25c} \approx 1.22 \quad (6.3)$$

Conversion from the leading edge sweep angle Λ_{LE} to the quarter-chord sweep angle $\Lambda_{0.25c}$ is derived from Torenbeek [1, p. 440], and yields $\Lambda_{0.25c}$ of 14.22 deg. This leads to a $C_{L_{max}}$ value of approximately 1.22, not accounting for the use of high-lift devices.

Stall speed

The next step is to identify the preliminary stall speed of HUGO. With a maximum lift coefficient $C_{L_{max}} \approx 1.22$ and $S = 0.49 \text{ m}^2$ at a mass $m = 50 \text{ kg}$ and sea level ISA conditions. The stall speed V_{stall} can be calculated using Equation 6.4.

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L_{max}}}} \approx 37 \text{ m} \cdot \text{s}^{-1} \quad (6.4)$$

Compared to the stall speed of the X-56A UAS of 33 m/s (25), the result appears reasonable for an initial estimation.

²⁸[NASA SC\(2\)-0412](#) (accessed on 11.05.2026)

²⁹[AirfoilTools](#) (accessed on 07.05.2026)

³⁰[NASA SC\(2\)-0010 airfoil](#) (accessed on 11.05.2026)

6.2. Aerodynamic modelling

In order to perform preliminary airplane sizing, including steps such as a matching diagram, it is vital to know the lift-to-drag-ratio and the breakdown between induced and parasite drag [22]. In order to achieve this, aerodynamic analyses using the AeroSandbox⁽³¹⁾ Python package focused on inviscid modelling to obtain the lift and induced drag, while the parasite drag was obtained via drag the build-up method.

6.2.1. Parasite drag estimation

As mentioned previously, the parasite drag (C_{D_0}) is estimated using the drag build-up method described by Raymer [23]. The subsonic parasite drag of each component is estimated separately, and then summed to obtain the total C_{D_0} . The component drag is a product of the wetted area S_{wet} , calculated flat-plate skin-friction drag coefficient C_f , form factor FF that estimates the pressure drag due to viscous separation, and interference component Q that estimates the increase in drag due to the interference between the different components, as shown in Equation 6.5 [23, p. 281].

$$(C_{D_0})_{\text{subsonic}} = \frac{\sum (C_{f_c} \cdot FF_c \cdot Q_c \cdot S_{wet_c})}{S_{\text{ref}}} + C_{D_{\text{misc}}} + C_{D_{L\&P}} \quad (6.5)$$

Here, $C_{D_{\text{misc}}}$ is miscellaneous drag due to flaps, un-retracted landing gear etc., and $C_{D_{L\&P}}$ is an estimate of the drag due to leakages and protuberances. S_{wet} is the wetted area, and S_{ref} is the reference area. The subscript c refers to a single component.

The C_f is estimated as a weighted average of the laminar and turbulent flow values, and can be calculated according to Raymer [23, p. 282]. In case the skin roughness is high, the C_f value can be expected to increase. An adjusted value can be calculated by introducing a cut-off Reynold number $Re_{\text{cut-off}}$ for subsonic flight [23, p. 282], for a given characteristic length and skin roughness.

Form factors FF for various components can be calculated according to Raymer, depending on the relevant geometric parameters and fineness ratio [23, p. 283]. Since the interference of the landing gear with the rest of the components can be assumed to be negligible, it was justified that a separate parasite drag estimation would be used. The landing gear C_{D_0} is estimated based on the method presented in Roskam [24]. Then, the leakage and protuberance drag of each component is estimated as a percentage of the total parasite drag. The exact values obtained from Raymer [23].

6.2.2. Aerodynamic model in Aerosandbox

The goals of the analyses in Aerosandbox were to obtain the lift and induced drag of the aircraft required for the later calculation of the L/D ratio, the lift curve slopes of all the lifting surfaces (both individually and combined with other lifting surfaces), the lift distribution along the span, the aerodynamic centre position, and the determination of the angle of attack required for trimming. For this purpose, the lifting-line theory (LLT) with modifications for nonzero sweep in AeroSandbox has been used. Since this analysis accounts for viscous effects, the viscous components of drag have been removed in the source code for the purpose of this project. To ensure that no errors are introduced by this modification, the results of the inviscid analysis have been compared with that of the standard viscous analysis⁽³²⁾. Also, the code corresponding to the rest of the performed simulations has been verified via unit tests⁽³²⁾.

The model has been created parametrically, with planform parameters, airfoils, and relative positions and orientations between the lifting surfaces used as parameters. A lifting surface is generated by specifying the properties of an arbitrary number of sections along its span. Each section is, in turn, specified by the position of its leading edge, chord length, the airfoil, and twist. The sections in this parametric model have been spaced uniformly to

³¹ [AeroSandbox](#) (accessed 18.05.2026)

³² [HUGO Project Repository](#) (accessed on 21.05.2026)

simplify the extraction of the lift distribution results. Once all lifting surfaces are specified, the aircraft model is generated by combining the surfaces in a single airplane object. This object can include a single lifting surface (the main wing in isolation, for example), as well as all the lifting surfaces combined (main wing, canards, empennage). The number of main wing sections used was 30, and verified against 100. Other surfaces were modelled with 5 sections due to their significantly smaller size.

To create the LLT-based aerodynamic model, the airplane model, atmospheric properties, and flight conditions (angle of attack, sideslip angle, airspeed and rotational rates) are required as inputs. Since the results provided by AeroSandbox exclude the lift distribution, the positions of vortex centres and vortex strengths have been extracted from the source code. Subsequently, the Kutta–Joukowski theorem ($L = \rho VT$) has been employed to compute the lift per unit span at each section. To find the lift curve slope, the analysis has been run for a range of angles of attack. Furthermore, since the trim analysis requires a specific angle of attack, additional steps have been employed. While the definition of the trim condition entails both vertical force and pitch moment equilibria, this analysis has only considered vertical force equilibrium due to the preliminary nature of the design (no elevators, empennage assumed fixed for simplicity). The required angle of attack has been obtained by finding the required lift coefficient for a specific flight condition, running the simulation for a range of angles of attack, and then calculating the angle of attack for obtaining the required lift, while assuming no lift at $\alpha = 0$ deg. Finally, the position of the aerodynamic centre has been extracted from the results provided by AeroSandbox.

Furthermore, model verification has been performed by comparing the obtained wing $C_L - \alpha$ curves with those obtained by Prandtl’s formula for high-aspect-ratio wings [25, p. 443] in the incompressible regime, and with those obtained by the DATCOM ⁽³³⁾ method in the compressible regime. Also, two different symmetric airfoils have been used: the supercritical NASA SC(2)-0012 ⁽³⁴⁾ the one actually used by the aircraft, and NACA0012 ⁽³⁵⁾ as the reference airfoil having the same maximum thickness-to-chord ratio. These results are shown in Figure 6.1. Clearly, the results in the compressible regime match with accuracy better than 0.04 up to $\alpha = 4$ deg for NASA SC(2)-0012 and with accuracy better than 0.05 up to $\alpha = 5$ deg for NACA0012. These differences are due to the different equations used by DATCOM (empirical) and Aerosandbox (LLT). Beyond $\alpha = 4$ deg for NASA SC(2)-0012 and $\alpha = 5$ deg for NACA0012, the Aerosandbox results become highly nonlinear, which is as expected. In the incompressible regime, the difference is lower than 0.1 for angles of attack up to $\alpha = 5$ deg for both airfoils. These differences are due to the fact that Prandtl’s formula assumes linear slope of the lift curve, whereas in the Reynolds regime of these wings, the lift curve slopes are nonlinear ⁽³⁴⁾⁽³⁵⁾. Due to analyses from the following sections assuming a linear lift curve, measurements of $C_{L\alpha}$ for those methods will be performed at angles of attack up to 4 deg.

Moreover, root-mean-square (RMS) error with respect to the reference (Prandtl, DATCOM) has been computed for a range of taper ratios and sweep angles for two different airfoils and in two different flight regimes, as shown in Figure 6.2. Clearly, maximum RMS in the incompressible regime does not exceed 0.06 for any of the airfoils, whereas in the compressible regime it remains below 0.1 for NASA SC(2)-0012 and below 0.032 for NACA0012. For the former airfoil in incompressible regime, the difference is the lowest for high sweep angles and low taper ratios, whereas the opposite is true for the latter airfoil. This is fully attributed to the reference methods not accounting for the features of supercritical airfoils. Opposite but similar trends can be observed for the compressible regime, which is again attributed to the limitations of the reference methods.

Finally, the calculation of the position of the aerodynamic centre as calculated by Aerosandbox has been verified by independently implementing the procedure. Namely, for a swept, tapered trapezoidal wing, the mean aerodynamic chord (MAC) of each wing section is first computed, and the aerodynamic centre of each section is

³³ [Scholz](#) (accessed on 18.05.2026)

³⁴ [Airfoil Tools, NASA SC\(2\)-0012](#) (accessed on 29.05.2026)

³⁵ [Airfoil Tools, NACA0012](#) (accessed on 29.05.2026)

assumed to be at 25% of the MAC. Subsequently, the weighted average of all the sections' aerodynamic centres is computed based on their surface area. This has been implemented as a part of unit tests ⁽³²⁾, by asserting the difference to remain below 1%.

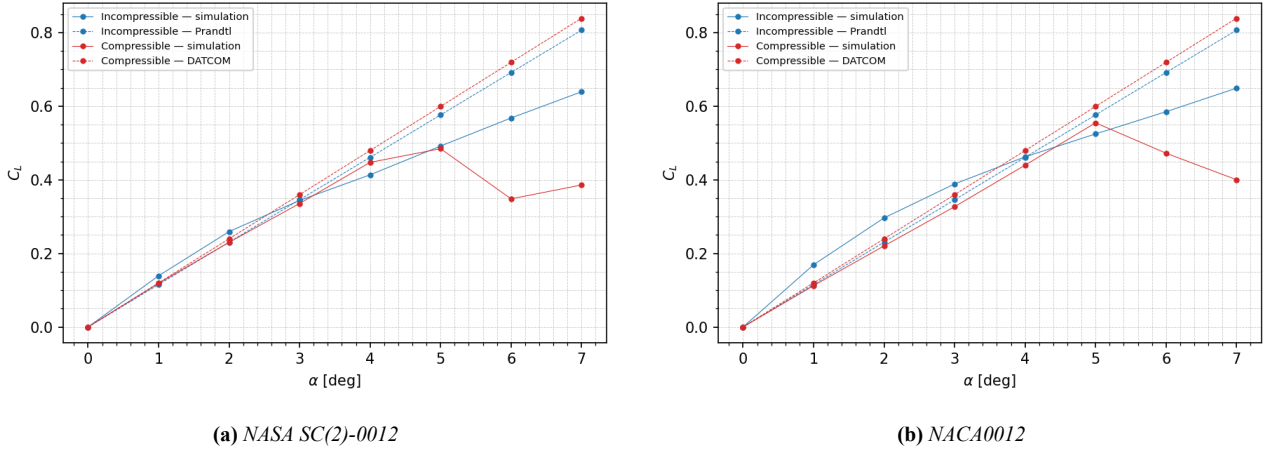


Figure 6.1: Lift coefficient as a function of angle of attack for two different airfoils. Both plots feature the simulated (Aerosandbox) and reference (Prandtl, DATCOM) lift coefficient results for incompressible (speed of $30.0 \frac{m}{s}$) and compressible (speed of $200.0 \frac{m}{s}$) flight regimes at standard sea level atmospheric conditions. Both wings feature the aspect ratio of 27.0, taper ratio of 0.5, span of 2.0m, and quarter-chord sweep of 15 deg.

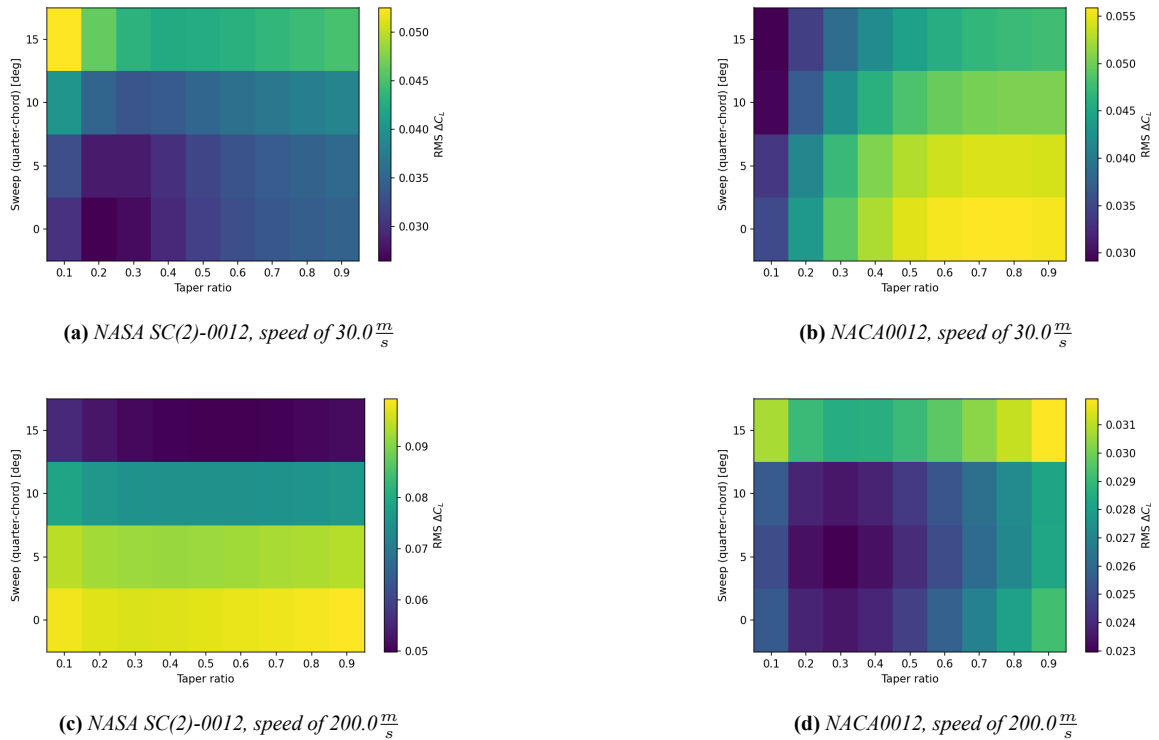


Figure 6.2: RMS error in lift coefficient for angles of attack between 0 deg and 5 deg for two different airfoils at two different flight regimes, as a function of sweep angle and taper ratio. Prandtl formula and DATCOM were used for computing reference values.

6.3. Fuel Mass Fraction Estimation

To estimate the fuel required for HUGO to perform its mission, one can express the speed and time requirements for different stages of the flight as range requirements. Then, a procedure for sizing the range can be followed, such as the one developed by Vos [22].

The total range R_{eq} can be expressed as a sum of the test segment range $R_{0.75}$ (distance flown at Mach 0.75), the distance flown at lower speeds $R_{0.4}$ (Mach 0.4), as well as the range required for go-around manoeuvres $R_{go-around}$ (for HUGO, up to four landing attempts can be expected ⁽³⁶⁾, as shown in Equation 6.6. Each of the aforementioned segments comprises the nominal (geometric) range, as well as lost range which accounts for energy loss on climb, turns, and acceleration.

$$R_{eq} = R_{0.4} + R_{0.75} + 4R_{go-around} \quad (6.6)$$

Estimations of each range segment follow from methodology described by Vos [22]. As the maximum allowed altitude of FL270 is below the optimal for jet aircraft flying at Mach 0.4 [26], it can be assumed that the flight segment at Mach 0.4 will be conducted at the maximum altitude as well. The go-around altitude is constrained at 1500 feet ⁽³⁷⁾. The equivalent ranges for the considered aircraft are around 1440 km.

Once the distances and glide ratios are established for each flight segment, the required fuel can be calculated according to Equation 6.7 [22] using the total propulsion system efficiency η_i and fuel energy density e_f .

$$\left(\frac{m_{final}}{m_{initial}} \right)_i = 1 - \exp \frac{R_i g_0}{\eta_i (L/D)_i e_f} \quad (6.7)$$

The mass fractions of the segments can be later combined into the total fuel mass fraction. To verify the implementation of the method, a fuel mass fraction of a test aircraft with equivalent range equal to that of HUGO, but a single glide ratio, and engine efficiency value has been computed both using the above method and the unmodified equations from Vos [22].

6.4. Structural Weight Penalty Analysis

The purpose of this analysis is to investigate the extent to which a configuration's versatility, via the inclusion of a canard port and the ability to shift the main wing forward and aft, impacts the empty mass of the aircraft. For this initial analysis, the canard and the main wing are assumed to be simple beams that bolt onto the fuselage, with a negligible mass penalty for the local reinforcement of the bolt holes on the fuselage. In the case of the variable mounting of the main wing, the mounting bolt holes are replicated in the x direction so the wing can be bolted on at the desired location before each flight. Furthermore, the mass of the canard wing itself is considered part of the tested planform, which is part of payload mass, and thus that penalty is neglected. Therefore, it was assumed that the most significant, and the only measured contribution to a change in mass would be the additional fuselage reinforcements required to maintain structural integrity for each configuration.

It is assumed for this analysis that the Young's modulus E is constant in all directions, i.e. the material used is isotropic. Quasi-isotropic properties of a carbon fibre reinforced polymer (CFRP) material from the Ansys GrantaEduPack database ⁽³⁸⁾ is used as a reference material. Furthermore, a minimum fuselage skin thickness of 0.3mm was determined for the optimiser, based on the range of minimum ply thicknesses from Galos [27].

The loads from the wing, canard, empennage and weight are placed as point loads in the direction transverse to the fuselage direction x . The simplified loading diagram is shown in Figure 6.3. L_c , L_w and L_t represent

³⁶Established in conversation with M.T.H. (Malcom) Brown

³⁷FAA air traffic control regulations (accessed on 22.05.2026)

³⁸Ansys Granta EduPack (accessed on 21.05.2026)

the lift components of the canard, wing and empennage, respectively. W is the weight, which is simplified as a point load acting through the centre of gravity located at x_{cg} . The main lifting loads are first determined from force and moment equilibrium. For the configurations with canard ports, an initial canard lifting fraction $L_c/L_w = 0.20$ was selected to minimise the possible impact of its wake whilst representing a significant load to the airframe [19].

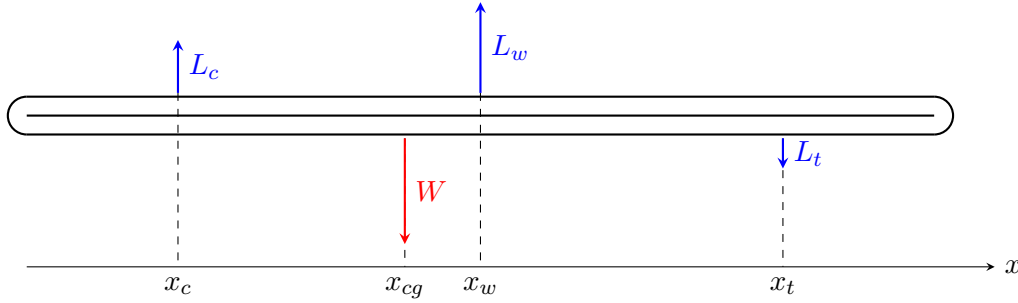


Figure 6.3: Simplified fuselage beam loading model in the x - z plane.

The point loads and point mass simplifications are suitable for the initial shear and bending loads estimation. However, more detailed distributed loads will be used further in the design stage for specific fuselage section design. The zero-lift pitching moment of the canard and tail are neglected in this initial analysis due to the reduced size of the canard and tail, whilst the zero-lift pitching moment of the main wing is also neglected due to the negligible pitching moment coefficient of the NASA SC(2)-0012 airfoil within the range of cruise angles of attack ⁽³⁹⁾.

For this preliminary analysis, the fuselage was simplified and modelled as a hollow cylinder with an outer diameter $d = 0.315 \text{ m}$ and length $l = 3.42 \text{ m}$ based on FLEXOP dimensions [11]. The model partitions the fuselage into discrete sections of length Δx along the fuselage length l based on an initial resolution value $i = 500$. The external loads were placed at the partition correspondent to their respective x values. The internal shear force at each station $V_z(x_i)$ is calculated by cumulatively summing all loads up to that point $F_z(x_j)$ and the internal moments $M_y(x_i)$ are obtained via numerical integration:

$$V_z(x_i) = \sum_{j=0}^i F_z(x_j); \quad M_y(x_i) = \sum_{j=0}^i V_z(x_j) \Delta x \quad (6.8)$$

For each fuselage station, the required skin thickness is found by checking shear stress, bending stress and buckling. The latter is considered to be the cylindrical shell buckling stress [28], with ν being the Poisson ratio, r shell radius, t shell thickness:

$$\sigma_{cr} = \frac{Et}{\sqrt{3(1 - \nu^2)}r} \quad (6.9)$$

The per-configuration weight penalty analysis begins by obtaining shear, moment and skin thickness for configuration HUG-CFG-302 by setting the canard lift fraction $L_c/L_w = 0$. This leads to no lifting loads applied at the canard location and thus no additional reinforcements are required at that location. Without a canard, the empennage is wholly responsible for the moment equilibrium around the centre of gravity, and the main wing will have to provide more lift to counter the aircraft weight.

³⁹NASA SC(2)-0012 (accessed on 20.05.2026)

The values obtained for shear, moment, and skin thickness are then plotted. From these charts, it is possible to see the structural reinforcement required. The diagrams for HUG-CFG-302, without a canard port and a static wing, are shown in Figure 6.4. Based on skin thickness, the mass of the fuselage is $2.69kg$.

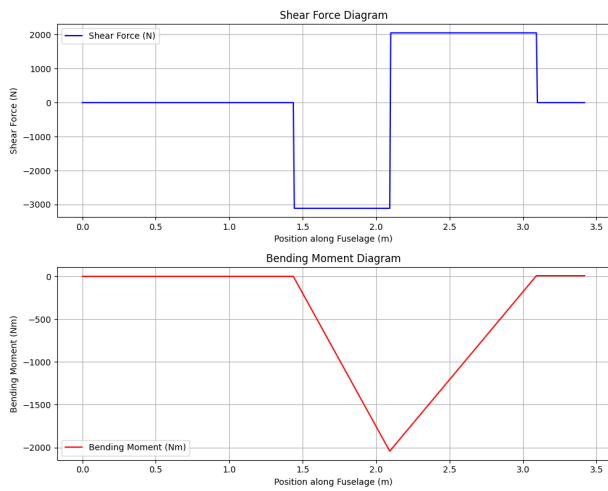
Furthermore, diagrams for HUG-CFG-303 and 301, shown in Figure 6.5, include flight conditions with and without a canard. Since the canard is intended to be optional, there will likely be scenarios where the drone flies without a canard and the moment loads are wholly counteracted by the empennage. The plotting selected the most constraining forces between both conditions for each stage. The estimate for fuselage skin mass of this configuration is $2.84kg$, representing a 5.5% increase in mass compared to HUG-CFG-302.

Lastly, a similar approach to the previous configuration was taken for HUG-CFG-304, presented in Figure 6.6. For this configuration, the main wing lift location was shifted by ± 1 the design C.G. excursion. Once again, the most constraining internal loading at each partition was selected and resulted in a fuselage skin mass of $3.17kg$, a 17.8% increase over HUG-CFG-302.

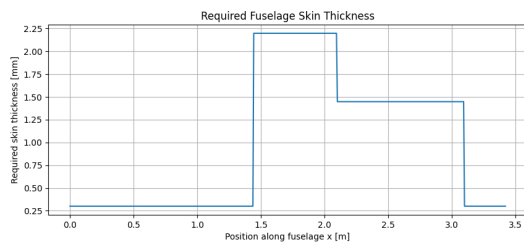
For the iterative sizing, a baseline empty mass fraction is obtained from Finger [29] and then increased by the weight penalty Δm :

$$\left(\frac{OEM}{MTOM} \right)_{baseline} = 0.699 MTOM^{-0.051}; \quad \frac{OEM}{MTOM} = \left(\frac{OEM}{MTOM} \right)_{baseline} + \frac{\Delta m}{MTOM} \quad (6.10)$$

With OEM meaning Operative Empty Mass. The mass penalty is incorporated into the sizing loop of the aircraft, and as such will affect its final Maximum Take Off Mass (MTOM).

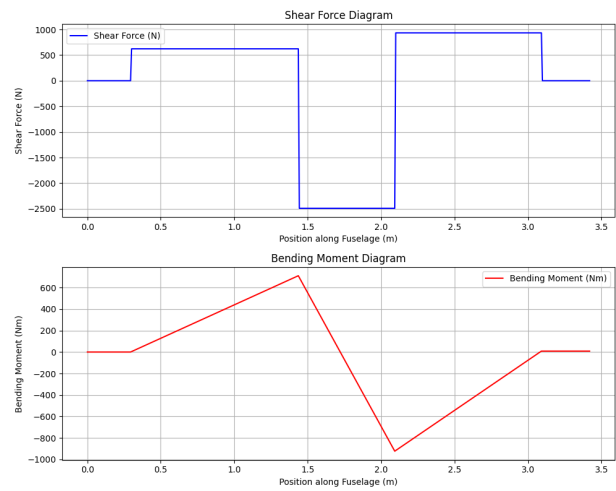


(a) Shear and moment diagram

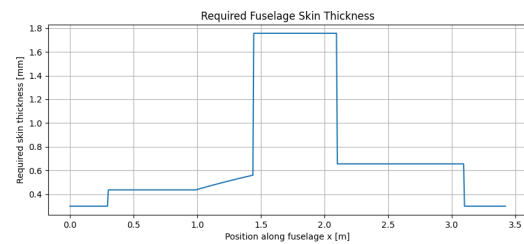


(b) Skin thickness diagram

Figure 6.4: Shear, moment, and skin thickness diagrams for HUG-CFG-302



(a) Shear and moment diagram



(b) Skin thickness diagram

Figure 6.5: Shear, moment, and skin diagram for HUG-CFG-303 and 301

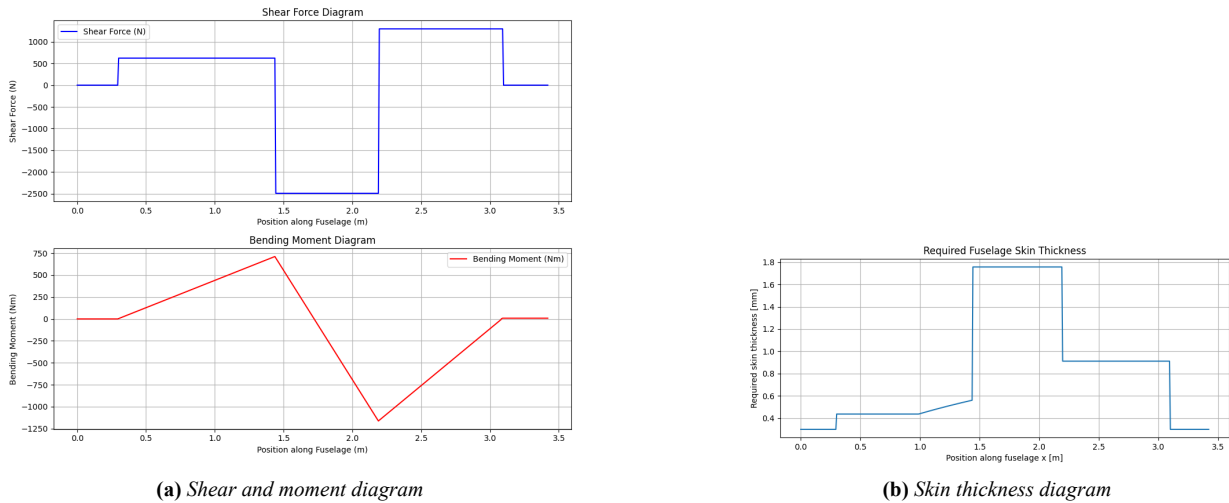


Figure 6.6: Shear, moment, and skin diagram for HUG-CFG-304

6.5. Matching Diagram for Wing and Engine Sizing

To specify the wing loading and thrust necessary for the aircraft, it is a standard practice to construct a matching diagram. This is done according to procedures outlined by Vos [22], but adapted for the aerobatic CS-23 and high-speed requirements of the designed aircraft.

Next to the cruise speed constraint, the constraint of providing thrust in the Mach 0.75 condition with half of the fuel weight is considered. Due to the replaceable wing requirement, HUGO cannot be sized to require high lift devices. This means all constraints are evaluated for a clean wing. Since there is no specified rate of climb and approach speed considered, those constraints are omitted. The climb gradient constraints from CS-23 still guarantee satisfactory climb performance, while the landing field length provide an upper bound on the wing loading. Both take-off and landing field lengths are constrained by runway length of Den Helder Airport (40). The set of matching diagram constraints is presented in Table 6.2.

Table 6.2: Matching diagram constraints

ID	Constraint	Value	LG Position	Altitude	Fuel Burned
HUG-CON-01	Runway length	1275 m	down	0 ft	0%
HUG-CON-02	Cruise speed	Mach 0.4	up	27000 ft	0%
HUG-CON-03	Maximum speed	Mach 0.75	up	27000 ft	50%
HUG-CON-04	Climb gradient single engine	0.01	up	0 ft	0%
HUG-CON-05	Climb gradient all engines	0.04	up	0 ft	0%
HUG-CON-06	Take-off field length	1275 m	down	0 ft	0%

The implementation of the matching diagram is verified by reproducing the example jet aircraft analysed by Vos [22]. Agreement with the textbook results is reached up to the precision of the values provided in the source.

6.6. Empennage Preliminary Design

Preliminary sizing of the empennage follows from an analysis of reference aircraft: DAST ARW-2, FLEXOP, and Boeing's Transonic Truss-Braced Wing (TTBW) [30]. The X56-A was not included in this part of the analysis due to being tailless. Relevant statistical data for the horizontal tail (HT) and vertical tail (VT) is provided

⁴⁰Den Helder - Information for pilots (accessed on 22.05.2026)

in Table 6.3. For parameters not included in official documentation, values were estimates from drawings and projected views.

Table 6.3: Tail parameters of reference aircraft [11] [13] [14] [15] [30]

Parameter	Tail	Symbol	DAST ARW-2	FLEXOP	TTBW
Tail configuration [-]	—	—	Conventional	V-tail	T-Tail
Span [m]	HT	b_h	1.71	1.27 (projected)	11.75
Arm [m]	HT	l_h	2.66	2.0	21.09
Non-dimensionalized arm [-]	HT	$l_h/\bar{c}$	4.75	2.67	7.25
Aspect ratio [-]	HT	AR_h	3.5	4.2	4.52
Taper ratio [-]	HT	λ_h	0.4	0.52	0.35
Dihedral [°]	HT	Γ_h	N/A	35	N/A
LE sweep [°]	HT	Λ_{LE_h}	45	23.55	21.3
Quarter-chord sweep [°]	HT	Λ_{qc_h}	41.27	19.83	15.82
Surface area [m^2]	HT	S_h	0.836	0.39	30.54
Half-span [m]	VT	b_v	0.83	N/A	9.64
LE sweep [°]	VT	Λ_{LE_v}	52	N/A	16.6

The following design approach for HUGO considers the conventional (i. e. non-canard) configurations featuring a T-tail. This is motivated by the fact that all configurations shall be able to perform operations without a canard, even if the ports for it are included in the design. The performance of HUGO is limited by two main factor which govern the allowable centre of gravity range: stability and controllability. The constraining relations are expressed by Equation 6.11 and Equation 6.12, respectively [20, p. 309].

$$\bar{x}_{cg, \text{stability}} = \bar{x}_{ac} + \frac{C_{L\alpha_h}}{C_{L\alpha_{A-h}}} \left(1 - \frac{d\varepsilon}{d\alpha}\right) \frac{S_h l_h}{S\bar{c}} \left(\frac{V_h}{V}\right)^2 \quad (6.11)$$

$$\bar{x}_{cg, \text{controllability}} = \bar{x}_{ac} - \frac{C_{mac}}{C_{L_{A-h}}} + \frac{C_{L_h}}{C_{L_{A-h}}} \frac{S_h l_h}{S\bar{c}} \left(\frac{V_h}{V}\right)^2 \quad (6.12)$$

The aircraft is controllable when $\bar{x}_{cg} > \bar{x}_{cg, \text{controllability}}$ and stable when $\bar{x}_{cg} < \bar{x}_{cg, \text{stability}}$ [20]. To determine the allowable x_{cg} range, we enforce $\bar{x}_{cg} = \bar{x}_{cg, \text{stability}}$ and $\bar{x}_{cg} = \bar{x}_{cg, \text{controllability}}$ = CGE, where CGE is the minimum required excursion between the controllability curve and the design point.

The aerodynamic parameters required for Equation 6.11 and Equation 6.12 are obtained by the aerodynamic analysis, as described in section 6.2. It is assumed that the tail velocity ration, $(V_h/V)^2 = 1$ and the downwash gradient, $d\varepsilon/d\alpha = 0.0$ due to the use of a T-tail [20]. That leaves the tail volume $S_h l_h$ as the only unknown. The horizontal tail position l_h can be estimated based on the ratio between the horizontal tail arm and surface area of the FLEXOP, effectively reducing the tail volume to a function of only surface area. This calculation is part of a larger iterative design loop.

6.7. Landing Gear Configuration

This section presents the selection analysis for the landing gear configuration of HUGO. Since the design features modular wings, the landing gear must be fully integrated into the fuselage structure. At this stage, two landing gear configurations are under consideration: buried retractable inside the fuselage and buried side podded. These have been chosen, as given the high testing speed of 0.75 a non-retractable landing gear would either

break or produce too much drag. This section will not mark either of the configurations as permanent, as the final design of HUGO is not yet selected.

6.7.1. Landing Gear Mechanism Concepts

The undercarriage concepts differ in both retraction mechanism and stowage arrangement. In the external side-pod configuration, the landing gear is housed in nacelles mounted outside the fuselage. This arrangement is expected to reduce the intrusion of the undercarriage into the main fuselage structure and thus simplify the internal layout design. However, the addition of side pods increases the wetted area, as well as introducing interference drag at the pod-fuselage junction. An advantage of this model is the load distribution due to the very short struts. This configuration example can be seen on the sketches of HUG-CFG-302 and HUG-CFG-304 in [Figure 7.1b](#) and [Figure 7.1d](#).

In the internal fuselage-bay configuration, the landing gear is retracted into the fuselage structure. This provides a cleaner external geometry, but requires dedicated internal volume, local structural reinforcement, landing gear doors, and a more integrated retraction mechanism. The placement of the undercarriage is also constrained by the need to avoid interference with the wing box, especially if a modular wing structure is used. Due to construction of this type, the main strut will experience all bending and compression stresses during landing, requiring reinforcements that increase weight. Therefore, the final landing gear arrangement will depend strongly on the selected HUGO configuration and on the available structural integration space. The fuselage-bay landing gear can be seen on the sketches of HUG-CFG-301 and HUG-CFG-303 in [Figure 7.1](#).

6.7.2. Design Constraints

The major design constraints related to operating HUGO on the runway, as well as pivoting during take-off and landing include the tip back angle, beta angle (the angle between the landing gear to the C.G. and from the landing gear vertically to the centre line), the turn over angle, and the wing-tip-to-landing-gear angle. The following restrictions must be obeyed by the landing gear [22, 31]:

- The scraping angle must be equal to, or larger than 15 deg (see: [Figure 6.7](#))
- The angle between the vertical line and the line through the main gear and the centre of gravity must not be less than the scraping angle (see: [Figure 6.7](#));
- The turn over angle must not exceed than 55 deg;
- The wing-tip-to-landing-gear-wheel angle must be greater than 7 deg;

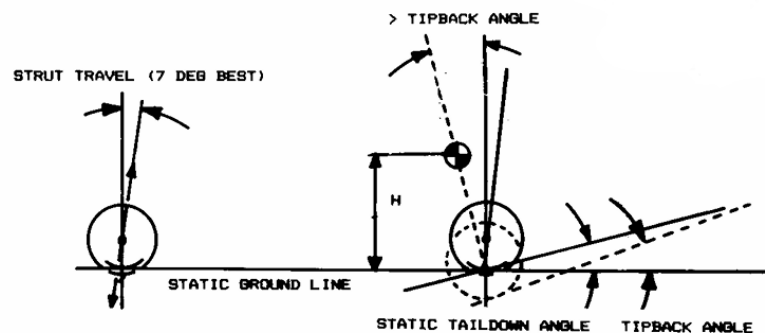


Figure 6.7: Landing Gear tip back angle and scraping angle [23]

6.7.3. Design Optimisation

By using trigonometric relations between the landing gear and critical points, such as the location of the centre of gravity, the position of the tail cone and the wing tips (as described in [31]), the location of the main and nose landing gear, landing gear length, and main landing gear separation can be optimised for the given fuselage geometric parameters. Using Sequential Least Squares Quadratic Programming, the euclidean norm of the main landing gear y position and height can be minimized. X positions of the main and nose landing gear, as well as the main landing gear y position and height themselves are taken as design variables.

For the optimising tool to converge to acceptable values, further constraints were imposed to limit the design space. The chosen airfoil has a stall angle of attack of 15° ⁽³⁹⁾. This value was used as a constraint for the tip-back angle, as the purpose of the latter is to prevent the aircraft from tipping back when it contacts the ground during landing at the maximum landing angle of attack, or during take off. This allows for the landing gear to be placed further back along the fuselage with increasing β . Other constraints were that the landing gear has to be taller than the fuselage radius and the nose landing gear has to be far enough from the aircraft nose to allow for retracting forward.

By allowing the landing gear to be housed in the tail cone, the lateral separation of the main landing gear remains close the diameter of the fuselage. This prevents unfeasible landing gear configurations, which would require complex mechanisms, more housing space, and add unnecessary weight. Based on the input parameters of fuselage length and diameter, up-sweep of the tail cone, and location of the C.G., the calculated values using the optimiser will help determine the kind of housing required for the landing gear of each configuration. The nose gear shall be positioned in the middle of the fuselage at $y = 0$.

Table 6.4: Landing gear dimensions and observations

(a) HUG-CFG-301		(b) HUG-CFG-302	
Parameter	Dimension	Parameter	Dimension
Nose gear x poz	0.386 m	Nose gear x pos	0.272 m
Fuselage gear x poz	2.147 m	Fuselage gear x pos	2.570 m
Fuselage gear y poz	0.332 m	Fuselage gear y pos	0.231 m
Height	0.281 m	Height	0.167 m

(c) HUG-CFG-303		(d) HUG-CFG-304	
Parameter	Dimension	Parameter	Dimension
Nose gear x pos	0.272 m	Nose gear x pos	0.273 m
Fuselage gear x pos	2.570 m	Fuselage gear x pos	2.570 m
Fuselage gear y pos	0.231 m	Fuselage gear y pos	0.231 m
Height	0.167 m	Height	0.168 m

Based on Table 6.4, the landing gear position does not pose an impediment for any of the configurations. HUG-CFG-301 has the longest landing gear, which is expected to incur the largest mass penalty. The rest of the configurations have similar dimensions, but the podded configurations will introduce additional weight to the design due to the extra material and added complexity. Based on the FLEXOP landing gear measured from the drawing by Luspay et al. [16] scaled down to the lower aircraft mass of HUGO, wheel diameter of 85 mm and width of 25 mm, as well as strut width of 25 mm were obtained. With the fuselage dimensions already found in subsection 6.1.1 to account for all the internal components analysed in section 9.2. A more detailed analysis and design of the landing gear will be performed in the final stage of the project.

6.8. Sensitivity Study

During the preliminary sizing, several assumptions have been made, whose impact needs to be assessed. The uncertainty in fuselage and landing gear dimensions (taken from FLEXOP geometry), the dimensions of landing gear pod with respect to the gear itself, as well as the target CG excursion between minimum controllability and neutral stability points are identified as driving assumptions. This is a basis for a sensitivity study, the details of which are shown in [Table 6.5](#). The results of the sensitivity study have been presented as MTOM plots for every configuration across different parameter values in [Figure 6.8](#).

Table 6.5: *Overview of the sensitivity study*

Parameter	Unit	Baseline Value	Lower Bound	Upper Bound	Evaluation Points
C.G. Excursion	%MAC	0.5	0.25	1.0	4
Fuselage Diameter	m	0.315	0.25	0.45	4
Fuselage Length	m	3.45	1.71	4.104	4
Landing Gear Diameter	m	0.085	0.05	0.12	4
Pod to Gear Diameter	—	2.0	1.5	3.0	4
Pod to Gear Length	—	1.25	1.0	1.5	3

The design iteration failed to converge for the shortest fuselage length, as well as for the largest fuselage diameter, due to infeasible landing gear. This is attributed to the combination of current landing gear requirements creating a heavily constrained design space for the landing gear. As the tail cone is replaceable, a tricycle configuration is strictly enforced over tail dragger.

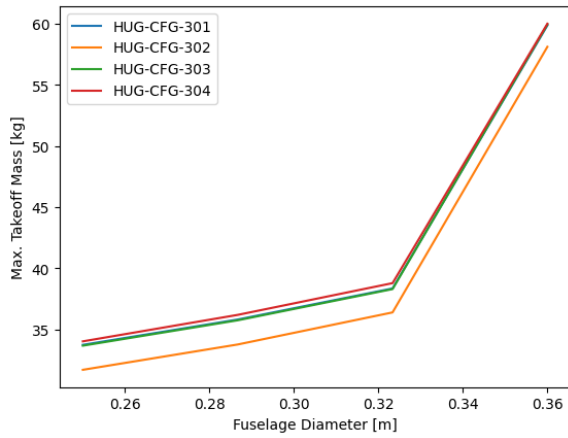
Due to the modular nature of the aircraft, the gear has to retract into fuselage or fuselage-connected pods. At the same time, a track width wider than the fuselage diameter is needed to satisfy the tip clearance and turnover angle constraints. The nose landing gear is also set to retract forward. While beneficial in case of actuator failure, this constraint might be reconsidered due to severely impacting the landing gear design freedom for shorter fuselages. As for the landing gear becoming infeasible for high fuselage diameter, it may be attributed to increasing the height of the CG without increasing the nose to main gear x distance, which worsens the turnover performance.

Regarding parameter values in the feasible design space, increasing the fuselage dimensions, especially the diameter, causes a rapid increase in mass (nearly double for the highest considered fuselage diameter), with the exception of the longer fuselage of HUG-CFG-302, which slightly decreases the total mass. It is thus recommended for the final design to avoid excessive enlarging of the fuselage, especially in diameter.

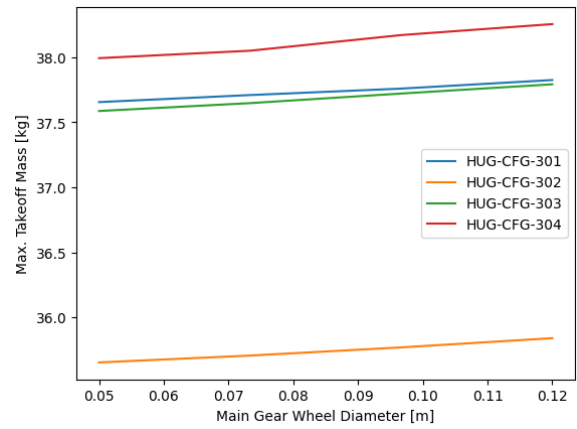
Changing the C.G. excursion used to size the tail has a much weaker effect. Options without movable tail port have their masses slightly decreased with larger C.G. excursions, while the HUG-CFG-304 experiences an increase in MTOM. This is due to the movable wig port having to account for larger C.G. excursion.

Changes to the landing gear bay width induce a small, roughly linear increase in mass, which is expected. Changes in the bay length have a significantly more minor effect. This can be explained by the fact that while a longer bay has a proportionally longer wetted area, its slenderness ratio is also proportionally larger. Changes in the wheel diameter of the landing gear affect the podded landing gear significantly more and similarly to increasing the bay width. This is because those options have a shorter main landing gear strut width, so the wheel diameter is more driving to their frontal and reference areas.

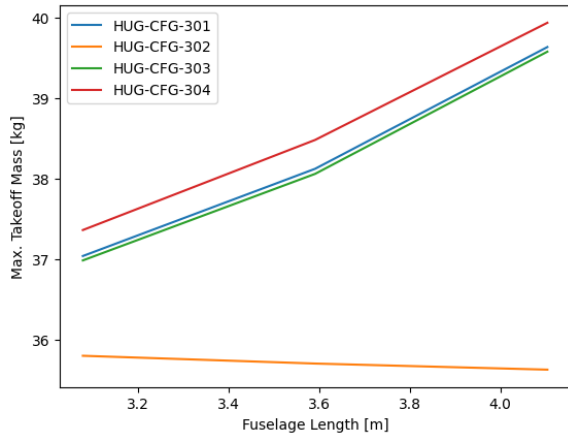
Overall, the results of the sensitivity study are plausible and highlight the importance of fuselage optimisation, as well as point to the challenging design space of the landing gear.



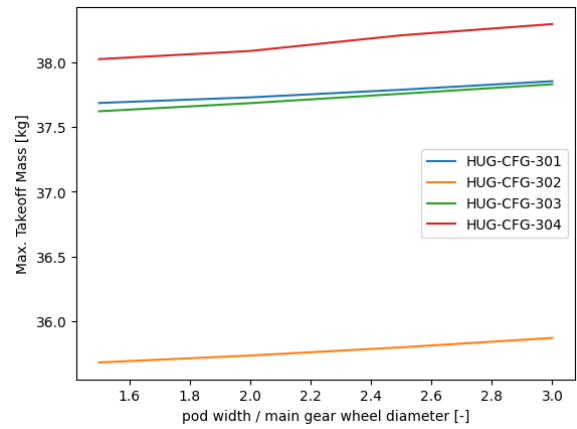
(a) Effect of altering the fuselage diameter



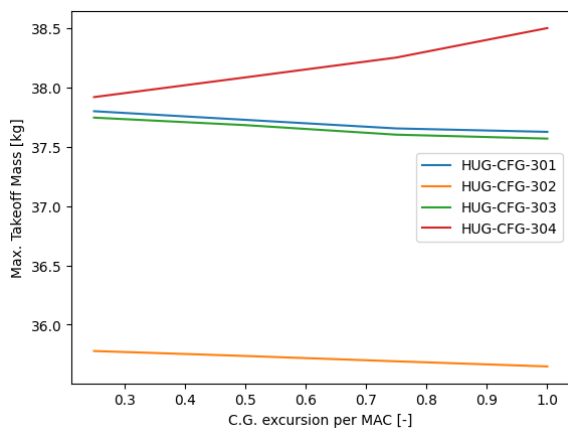
(b) Effect of altering the landing gear wheel diameter



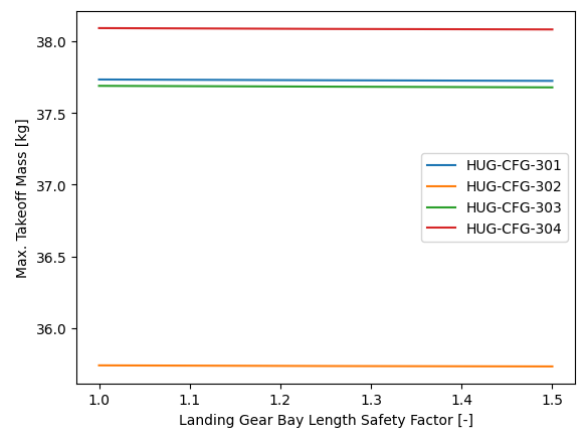
(c) Effects of altering the fuselage length



(d) Effect of altering the width of landing gear pods



(e) Effect of altering the design C.G. Excursion



(f) Effect of altering the landing gear pod length

Figure 6.8: The results of the sensitivity study

7. Design Trade-Off

In this chapter, the preliminary HUGO configurations developed in previous design stages are compared by means of a trade-off. Suitable trade-off criteria are defined in [section 7.1](#), followed by a definition of weight factors in [section 7.2](#). The results of the trade-off analysis are presented in [section 7.3](#), and the final configuration to be further developed can be found in [section 7.4](#). Additionally, a material selection trade-off is performed in [section 7.5](#).

[Table 7.1](#) lists the design configurations considered for the trade-off. Additionally, each design option features a static port for the tail, fuselage-mounted fuel tanks and fuselage-mounted engines ⁽¹⁾. The low-wing configuration now incorporates a static wing port, as it was found that a variable wing port would significantly compromise landing gear integration and reduce overall stability.

Table 7.1: Final integrated candidate architectures for detailed trade-Off⁽¹⁾

ID	Wing Configuration	Wing Modularity	Canard Modularity
HUG-CFG-301	Low Wing	Static Port	Static Port
HUG-CFG-302	High Wing	Static Port	No Port
HUG-CFG-303	High Wing	Static Port	Static Port
HUG-CFG-304	High Wing	Variable Port (X)	Static Port

As the changes between considered configurations lie mostly in the modularity details, a separate trade-off will be performed concurrently to determine the most suitable materials for different subsystems of the final configuration. Candidate will be discussed in [section 7.5](#).

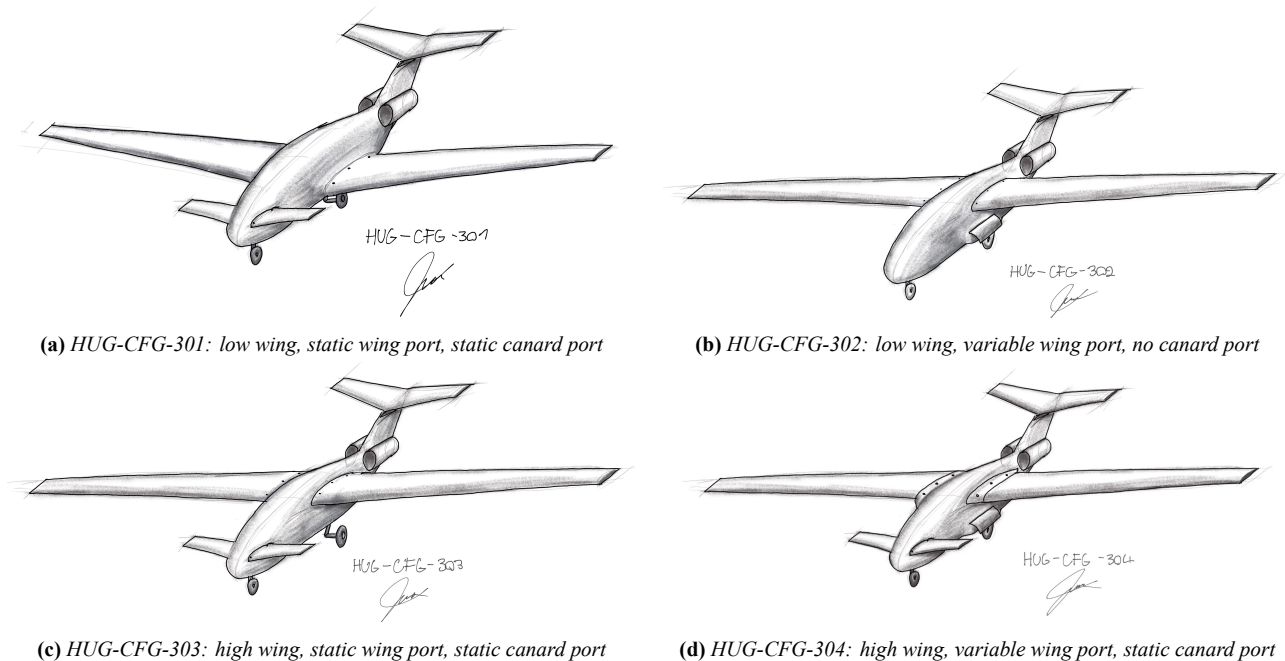


Figure 7.1: Initial HUGO configuration concepts

7.1. Trade-Off Criteria Selection

Meaningful trade-off criteria should be developed such that all available technical specifications and key performance metrics are considered. Moreover, it is recommended to identify criteria where one or more configurations perform poorly, as these are often the factors that distinguish them. In contrast, using criteria where all configurations score similarly is less useful, since such criteria do not provide meaningful differentiation

[32]. Lastly, the key requirements can be used to ensure that the developed criteria encompass all crucial design aspects.

Taking all of the above into consideration, the following criteria were selected for the trade-off of HUGO designs: Versatility, Reliability, Cost, Performance, and Sustainability. Table 7.2 lists the rationale behind selecting each criterion for the configuration trade-off, while subsection 7.5.3 lists those used for the material trade-off.

Table 7.2: Trade-Off criteria for HUGO configurations

Criterion	Rationale for criteria selection
Versatility	HUG-CFG-301 is a low-wing concept that prevents the testing of truss-braced wing concepts, which is considered a stakeholder requirement specified by NLR (41). Similarly, HUG-CFG-302 prevents the testing of canard-dependent configurations, which limits the range of possible test scenarios.
Reliability	Static canard ports introduce an additional potential failure mode. In addition, the variable wing ports in HUG-CFG-304 increase fatigue exposure at modular interfaces, which raises operational risk. Furthermore, the HUGO design must comply with specific reliability requirements (see HUG-STR-003 (1)).
Cost	Variable ports in HUG-CFG-304 substantially increase design complexity, as well as development, maintenance, and repair costs, compared to configurations without ports. Static canard ports have a similar effect, although to a lesser degree. Moreover, the design of HUGO is subject to specific cost requirements (see: HUG-FIN-001 (1)).
Performance	Complex design options featured are expected to result in higher design weights, which affect performance and fuel consumption. Moreover, the design of HUGO is subject to specific performance requirements (see: HUG-OPS-005, HUG-OPS-006 (1)).
Sustainability	Sustainability is a defined project objective (HUG-OPS-008 (1)). While variable wing ports and additional canard ports increase material usage and require more complex manufacturing processes, the primary drivers of emissions remain fuel consumption during operation and the production of sustainable aviation fuel (SAF).

The preliminary configurations and materials shall be evaluated and assigned a value ranging from 1 to 5 in each trade-off criterion. The definition of specific scores for different criteria can be found in Appendix B. Table 7.3 presents the rationale used to quantify the scores for each criterion across the different configurations.

Table 7.3: Trade-Off scoring for HUGO configurations

Criterion	Rationale for scoring
Versatility	Versatility is quantified as the range of experiments and stability conditions that can be tested with a given configuration. A low-wing configuration that excludes the evaluation of truss-braced wings receives a deduction of 2 points from the maximum score of 5, as this capability is a stakeholder requirement from NLR. A configuration that does not support canard-dependent designs receives a 2-point deduction, as it significantly restricts the available test envelope. Similarly, an inability to integrate variable wings results in a 1-point deduction.

Continued on next page.

⁴¹ In conversation with NLR

Criterion	Rationale for assigning scores
Performance	Performance is evaluated based on how effectively the maximum Mach number at test altitude can be achieved while carrying the full instrumentation payload, with minimal drag and fuel consumption. Configurations that are lighter, less complex, and feature fewer protruding components will receive higher scores.
Reliability	Reliability is evaluated as the likelihood that the testbed successfully performs its intended function without failure under specified operating conditions throughout its lifetime. Mainframe fatigue at modular interfaces is identified as a primary reliability driver. A variable wing port incurs a 2-point deduction due to increased failure risk associated with its higher complexity, as well as the added aerodynamic panels required to cover unused mounting locations. A static canard port results in a 1-point deduction, as it introduces an additional potential failure point.
Cost	Cost is quantified as the manufacturing effort, materials, development, and maintenance required for each configuration. More complex and heavier designs will score lower in this category. Variable ports substantially increase design complexity as well as development, maintenance, and repair costs compared to configurations without ports, while static canard ports have a similar but less severe effect.
Sustainability	Sustainability is a defined project objective (HUG-OPS-008 ⁽¹⁾). While variable wing ports and additional canard ports increase material usage and require more complex manufacturing processes, the primary drivers of emissions remain fuel consumption during operation and the production of sustainable aviation fuel (SAF).

Following the criterion definition, no-go values were established for which a design would be immediately deemed as unfeasible and rejected. These are shown in [Table 7.4](#).

Table 7.4: No-Go values and rationale

Criterion	No-go	Rationale
Versatility	1	Versatility is a key feature of a flying testbed. A configuration with significant limitations to its test capability is failing its purpose. Moderate limitations are acceptable.
Performance	2	If a configuration cannot effectively achieve the required test conditions, the resulting test data has limited value. Therefore, moderate penalties in drag or weight are considered acceptable.
Reliability	1	An unreliable testbed will produce corrupted data or result in large periods of Aircraft on Ground. Multiple failure modes are acceptable, so long as proper mitigation and contingency plans are defined.
Cost	1	Maintaining low costs is not design-driving, but remains beneficial for economic sustainability and profitability.
Sustainability	—	Not a mission-critical criterion.

Although performance was initially considered a viable trade-off criterion, preliminary mass and drag estimates and fuel fraction data obtained from the parametric model showed comparable values across all configurations as can be seen in [Table 7.5](#). Therefore, performance was excluded from the trade-off analysis.

Table 7.5: Performance parameters for configurations

ID	MTOM [kg]	T/W	Fuel Fraction	CD0 Cruise	CD0 Mach Max	CD0 Go Around	CD0 Takeoff
HUG-CFG-301	37.728	0.346	0.273	0.0232	0.0227	0.0233	0.1896
HUG-CFG-302	35.736	0.362	0.276	0.0241	0.0237	0.0243	0.2874
HUG-CFG-303	37.683	0.345	0.273	0.0231	0.0226	0.0232	0.2724
HUG-CFG-304	38.085	0.341	0.272	0.0229	0.0224	0.0230	0.2695

7.2. Trade-Off Weight Factors

Determining the weight factors for the various criteria was developed independently for the analysis of configurations and materials, based on the specific selection criteria. The configuration selection followed the analytic hierarchy process (AHP) approach, while the material weights were assigned arbitrarily based on an in-depth analysis of relative criterion importance.

The AHP was applied as described by Pant et al. [33]. This process encompasses pairwise comparison of the trade-off criteria. To form a pairwise comparison matrix (PCM) for n distinct criteria, $n(n-1)/2$ comparisons are required. The diagonal entries of the PCM are all equal to 1, and the remaining values are reciprocal pairs of the $n(n-1)/2$ comparison values.

For the pairwise comparison, a scale from 1 to 9 was used, as proposed by Saaty [34]. These scores are defined as follows: 1 - equal importance; 3 - moderate importance, 5 - strong importance; 7 - very strong importance; 9 - extreme importance. Assigning weights from 1 to 9 to the trade-off criteria, the PCM is constructed so that the rows represent the ratios of a criterion's score to the scores of the other criteria [34]. The resulting PCM is shown in Table 7.6. Derivation of specific importance scores is further described in Appendix B.

Table 7.6: Pairwise comparison matrix

	Cost	Versatility	Reliability	Sustainability
Cost	1	1/6	1/4	2
Versatility	6	1	3	7
Reliability	4	1/3	1	4
Sustainability	1/2	1/7	1/4	1

The next step is computing the normalised principal eigenvector (priority vector) $\mathbf{w}$ (42). The priority vector shows relative weights among the compared parameters; the higher the entry value, the higher the importance of its respective criterion. Individual components of $\mathbf{w}$ can be obtained according to Equation 7.1 for a matrix with n criteria. The normalised PCM and resultant $\mathbf{w}$ are shown in Table 7.7.

$$w_i = \frac{1}{n} \sum_{j=1}^n \text{PCM}_{ij} \quad (7.1)$$

To ensure that the comparative weights are reliable, a consistency study has to be performed. A consistency index (CI) can be defined for the PCM matrix using Equation 7.2 [34], based on the number of parameters and the principal eigenvalue $\lambda_{\max}$, which for the PCM shown in Table 7.6 attains a value of 4.1475.

$$CI = \frac{\lambda_{\max} - n}{n - 1} \approx 0.0492 \quad (7.2)$$

⁴² Analytic Hierarchy Process (AHP) Tutorial (accessed on 11.05.2026)

Table 7.7: Normalized PCM with priority vector

	Cost	Versatility	Reliability	Sustainability	w
Cost	0.0870	0.1014	0.0566	0.1429	0.0967
Versatility	0.5217	0.6087	0.6667	0.5000	0.5743
Reliability	0.3478	0.2029	0.2222	0.2857	0.2647
Sustainability	0.0435	0.0870	0.0556	0.0714	0.0644

For a fully consistent matrix $\lambda_{max} = n$, which is not true for the considered case. However, according to Saaty [34], a consistency ratio (CR) below 0.1 indicates that the estimation of **w** is sufficiently accurate. The CR is defined as a ratio of the CI to the consistency ratio of a random $n \times n$ matrix (RI). For the considered case, the estimate of RI is 0.9 [35]. The corresponding CR is obtained with Equation 7.3.

$$CR = \frac{CI}{RI} \approx 0.0546 \quad (7.3)$$

The obtained CR implies that the evaluation of **w** is sufficiently consistent. Therefore, the entries of **w** will be used as weight factors for the final configuration trade-off.

7.3. Trade-Off Matrix and Analysis

Table 7.9 shows the final trade-off matrix constructed for the candidate configurations for HUGO. The criteria have been weighted according to the priority vector defined in section 7.2, and specific scores were assigned for each criterion-configuration pair according to the definitions provided in Appendix B.

The clear overall winner is **HUG-CFG-303**, achieving the highest total score of **3.84**. HUG-CFG-304 is eliminated early in the evaluation due to failing to meet the minimum acceptable threshold in cost.

From a cost and manufacturability perspective, HUG-CFG-302 performs best due to its relatively low structural and production complexity. HUG-CFG-301 and HUG-CFG-303 also perform well in this category, as they include only an additional static canard port. In contrast, HUG-CFG-304 features a higher degree of modularity, which substantially increases both design and manufacturing complexity.

Regarding versatility, the configurations with higher modularity generally perform better, as they allow for a wider range of testing options. As explained in Table 7.3, HUG-CFG-301 and HUG-CFG-302 each receive a 3-point penalty, as the former excludes testing of truss-braced wing designs and variable wings, while the latter excludes canard-based designs and variable wings. Similarly, HUG-CFG-303 receives a 1-point deduction due to the absence of a variable wing port. In contrast, the fourth configuration includes full modularity and therefore achieves the maximum score of 5.

Reliability is inversely correlated with system complexity. Consequently, configurations with variable ports and increased modularity tend to perform worse in this category. HUG-CFG-304 receives the lowest reliability score, with a 2-point penalty for the variable wing port and the associated aerodynamic panels, and a further 1-point penalty for the static canard port, as specified in Table 7.3. Similarly, HUG-CFG-301 and HUG-CFG-303 each receive a 1-point deduction due to the additional failure mode introduced by the static canard port.

Finally, sustainability is assessed based on fuel mass predictions obtained from the parametric model. Although it was previously stated that the fuel fractions are comparable across all configurations, fuel burn and SAF production are still taken into account, as they are the primary contributors to emissions. The fuel consumption values for all five configurations are presented in Table 7.8.

Table 7.8: *Fuel consumption of HUGO configurations*

	HUG-CFG-301	HUG-CFG-302	HUG-CFG-303	HUG-CFG-304
Fuel Mass	10.29 kg	9.87 kg	10.26 kg	10.34 kg

In terms of fuel efficiency, HUG-CFG-302 performs best with the lowest fuel mass. The other three configurations exhibit the highest fuel consumption, with differences of approximately 0.5 kg relative to the most efficient design, and therefore receive the lowest sustainability scores. This is further attributed to the simpler manufacturing process and lower material usage of configuration 2, whereas the alternatives would require more complex manufacturing and greater material usage, particularly for reinforcing the variable and static ports.

Table 7.9: *Trade-Off matrix for configurations*

Configuration	Cost	Versatility	Reliability	Sustainability	Total
Weight	0.10	0.57	0.26	0.06	1
HUG-CFG-301	3	2	4	1	2.56
HUG-CFG-302	5	2	5	5	3.28
HUG-CFG-303	3	4	4	1	3.71
HUG-CFG-304	1	5	2	1	3.56

Excellent
 Good
 Average
 Poor
 Very Poor

7.4. Final Design Configuration

A simplified sketch of the final selected configuration, HUG-CFG-303, is presented in [Figure 7.1c](#). The design adopts a high-wing layout, which is essential for enabling the testing of truss-braced wing configurations.

The wing interface is implemented as a static port. Compared to a variable port architecture, this approach significantly reduces design and manufacturing complexity. While variable ports offer the advantage of adjusting the aerodynamic center and center of gravity, similar effects can be achieved through simpler means, such as redistributing payload within the fuselage.

The configuration also includes a static canard port, which substantially increases the range of achievable test configurations and expands the envelope of stability conditions that can be investigated.

The propulsion system is fuselage-mounted. This choice avoids direct interference with the modular wing requirement and prevents significant complications in routing fuel lines, hydraulic systems, and avionics through the wing structure. Additionally, wing-mounted engines would require quick-release mounting systems and pylons capable of withstanding high dynamic loads, both of which would introduce considerable structural and maintenance complexity.

The landing gear is a retractable configuration to minimise aerodynamic drag at high subsonic Mach numbers. During the final design, a selection will be made between buried retractable landing gear inside the fuselage or buried side podded landing gear.

Finally, a T-tail configuration is employed to ensure that the horizontal stabilizer remains outside the wake of the fuselage-mounted engines. As noted earlier in this chapter, the empennage is also designed with a static port system, enabling a broader range of tail configurations to be tested. However, it should be noted that fuselage-mounted engines impose limitations on certain empennage layouts, such as conventional tail configurations, due to potential interference from engine wake effects on the stabilizing surfaces.

7.5. Material Selection

This section addresses the selection of aerospace materials for various subsystems of HUGO. Key factors considered in this analysis are: structural properties, material cost, manufacturing cost, durability, sustainability, and special properties. For the purpose of a preliminary material selection, the aircraft was divided into the following subgroups:

- **Fuselage structure:** The fuselage is the main structure of the aircraft, which acts as a tethering point for the wings, empennage, and propulsion [36]. Additionally, it contains the major control system and payload. The main loads acting on a fuselage are compression, tension, and bending, and main design constraints are based on phenomena such as buckling, yielding, and deflection [37]. For HUGO, the largest loads are expected to occur at a flight speed of Mach 0.75 and during high-g manoeuvres.
- **Wing structure:** The wings of an aircraft are subjected to varying types of loads during flight, including bending, tension, compression, twist, vibration, and fatigue [37]. The main design constraints are related to an analysis of buckling, vibrations, yielding, and deflection. The material for HUGO's wing planform shall therefore be chosen to provide a sufficiently stiff and strong structure (e.g. to withstand the load factors defined in the flight envelope shown in [Figure 5.3](#)). Depending on the nature of the test, and especially the required stiffness, the choice of material for this subsystem is expected to vary.
- **Empennage and control surfaces:** The empennage will experience loads of varying degrees during the flight. The structure needs to act as a moment-arm for the generated forces at the tailplane. This makes it possible to generate moments around the centre of gravity both in the vertical and horizontal plane. This is not only to ensure straight and steady flight but also to make manoeuvring (change of heading, flight direction and etc.) possible for the aircraft.
- **Landing gear:** The landing gear must be able to carry the weight of the aircraft, as well as withstand take-off and landing loads. According to Albuquerque [38], the key loads acting on the undercarriage can be split into: ground loads (static load, braking, braked roll, ground turns, reversed braking, nose wheel yaw and steering, pivoting, towing etc.), landing loads, and take-off loads (associated with retraction of the landing gear and gyroscopic effects of changing wheel direction). All of these cases are relevant for HUGO, as the in its final configuration, the landing gear retracts into the aircraft during flight (see: [section 7.4](#)).
- **Pylons:** For the design of HUGO, it is assumed that an already existing engine will be chosen, together with a nacelle provided by the manufacturer. However, the design of pylons still has to be considered. The main loads acting on the pylons come from the engines, specifically the associated vibrations and thermal loads [39]. Therefore, key considerations for this structural element are related to fatigue life and resistance to high temperatures.

It is important to note that at this design stage, the materials are only chosen for the primary components of each structural subsystem. A more detailed design of elements such as antenna mounts, access panel hardpoints, fastener inserts etc., as well as the details of internal supporting structure, will be analysed in more detail once the final external configuration is established for HUGO.

7.5.1. Material Analysis of Reference Aircraft

The material selection shall begin with an analysis of reference aircraft, X56-A and DAST ARW-2 previously introduced in [subsection 6.1.1](#), as well as existing trends in the aerospace industry. The purpose of this research is to ensure that HUGO can model the behaviour of contemporary aircraft as accurately as possible.

X56-A, the high aspect ratio testbed developed by NASA in the 2010s, features a carbon fibre composite centrebody with honeycomb bulkheads [40]. Carbon fibre composites were also used for the wing spars and ribs, as well as the skin for stiff wings. Flexible wing configurations featured glass fibre skins due to their lower

stiffness. It also features a stiff landing gear designed for parachute landing loads. While specific materials used for the landing gear do not seem to be publicly available, accessible footage suggests that the structure is metallic.

The DAST ARW-2, a slightly older concept from the 1980s, featured a wing using 7075-T73 aluminium spars and ribs covered with a fibreglass honeycomb sandwich skin material (36 layers at inboard end, 27 at outboard, with 25-percent of the layers at ± 45 degree orientation) [41]. High stiffness was achieved by incorporating a thick aluminium plate in the centre section of the wing, and strain gauge bridges were mounted on the spars and ribs.

The prevalence for composite usage in external structures is also present in modern commercial aircraft. Leading aircraft manufacturers, such as Boeing and Airbus, both incorporate carbon laminates into the airframes of their products as a lighter alternative to metallic alloys (43)(44). Additionally, Airbus A330neo nacelles are made with 60% composites, together with a honeycomb structure for acoustics (45), while Boeing 777x uses fine metallic alloys instead (46). In the recent years, NLR has also began experimenting with implementing composites into upper spars of pylons (47). Aluminium, steel, and titanium alloys are also among the most commonly used materials for typical landing gear systems [42], as well as engine pylons (specifically Ti and Al-Li alloys)[43].

Table 7.10 showcases the usage of aerospace materials for two exemplary commercial aircraft: Airbus A350 XWB, and B787 Dreamliner. These models were chosen due to data availability, and were deemed as representative of modern aircraft systems.

Table 7.10: Materials used in external structures of A350 XWB[44] and B787[45]

Material	A350 XWB	B787
Carbon laminate	Fuselage skin, wings, centre wing box, keel beam, tail cone, skin panels, frames/stringers/doublers, doors	Fuselage skin, ailerons, flaps
Fibreglass	—	Wing skin, wing-to-fuselage joints
Al alloys	Cockpit, wing leading edges, engine inlets	Wing leading edges, engine inlets
Metallic alloys	Pylons (Ti)	Pylons (Al/Steel/Ti)
Carbon sandwich	—	Nacelles, elevators, rudders

7.5.2. Candidate Aerospace Materials for HUGO

According to Mouritz [46], there are more than one hundred materials with properties suitable for aerospace applications. The main materials used for airframes and engines include: aluminium, titanium and steel alloys, and composites. Suitable representatives of each material group selected for the material design trade-off for HUGO are described in Table 7.11. A quantitative analysis of the candidate materials can be obtained via an analysis of the base material properties of each candidate (listed in Table 7.12), as well as deriving the weight-specific properties, such as the specific modulus, specific strength, specific fracture toughness, column buckling efficiency, and panel/skin buckling efficiency (listed in Table 7.13). The weight-specific properties can be further normalised and summed to provide an overall index for the mechanical properties of the materials,

⁴³ [Airbus - Aviation's material evolution](#) (accessed on 13.05.2026)

⁴⁴ [Boeing - 787 Dreamliner Difference](#) (accessed on 13.05.2026)

⁴⁵ [Airbus A330neo nacelles](#) (accessed on 13.05.2026)

⁴⁶ [Boeing 777x exhaust system](#) (accessed on 13.05.2026)

⁴⁷ [Automated Production of Large and Thick Thermoplastic Composite Aircraft Parts](#) (accessed on 18.05.2026)

which can be used to rank the materials in the trade-off. Specific costs of each material used in [Table 7.12](#) were obtained from the Spot materials database [\(48\)](#). All calculations made to determine the derived properties and assist with the trade-off can be accessed via the Project HUGO GitHub [\(49\)](#).

Table 7.11: *candidate materials for detailed Trade-Off*

Material ID	Material	Description
HUG-MAT-01	Sitka spruce	Wood has played a part in aerospace engineering since the first powered flight. The softwood Sitka spruce was chosen based on its historical use in the construction of aircraft stemming from high specific strength and stiffness (50) . The material is additionally characterised by its low density and low cost.
HUG-MAT-02	Ti-6Al-4V	Primarily used for manufacturing of engine subsystems and landing gear, titanium is known for its high strength, long fatigue life, fracture toughness, creep strength, and excellent resistance to corrosion and oxidation [46] . Ti-6Al-4V is considered a traditional alloy for commercial airliners [47] .
HUG-MAT-03	Al 7075-T76	Aluminium has been widely used in aviation since the 1930s, and is a proven material in the industry. The specific alloy was chosen due to its outstanding anti-corrosive properties and stress corrosion cracking resistance [48] .
HUG-MAT-04	Mg WE46-T6	Magnesium has traditionally been used in aircraft due to its extremely low density (similar to that of carbon-fibre composites) [46] . However, its popularity has declined due to poor corrosion resistance and high cost. The specific alloy was chosen based on its proven applications in aerospace industry (51)
HUG-MAT-05	Steel 18Ni (350)	Steel is primarily applied to safety-critical structural components due to its very high specific strength [46] . The specific steel type was chosen due to its high fracture toughness, as well as demonstrated reliability in aerospace applications [49] .
HUG-MAT-06	Carbon/epoxy Composite	The widespread use of CFRP in contemporary aviation stems from its remarkably high strength-to-weight ratio, as well as high stiffness and corrosion resistance [46] . Depending on the application, uni- or quasi-directional fibre placement can be selected to suit the envisioned loading mode of a structure (52) .
HUG-MAT-07	Glass/epoxy Composite	Glass composites present similar properties to their carbon counterparts, with the most notable exceptions of a slightly lower strength and stiffness [46] . This makes them a viable design option for cases where flexibility is desired. Similarly to the CFRP, different fibre orientations can be chosen depending on the purpose of the designed part or structure.

⁴⁸ [Spot material](#) (accessed on 20.05.2026)

⁴⁹ [Midterm Material Selection](#) (accessed on 20.05.2026)

⁵⁰ [Wood properties and uses of Sitka spruce in Britain](#) (accessed on 19.05.2026)

⁵¹ [MatWeb - Magnesium WE54-T6](#) (accessed on 19.05.2026)

⁵² [Fundamental Behavior of Carbon Fiber Epoxy Resin Composite Laminates](#) (accessed on 19.05.2026)

The sustainability of a material is a measure of: the emissions released during its production, its impact on the environment and its recycle-ability. To systematically assess the sustainability of a material it is necessary to perform a preliminary LCA-analysis of the aircraft using each chosen material. To quantify the emissions the unit of global warming potential (GWP) is used for the emissions. In this stage it makes sense to make a streamlined LCA where the focus is the emissions related to the production of the materials.

For the LCA-analysis the software openLCA was used with the added databases: Swiss Federal Administration LCI Database (BAFU:2025) and the openLCA LCIA Methods package ⁽⁵³⁾. For the streamlined assessment the focus is on the construction of fuselage, empennage and wings. The preliminary mass together with a weight fraction can be used to estimate the weight of wings, fuselage, empennage and landing gears (the turbine is not taken into consideration since this is not part of the material selection). Different materials will lead to different weights (as the specific structural characteristics are different for each material). The complete graphs and data files for each material can be found at ⁽⁵⁴⁾. Finally it is necessary to assess the recycle-ability of the different materials. Wood can be relatively easily be reused or can be burned as fuel. As is often the case with metals it is possible for the recycled metal to retain most of the structural performance of the virgin material ⁽⁵⁵⁾. Recycling of glass and carbon- composites is typically very difficult and the large majority of these materials ends in landfills or are incinerated ^[50].

Table 7.12: Base mechanical, economic, and environmental properties of selected aerospace materials (mechanical data derived from ^[46])

ID	Density (ρ) ($kg\ m^{-3}$)	Young's Modulus (E) (MPa)	Specific Cost (Euro kg^{-1})	GWP ($kg\ CO_2\ eq$)
HUG-MAT-01	400	9000	0.85-1.7	-4.17454
HUG-MAT-02	4600	108100	21-25	52.51
HUG-MAT-03	2700	69930	2.55-5.1	90.2384
HUG-MAT-04	1850	49025	2.975-5.95	1019.0540
HUG-MAT-05	8100	199260	8-20	65.02890
HUG-MAT-06	1700	49980	25.91-32.30	389.27245
HUG-MAT-07	2000	57400	6.5-8.7	67.81310

⁵³ [openLCA Nexus](#) (accessed on 06.05.2026)

⁵⁴ [Streamlined LCA](#) (accessed on 18.05.2026)

⁵⁵ [Metal Recycling Factsheet](#) (accessed on 18.05.2026)

Table 7.13: Derived specific material indices and economic factors for aerospace structural design ⁽⁴⁹⁾

ID	Specific Modulus (E/ρ) (GPa / $kg\ m^{-3}$)	Column Buckling ($\sqrt{E}/\rho$)	Plate Buckling ($\sqrt[3]{E}/\rho$)	Specific Strength (σ/ρ) $\times 10^3$ (N m kg ⁻¹)	Specific Fracture Toughness (K_{Ic}/ρ) $\times 10^3$ (N m ^{1.5} kg ⁻¹)	Total Normalized Economic Factor
HUG-MAT-01	22.5	0.24	0.05	138	10.0	0.21
HUG-MAT-02	23.5	0.07	0.01	217	21.7	4.34
HUG-MAT-03	25.9	0.10	0.02	174	10.7	0.68
HUG-MAT-04	26.5	0.12	0.02	117	11.8	0.92
HUG-MAT-05	24.6	0.06	0.01	295.6	4.9	3.80
HUG-MAT-06	29.4	0.13	0.02	450	22.4	3.35
HUG-MAT-07	28.7	0.12	0.02	384	31.0	0.94

The material cost per kg does not paint a clear picture, as it does not account for differences in mechanical properties. Therefore, the total normalized economic factor shown in Table 7.13 is determined by dividing the material cost per kilogram by its material indices, multiplying them by their weights, and then normalizing the results. Lower values of the normalized economic factor point towards more cost-effective solutions. Detailed calculations regarding the economic factor of HUGO's material selection can be found in the Project HUGO Git ⁽⁴⁹⁾.

7.5.3. Criterion and Weight Factors for Material Selection trade-off

For the selection of material weight factors, a more holistic approach was applied, resulting in values from 0.1 to 1.0, where 1.0 warrants the highest priority. The following rationale was applied to justify the choice of specific values:

- **Structural Properties (Weight 1.0):** It is in our best interest to minimize the weight of HUGO's structural components. Choosing materials with high specific modulus (E/ρ) and plate buckling efficiency ($\sqrt[3]{E}/\rho$) are required to withstand aeroelastic effects at Mach 0.75 while minimizing weight. Concentrated load structures (landing gear, pylons) are driven by high specific yield strength (σ_y/ρ) and fracture toughness (K_{Ic}/ρ) to safely absorb dynamic landing loads. Therefore, structural performance directly dictates the aerodynamic and operational feasibility of the platform. It warrants the highest trade-off weight, 1.0.
- **Durability (Weight 0.85):** HUGO is designed for a 15-year operational lifecycle with several flight days per year. The selected materials must withstand continuous high-impact cyclic loading, runway debris abrasion, and environmental degradation - including UV exposure for composites and galvanic corrosion at hybrid metallic-composite interfaces. While these issues can be solved using protective coatings and modern construction methods like using patches of fibreglass between the aluminium and carbon fibre components, the durability of the materials used is a large factor in the material selection process, warranting a weight of 0.85.
- **Material Cost (Weight 0.1):** The cost of the raw materials of the structure of HUGO has a small overall impact on the financial feasibility of the project, as found in the market analysis during the previous design phase ⁽¹⁾. As explained in section 7.1, the overall costs are expected to be dominated by manufacturing. Therefore, this criterion was assigned the lowest weight of 0.1.
- **Manufacturing Cost (Weight 0.5):** Manufacturing complexity dictates the primary financial burden of

the project due to the low-volume production model. For aerodynamic surfaces, high Non-Recurring Engineering (NRE) costs associated with custom metallic tooling (e.g., stretch forming blocks) must be balanced against the high labour and autoclave operational costs of composite layups. For localized structures, direct CNC machining of metallic billets offers a highly efficient, low-NRE production route. Manufacturing methodology directly controls the economic viability of unit scaling. However, as shown in the previous design phase ⁽¹⁾, there is a wide margin in the unit cost in the overall financial viability, which is why the criterion was deemed as one of medium importance, and assigned a weight of 0.5.

- **Design Freedom (Weight 1.0):** Project HUGO functions as a flying testbed, rather than a mass-production vehicle. Maximizing design freedom allows for the implementation of complex, aerodynamically optimized profiles and fuselage geometries, that would be impossible or prohibitively heavy if restricted to simple, developable surfaces. Because the ability to manufacture advanced aerodynamic shapes directly governs the quality of the research data the platform can generate, design freedom is classified as a primary requirement and assigned the maximum weight of 1.0.
- **Sustainability (Weight 0.5):** Being at the forefront of modern aviation research, HUGO shall embrace the industry's movement towards sustainable solutions. At the same time, the main design focus is the ability of HUGO to perform its mission. Therefore, this criterion was assigned a weight of 0.5.

7.5.4. Trade-off table

Based on the trade-off criteria and weights selected in [subsection 7.5.3](#) and, the following trade-off matrices were generated. Since the requirements and manufacturing methods and complexity varies significantly between the aerodynamic components (fuselage, empennage and wings), and the functional components (landing gear, pylons etc.), separate trade-off matrices were required. [Table 7.14](#) shows the trade-off matrix for the aerodynamic components, and [Table 7.15](#) shows the trade-off matrix for the functional components.

Table 7.14: Trade-Off matrix for fuselage, wings and empennage

Material	Structural properties	Durability	Material cost	Manufacturing cost	Design freedom	Sustainability	Weighted Total
Weight	1	0.85	0.1	0.5	1	0.5	
HUG-MAT-01	4	1	5	3	1	5	10.35
HUG-MAT-02	3	5	1	1	3	2	11.85
HUG-MAT-03	3	4	4	2	3	3	12.30
HUG-MAT-04	3	3	4	1	3	1	9.55
HUG-MAT-05	3	2	1	2	3	3	10.30
HUG-MAT-06	5	3	2	2	5	1	15.10
HUG-MAT-07	5	3	4	2	4	1	14.30

Excellent
 Good
 Average
 Poor
 Very Poor

Table 7.15: Trade-Off matrix for landing gear, pylons, and connection ports

Material	Structural properties	Durability	Material cost	Manufacturing cost	Design freedom	Sustainability	Weighted Total
Weight	1	0.85	0.1	0.5	1	0.5	
HUG-MAT-01	4	1	5	5	1	5	11.35
HUG-MAT-02	3	5	1	3	5	2	14.85
HUG-MAT-03	3	4	4	4	5	3	15.30
HUG-MAT-04	3	3	4	3	5	2	13.45
HUG-MAT-05	3	2	1	4	5	3	13.30
HUG-MAT-06	5	3	2	1	3	1	12.60
HUG-MAT-07	5	3	4	1	2	1	11.80

Excellent
 Good
 Average
 Poor
 Very Poor

7.5.5. Material Trade-Off Results

The CFRP emerged as the clear optimal choice for the primary lifting surfaces and fuselage. Its advantage in specific modulus and plate buckling efficiency ($\sqrt[3]{E}/\rho$) is critical for a high-aspect-ratio (HAR) aircraft operating at Mach 0.75. This is what drove its score in the structural properties section. Furthermore, CFRP offers maximum design freedom, allowing for the design of complex aerodynamic profiles necessary for a flying laboratory. It also allows for the development and construction of new wing designs without large tooling costs. Utilizing CFRP ensures that weight of the fuselage and wings can be minimized, while retaining the maximum payload volume required for the 5 kg experimental payload.

For localized, high-stress components such as the landing gear struts, engine pylons and modular port connection points, Aluminum 7075-T76 outperformed the composite alternatives. These components require a balance of high specific fracture toughness to absorb dynamic landing impacts and high manufacturability. Aluminum 7075-T76 provides excellent specific strength (σ_y/ρ) while keeping manufacturing and material costs significantly lower than titanium or composite alternatives. The selection of high-strength aluminium heavily supports the project's modularity requirement.

8. Manufacturing, Assembly, and Integration Plan

This chapter outlines the Manufacturing, Assembly, and Integration (MAI) plan for HUGO. Based on the material trade-off analysis performed in [chapter 7](#), the final configuration features parts made out of CFRP and aluminium. The manufacturing of individual components from the raw materials is described in [section 8.1](#). Selected joining methods are outlined in [section 8.2](#), followed by a discussion of the integration of the final assembly in [section 8.3](#).

A preliminary version of the MAI plan for HUGO is presented in [Figure 8.1](#). It shall be noted that this iteration does not address a complete design and is expected to change as more details regarding the external and internal configuration of HUGO are established.

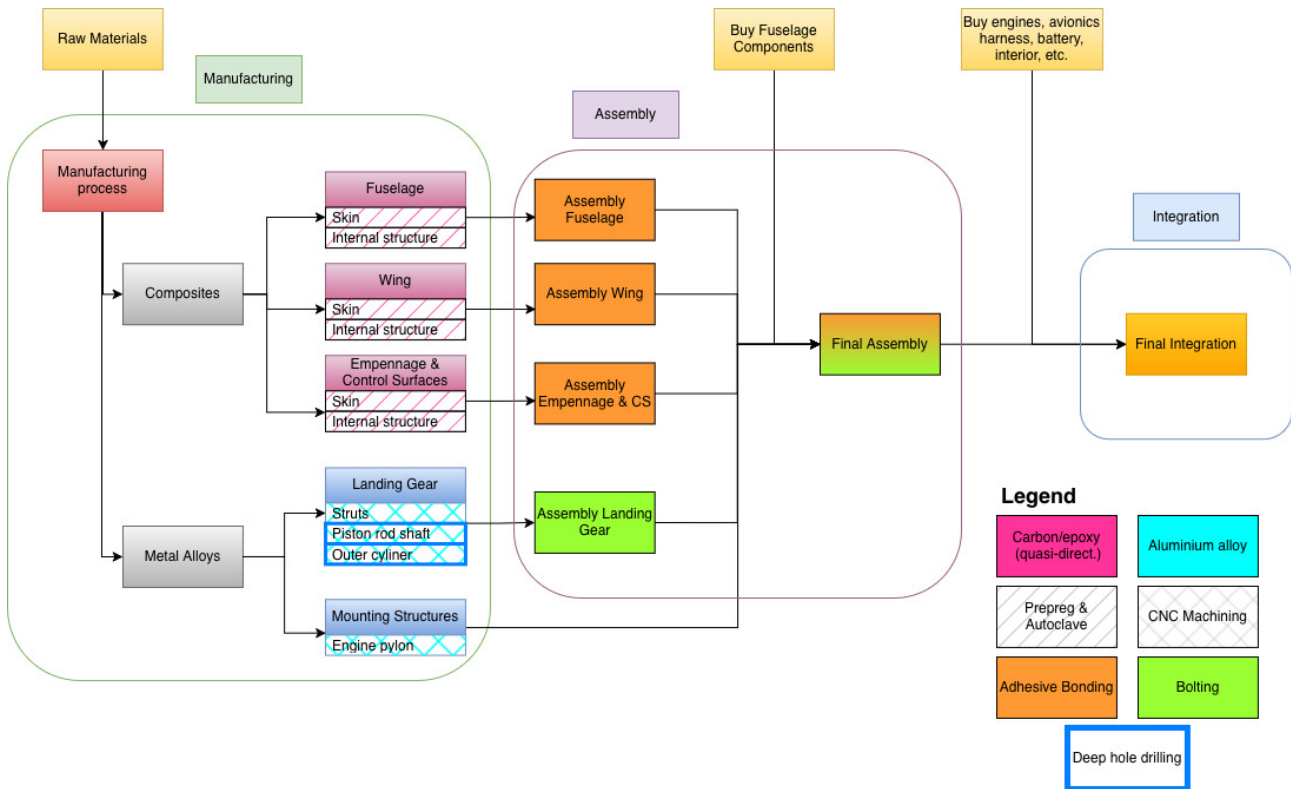


Figure 8.1: Preliminary MAI plan for HUGO

8.1. Manufacturing

This section addresses the manufacturing methods for carbon fibre composites and aluminium alloys, based on the final design presented in [section 7.4](#). The key factors influencing the fabrication method of aircraft parts include: part size, raw material type, labour skill and cost, number of units, and desired performance [51].

8.1.1. Composite Manufacturing

The final configuration of HUGO features CFRP wings, fuselage, and empennage. [Table 8.1](#) lists the main loading modes and design constraints applicable for these structural elements. The composite layers shall be manufactured in a way that results in a strong and stiff structure resistant to buckling induced by pre-, in-, and post-flight loads.

Table 8.1: *Aerospace applications/idealizations, adapted from Arnold et al. [37, p. 6]*

Application	Loading	Constraints
Truss	Compression, Tension	Strength
Fuselage	Compression, Tension, Bending	Buckling, Strength, Deflection
Wing	Bending, Tension, Twist, Vibration, Fatigue	Buckling, Vibration, Strength, Deflection

A study of available composite manufacturing methods made by Jabbar and Nasreen [51], along with an analysis of industry trends ⁽⁵⁶⁾⁽⁵⁷⁾⁽⁵⁸⁾, point towards prepreg manufacturing with autoclave curing as a viable composite manufacturing method for HUGO. This method has proved successful in the production of complex geometries with tight tolerances due to precise fibre placement with accurate orientation control which can be placed over curved surfaces.

Autoclave prepreg delivers significant process efficiencies even at low-volume production compared to methods like compression moulding or resin transfer moulding (RTM). It has been shown to reduce labour costs by up to 40% through less hands-on processing time, cut rework by 60% due to fewer defects from the outset, and achieve 30–50% faster cycle times by eliminating the waiting associated with resin infusion ⁽⁵⁹⁾. Moreover, structures made with autoclave prepreg have excellent structural properties, which are required for the successful operation of HUGO under test conditions ⁽⁵⁹⁾.

8.1.2. Metal Alloy Manufacturing

As discussed in [section 7.5](#), AL 7075-T76 was chosen as the primary material for structural components incl. landing gear, pylons, and connection ports. A study of aluminium manufacturing methods conducted by Francis [52], supported by an analysis of industry trends ⁽⁶⁰⁾, the following manufacturing steps were defined for HUGO's aluminium components:

1. **CNC Machining and Turning** shall be applied to produce three-dimensional component profiles and axial geometries. To ensure dimensional stability during and after material removal, stress-relieved stock (such as stretcher-stressed plate or extrusions designated with the T7651/T76511 suffix) must be specified to ensure the material remains flat and stable [53].
2. **Deep hole drilling** shall be applied to long, slender parts (e. g. piston rods, outer cylinders, axles) to achieve the required dimensional accuracy and surface quality where conventional turning methods may not yield sufficiently precise results ⁽⁶⁰⁾⁽⁶¹⁾.
3. **Heat treatment** shall be applied to relieve residual stresses, enhance component strength and ensure that it can perform under extreme loading conditions. Depending on the component, one of the following techniques will be applied: annealing, solution heat treatment, natural and artificial ageing, or homogenising ⁽⁶²⁾.
4. **Surface Treatment and Protection:** As 7000-series Al alloys are susceptible to environmental degradation and corrosion, electrochemical treatments and chemical conversion coatings shall be applied to

⁵⁶Composites: Airbus continues to shape the future (accessed on 22.05.2026)

⁵⁷Boeing Composite Manufacturing Center Begins Fabricating 787 Vertical Fin (accessed on 22.05.2026)

⁵⁸Successful Hydraulic Test at Airbus (accessed on 22.05.2026)

⁵⁹The Complete Engineer's Guide to Prepreg Composites: What You Need to Know (accessed on 13.05.2026)

⁶⁰The main manufacturing technology of landing gear (accessed on 19.05.2026)

⁶¹Landing gear machining struts its stuff! (accessed on 18.05.2026)

⁶²4 Types Of Heat Treatment For Aluminium Alloys (accessed on 18.05.2026)

the strengthened components. This process is expected to maximise resistance to wear, exfoliation, and stress-corrosion cracking in operational environments ⁽⁶³⁾. Additional protection from the influence of CFRP interface (incl. galvanic corrosion) shall be applied via means such as: a galvanic isolation layer, anodising and priming, fastener isolation, or implementing a sacrificial anode ⁽⁶⁴⁾.

8.2. Assembly

This section describes the joining methods selected for CFRP and aluminium components for HUGO's main structural subsystems outlined in [section 7.5](#). The primary components shall be assembled separately, and then assembled into the final structure together with the remaining parts of the UAV.

CFRP components shall be connected using adhesive bonding, which is the common practice in aerospace industry, and has been proven to comply with certification requirements ⁽⁶⁵⁾ [54]. This process is expected to provide components with a smooth surface and good corrosion resistance, as well as joining of dissimilar materials (incl. CFRP-aluminium joints). An exception has to be made in case of pylons, which cannot be attached to the fuselage with an adhesive joint due to integrity and structural concerns ⁽⁶⁶⁾. The alternative solution to joining the CFRP and aluminium surfaces may include using bolts, disks, pucks, lug fittings, bosses, trunnions, support beams, and/or caps, depending on the detailed joint structure.

The landing gear, pylon, and connection port subsection shall be assembled using bolted joints in favour of bonded joints. In particular, bolted and lockbolted joints shall be applied for rigid structural connections, and pin joints wherever relative motion is required (e. g. attachment points for the torque link or retraction actuator). Welding is excluded as a joining method for this application, as the local melting of material alters the microstructure and degrades the mechanical properties of the high-strength aluminium alloy ⁽⁶⁷⁾.

8.3. Integration

Following structural assembly, integration of the aircraft systems proceeds in a defined sequence driven by accessibility - items that are hardest to reach or that other systems depend on are installed first. The avionics wiring harness is installed first, before any interior or equipment obstructs access to the airframe structure. The wire harness connects every critical system from avionics to landing gear, and can be found on gas turbine engines, nacelles, and airframe fuselages - with modern fly-by-wire aircraft relying critically on these interconnects. The battery is installed next as part of the electrical system, followed by the avionics equipment itself - flight computers, sensors, and displays - which are connected to the pre-routed harness. The engines are then installed and connected to the fuel, hydraulic, and electrical systems. The interior is installed last, as it covers and protects the underlying systems while also being the most easily removable for any rework that may be required during systems testing. Arranging the integration in the proper sequence is an important objective, as improper sequencing causes rework, time delays, cost increases, and potential safety risks.

⁶³ [Surface Treatment Of Aluminum And Aluminum Alloys](#) (accessed on 18.05.2026)

⁶⁴ [How Do You Prevent Galvanic Corrosion Between Aluminum and Carbon Fiber?](#) (accessed on 20.05.2026)

⁶⁵ [One way to make composite aircraft lighter: stop riveting and start bonding](#) (accessed on 22.05.2026)

⁶⁶ [US Patent Application Publication - Method for Mounting a Pylon to an Aircraft](#) (accessed on 22.05.2026)

⁶⁷ [How Welding Affects Material Properties: A Metallurgist's Perspective](#) (accessed on 22.05.2026)

9. Configuration Layout

This chapter outlines the details of the external and internal configuration of HUGO. [section 9.1](#) showcases sketches of external geometry for each of the configurations considered in this design phase. Details regarding the internal components and their functions, along with a brief description of the electronic units carried on-board, are provided in [section 9.2](#). This section is not included into the trade-off process of the configurations because the internal components are the same for all designs, and layouts are similar to each other. It will serve as an analysis backbone for the final internal design, identifying necessary subsystems and their contributions. The final internal layout will be created in the next stage of the project realisation.

9.1. External Configuration

The final design of the plane will have an external configuration that is optimised for the test section of the flight. It has a minimum number of outer ports, focusing on a smooth exterior to prevent as much as possible flow separation.

Regardless of the smooth surface criteria, HUGO still has some outside ports, such as a pitot static tube placed under the nose cone, away from disturbed flow, the lidar port that will slightly protrude outside to scan the surrounding atmosphere, antennas for transmitting and receiving as well as the outside lights for visibility.

Other outer characteristics include the doors for the internal components, placed in top of the fuselage, doors for the landing gear, both nose and main, camera ports and the fuel port. A sketch of the external configuration can be found in [Figure 9.1](#).

As per calculations in [chapter 6](#), the aircraft requires a wingspan of 2.667 m at the maximum aspect ratio of 27. This should not cause issues with transporting the aircraft, as per van dimensions from the Baseline Report ⁽¹⁾.

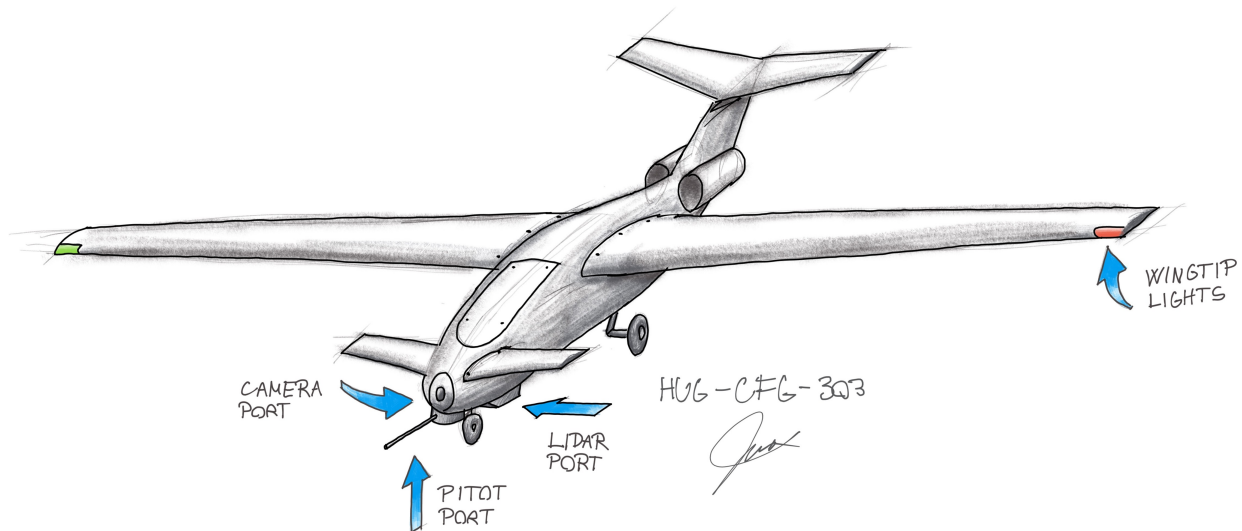


Figure 9.1: HUG-CFG-303 external configuration sketch

9.2. Internal Configuration

This section evaluates a possible internal configuration of HUGO, mapping the layout of the electronic system, propulsion system and fuel system, along with the possible sensors to measure the aero-elastic effects during the 0.75 Mach manoeuvre.

9.2.1. Electronic System

Along the payload, the fuselage will house the electronic system, including the sensors and supply the payload. The engine requires an outside source of power for the start-up procedure, as such, the battery will provide

the additional power needed. The margin left from the battery will power the payload, and other secondary subsystems.

A list of components included in the electronic system of HUGO can be seen in [Table 9.1](#). For the internal configuration of HUGO, the dimensions and weight of each electronic component need to be catalogued, and in the next stage of the project, a detailed power system analysis can be conducted. For now, the numbers in [Table 9.1](#) represent a first-order approximation to enable calculation of dynamic stability and C.G. properties of the UAV. The components can be found in the project repository on GitHub [\(68\)](#).

For the purpose of this application, autopilot VECTOR-600 [\(69\)](#) was chosen as it incorporates many of the basic air sensors needed to conduct the flight. Working along a flight companion, preferably made by the same company for compatibility, the system has the capability to collect the sensor data, store it, use it for flight calculations and ultimately transmit it back to the base of operations. Additionally, this autopilot has been qualified by the industry, being tested against vibrations and crash loads. It incorporates: Dual Inertial Measurement Unit (IMU), redundant inertial sensing for altitude, magnetometer, gyroscope, inertial magnetometer, altimeter, airspeed sensor, Global Navigation Satellite System (GNSS) receiver for gps signal. One limitation of this system is the maximum air speed calculator, which can only reach speeds of 450 kts, while HUGO can fly at a maximum speed of 500 kts, as such an additional speed sensor is needed. Alongside the autopilot, the design team has decided to include additional data sources such as: pitot static tube, radar system, temperature sensors, humidity sensors, cameras (one for piloting and 2 more for the view of the fuselage), and an angle of attack sensor. Storage of the data for further processing is done on a SATA SSD, with a capacity of 1TB, as it is robust and will offer enough storage space for a 30-minute flight.

Table 9.1: *Electronic system components and approximate weight and volumes*

Component	Weight [g]	Dimensions [mm]
Battery	4000	206 x 124.3 x 70.5
Wires	0.33 kg/1000 ft	-
Rotary actuators (x10)	19	28.4 x 38.0 x 13.3
Linear actuators (control surfaces and landing gear) x 5	200	22.6 x 30.4 x 150
Flight Controller/Autopilot	180	45.0 x 68.0 x 74.5
Flight companion	160	45.0 x 68.0 x 74.5
SATA SSD	50	64.5 x 42
ADS-B Transceiver and FLARM Receiver system	120	57 x 82 x 30
Electric switch	-	-
Lights (x4)	-	-
Power distribution system (70)	250	180 x 100 x 40

The electronic system is complemented by a multitude of sensors present inside and on the wings and tail to study the aeroelastic behaviour of structures. A list of sensors that have been considered for the aeroelastic analysis can be found in [Table 9.2](#).

The nose cone will house a weather radar and a lidar sensor specific for atmospheric measurements, giving HUGO weather prediction capabilities, aiding in measuring the atmospheric parameters.

⁶⁸ [Electronic Component List](#) (accessed on 26.05.2026)

⁶⁹ [Vector-600](#) (accessed on 13.05.2026)

⁷⁰ [Lithium Ion Battery Management System](#) (accessed on 18.05.2026)

Table 9.2: *Candidate sensors for aeroelastic response measurements*

Sensor	Use on wing
Foil strain gauges	Estimate wing-root bending moment, spar loads, and local stress through calibrated strain measurements.
Strain gauge rosettes	Measure torsion, shear, and principal strain near spars, wing roots, cut-outs, and complex load paths.
Wheatstone bridge strain gauge circuits	Convert strain measurements into bending moment, shear force, or torsional load through calibration.
Fibre Bragg Grating sensors	Measure spanwise strain along the spar and support distributed load estimation and wing deformation reconstruction.
FBG-based shape sensing	Reconstruct wing bending and twist from distributed strain measurements along the wing structure.
Linear Variable Differential Transformers (LVDTs)	Measure static wing deflection during structural tests and calibration loading cases.
String potentiometers / draw-wire sensors	Measure wing-tip deflection during ground tests or structural calibration tests.
Inclinometers / tilt sensors	Support wing twist or slope reconstruction when placed at selected spanwise locations.
Small IMUs along the wing	Estimate dynamic wing motion, bending response, and torsional behaviour during flight.
Accelerometers	Measure modal response, gust response, flutter behaviour, and aeroelastic vibration.
MEMS IMUs	Measure wing-tip motion, torsional response, and flexible-body dynamics during flight.
Piezoelectric patches	Monitor flutter, vibration, and dynamic aeroelastic response; can also support active aeroelastic control concepts.
Acoustic emission sensors	Detect crack initiation, delamination, or local structural damage events for structural health monitoring.
Control-surface position sensors	Correlate structural response with commanded control input and real control-surface motion.
Actuator current sensors	Provide an indirect indication of hinge moment, actuator loading, binding, or overload during manoeuvres.
Lidar Sensor	Provides 3D measurements and real-time view of the wing through a laser-based measurement system.

9.2.2. Fuel and propulsion system

Systems related to propulsion, including the turbojet engine, fuel tanks, fuel management, and distribution systems are included in this category. The propulsion system is integrated into the fuselage group to allow for wing versatility, provide clean airflow over the wing and prevent aerodynamic interference of the engine with the wings. This has been determined for all of the configurations in [chapter 7](#).

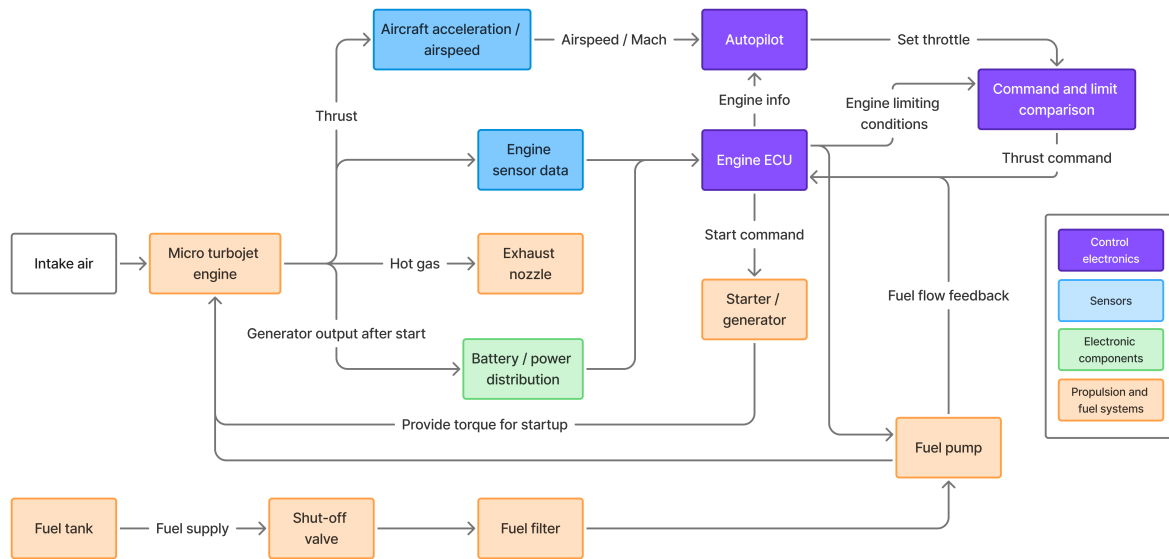
The initially selected VULCAN engine is in an early development stage, and the figures regarding its overall propulsive efficiency are uncertain. Furthermore, small turbojet engines on the market can be easily integrated due to their known characteristics and availability of the supporting subsystems. Therefore, based on the approximate thrust requirement found in [section 6.5](#), a pair of Jetcat P100-RX-BL engines ⁽⁷¹⁾ are selected as the candidate engines for HUGO and are used as a reference component in [Table 9.3](#).

⁷¹ [Jetcat P100-RX-BL engine](#) (accessed on 19.05.2026)

Table 9.3: Propulsion system components and approximate weight and volumes

Component	Weight [g]	Dimensions [mm]
Engine: JetCat P100-RX-BL (X2)	1080	97 (diameter) x 241 (length)
ECU (V10.0) (X2)	26	55 x 36 x 18 (length x width x height)
Fuel tank	0.57	N/A (fuel bag, 3 gal capacity) ⁽⁷²⁾
Fuel pump (Bus-system BL) (X2)	110	31.5 (diameter) x 77 (length)
Fuel filter (Standard Inline) (X2)	15	20 (diameter) x 45 (length)
Fuel hoses	170	8 x 4 x 2000 (outer x inner diameter x length)

The turbojet engine needs additional components to become operational. An Engine Control Unit (ECU) is necessary to control combustion. The fuel system consists of the fuel pump, fuel filter, shut-off valve and fuel distribution hoses ⁽⁷³⁾. An overview of the propulsion system components is presented in Table 9.3 and a simple diagram of the subsystem is shown in Figure 9.2. The weight and dimensions of the power distribution system were estimated based on a universal Battery Management System (BMS) ⁽⁷⁴⁾. Furthermore, 2m of Polytetrafluoroethylene (PTFE) tube for fuel distribution is added and a 3 gal fuel bag will be used for fuel storage ⁽⁷⁵⁾. This corresponds to the fuel mass estimated in section 7.3.

**Figure 9.2:** HUGO propulsion system diagram

⁷² Armadillo Bag (accessed on 22.05.2026)

⁷³ JetCat fuel system accessories (accessed on 18.5.2026)

⁷⁴ Lithium Ion Battery Management System (accessed on 18.5.2026)

⁷⁵ PTFE tube 4 mm x 8 mm (accessed on 19.5.2026)

10. Sustainable Development Strategy

The current strategy for sustainable development is a natural continuation of the approach presented in the baseline report ⁽⁷⁶⁾. For all major decisions throughout this process sustainability has been a trade-off criterion and been carefully evaluated. At this point the design is defined in enough detail that it is possible to make a more detailed analysis of the impact of HUGO. To access the environmental impact of the system, a preliminary Life Cycle Assessment (LCA) analysis was set up using the open-source software openLCA. Contrary to the streamline used for the material selection the preliminary LCA will take flight operations and end-of-life of the material into account. The files for the preliminary LCA can be found at ⁽⁷⁶⁾.

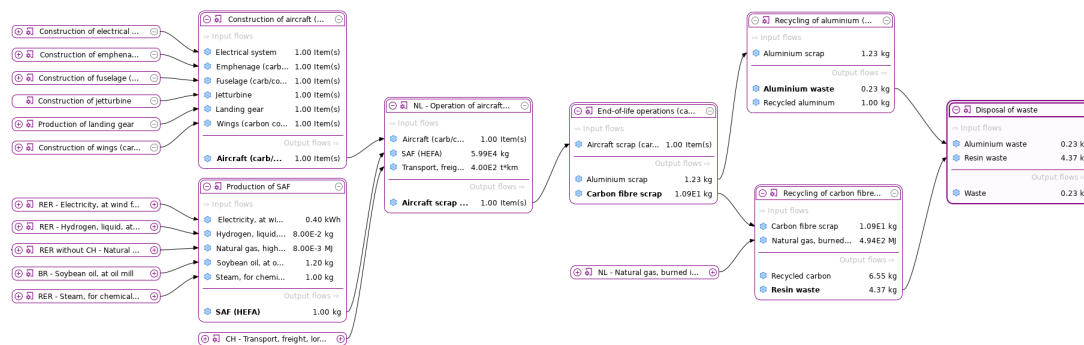


Figure 10.1: Preliminary LCA-analysis of HUGO

Figure 10.1 is the preliminary LCA-analysis of the project and shows the interdependencies between the different processes. Having the preliminary LCA and using the Intergovernmental Panel on Climate Change (IPCC) 2021 (ISO 14067) method it is possible to calculate the GWP of the entire lifespan of HUGO $1.92794 \cdot 10^6$ [kg CO₂ eq]. The six largest contributors are shown in Figure 10.2 below.

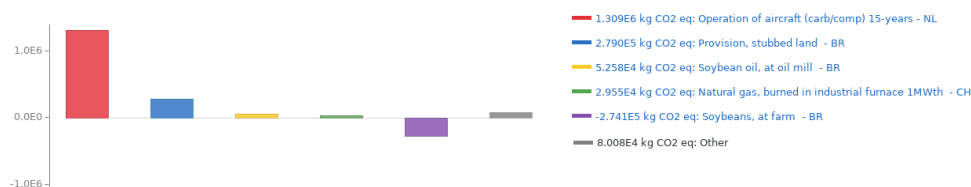


Figure 10.2: The six biggest contributors to the overall emissions in [kg CO₂ eq]

Having identified these it is possible to focus on the processes which has the greatest influence on emissions. In this case it is apparent that it is the operation of HUGO (the burning of fuel) and the production of the fuel. This should be the focus for the subsequent development of HUGO to implement methods to mitigate the fuel-burn emission and find production methods of SAF with lower emissions.

Care has been put into this project to ensure that the production and operation of the aircraft is performed in such a way that no harm comes to anyone both physically and mentally. This is achieved by carefully planning the operation of the aircraft and ensuring safety for all stakeholders involved (see chapter 4) and by adhering to the regulation stipulated by the European commission regarding the emissions and noise generated during flight (see section 5.5).

The economic sustainability has already been addressed extensively in the market analysis which can be found baseline review ⁽⁷⁷⁾. The main conclusion was that the aircraft could be produced and operated profitable.

⁷⁶ Preliminary LCA (accessed on 21.05.2026)

⁷⁷ Baseline report (accessed on 21.05.2026)

11. Verification and Validation Plan

The methods used to size the design need to be verified and validated. This means ensuring the method is implemented according to its specification (verification) and assessing the relevance of the method and its ability to accurately reproduce physics (validation).

The verification is highly dependent on how the method is implemented. For project HUGO, the design options are sized using iterative code (32). The code executes several independent analyses in a loop. After each analysis, parts of a global single source of truth object representing the aircraft are updated. The loop runs for a pre-determined amount of steps and values of selected parameters are saved at each iteration.

This setup allows to verify the particular analyses themselves. Per-analysis verification is described in the relevant sections of [chapter 6](#). As a guideline, particular analyses are expected to be general enough to model both HUGO and the verification case.

It is also necessary to ensure the iterative sizing converges. This is done by tracking the history of aircraft MTOM for each iteration step for each configuration. The results of the convergence study are shown in [Figure 11.1](#). Based on that, eight iterations were chosen for the Midterm Report trade-off, as from that point the difference in mass between the design options stays constant and the simulation runtime is acceptable for conducting a sensitivity study on the four configurations (approximately 20 minutes per parameter). For the sizing of the final design, a similar convergence study shall be conducted. Nevertheless, due to higher precision required for the final design, a larger number of iterations may be required.

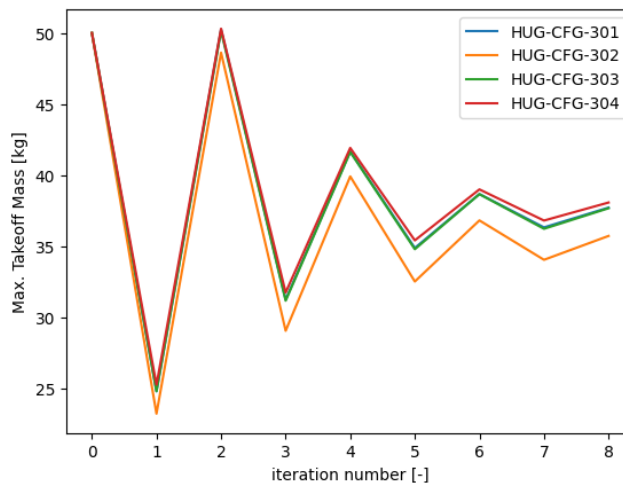


Figure 11.1: Convergence of the maximum take-off mass of the four considered configurations

As for the validation, the main goal for the analyses of the Midterm Report is to provide numerical data for the trade-off. As such, a sensitivity study is conducted in [section 6.8](#) to assess whether the analyses' predictions will not change to an extent that affects the trade-off with varying their assumptions.

For the Final Report, the goal of the analyses is to provide accurate numerical values used for subsystem design. As such, it can be verified by employing the same analysis to evaluate an existing aircraft, and compare its results with real aircraft data. This will require detailed information on the reference aircraft of choice. The analysis will also have to be general enough to accommodate both HUGO and the reference aircraft. The need for validation may thus be driving for the development of the Final Report methods. Depending on the availability of statistical data, it might be necessary to validate different parts of the method against different reference aircraft.

A sensitivity study shall be conducted for the analyses from the Final Report as well. In that case, the question to answer will be if changes to assumptions significantly affect subsystem design or final system performance.

12. Conclusion

The midterm report aims to converge on a final HUGO configuration suitable for detailed design by establishing the operational, regulatory, and engineering foundations necessary to make the selection. The team has systematically evaluated five candidate architectures against HUGO's core mission requirements: the testing of aeroservoelastic phenomena on high aspect ratio wings at high subsonic speeds, within the constraints of certifiability, operability, and sustainability.

Den Helder Airport is confirmed as the base of operations, offering a controlled airspace environment, paved runway, and proximity to the designated North Sea test area. The certification pathway demonstrates that a Type Certificate supported by a Design Organisation Approval is both legally required and practically attainable, with the DOA mechanism allowing major structural reconfigurations to be classified as research procedures internally, avoiding full re-certification for each test campaign.

The engineering analyses in [chapter 6](#) have yielded a set of preliminary design parameters. The NASA SC(2)-0012 airfoil was selected, with an estimated stall speed of approximately $37 \text{ m} \cdot \text{s}^{-1}$ and a maximum lift coefficient of 1.22. Although possible limitations in the AeroSandbox software and potential issues with the model were observed, the results are considered sufficiently accurate for distinguishing between configurations, particularly at low angles of attack. The fuselage structural analysis demonstrates that the weight penalty of versatility features such as variable wing ports and canard ports is quantifiable and manageable. The flight envelope, constructed to EASA CS-23 aerobatic category requirements, shows that manoeuvre loads are the governing structural case, with gust loads remaining non-critical. The material and configuration trade-offs in [chapter 7](#) collectively indicate a clear design direction. Carbon fibre composite emerges as the primary structural material for the fuselage, wings, and empennage, while Aluminium 7075-T76 is preferred for localised high-stress components such as landing gear struts, pylons, and connection ports. The trade-off analysis eliminated HUG-CFG-304 for failing the cost no-go threshold, with HUG-CFG-303 emerging as the clear winner at a total score of 3.84 - a high-wing configuration with static wing and canard ports, and retractable landing gear selected for its balance of experimental versatility, cost efficiency, and reliability. The manufacturing, assembly, and integration plan in [chapter 8](#) confirms that the selected material combination is feasible using established aerospace processes - prepreg autoclave curing for composites and CNC machining for metals - at the low production volumes expected for HUGO. A preliminary internal and external configuration layout has been established for HUG-CFG-303, and a lifecycle assessment confirms that the primary environmental impact arises from flight operations, with material production and end-of-life contributing comparatively minor shares.

Several assumptions limit the precision of the conclusions reached at this stage. Validation of the aerodynamic model against experimental data specific to HUGO was not possible due to the lack of available data, and known discrepancies with DATCOM predictions at higher angles of attack suggest that lift estimates may carry non-negligible uncertainty, particularly for the swept wing configuration. The structural weight penalty analysis relies on simplified beam models with point loads, which will require refinement once the final configuration geometry is fixed. These limitations are consistent with a midterm design stage and will be addressed systematically in the detailed design phase.

For the final report, the priority recommendations are: converging the weight and centre-of-gravity iteration loop, refining the aerodynamic model through higher-fidelity analysis at high subsonic conditions, completing the detailed structural design of the wing-fuselage and tail-fuselage static ports, and finalising the internal layout and fuel system sizing for HUG-CFG-303. On the organisational side, initiating preliminary contact with EASA regarding the DOA application and confirming hangar availability at Den Helder are advised as early actions that can be pursued in parallel with the ongoing design work.

Bibliography

- [1] Egbert Torenbeek, *Synthesis of Subsonic Airplane Design*, 1st ed. Springer Dordrecht, 1982. [DOI](#).
- [2] William Spitz, Frank Berardino, Richard Golaszewski, and Jesse Johnson, “Development Cycle Time Simulation for Civil Aircraft,” Contractor Report NASA/CR-2001-210658, NTRS ID: 20010032406, Gellman Research Associates, Inc., Jenkintown, PA, United States, Jan. 2001. [Url](#).
- [3] European Union Aviation Safety Agency, “Easy Access Rules for Initial Airworthiness and Environmental Protection (Regulation (EU) No 748/2012),” Nov. 2025, [Url](#) (accessed 05/11/2026), EASA eRules, Revision from November 2025.
- [4] European Aviation Safety Agency, “Easy Access Rules for Continuing Airworthiness (Regulation (EU) No 1321/2014),” European Aviation Safety Agency, Cologne, Germany, Sep. 2025. [Url](#).
- [5] International Civil Aviation Organization, “Unmanned Aircraft Systems (UAS),” 328 AN/190, ICAO, Montreal, Canada, 2011.
- [6] Den Helder Airport, DHA Nieuwsbrief, Yumpu, Published online 29 July 2013. Accessed 12 May 2026, Jul. 2013. [Url](#).
- [7] European Union Aviation Safety Agency, Special condition SC Light-UAS High Risk (Issue 01), Dec. 2021. [Url](#).
- [8] European Commission, Commission Implementing Regulation (EU) 2019/2153 of 16 December 2019 on the fees and charges levied by the European Union Aviation Safety Agency, and repealing Regulation (EU) No 319/2014, Dec. 17, 2019.
- [9] European Aviation Safety Agency, *Certification specifications for normal, utility, aerobatic, and commuter category aeroplanes (cs-23), amendment 3*, Amendment 3, European Aviation Safety Agency, Cologne, Germany, Jul. 2012. [Url](#).
- [10] Raymond E. Mineck and Pierce L. Lawing, “High Reynolds Number Tests of the NASA SC(2)-0012 Airfoil in the Langley 0.3-Meter Transonic Cryogenic Tunnel,” NASA Technical Memorandum 89102, NASA Langley Research Center, Jul. 1987.
- [11] Fanglin Yu, Julius Bartasevicius, and Mirko Hornung, “COMPARING POTENTIAL FLOW SOLVERS FOR AERODYNAMIC CHARACTERISTICS ESTIMATION OF THE T-FLEX UAV,” 2022.
- [12] Jacob Schaefer, Peter Suh, Matt Boucher, Jeff Ouellette, Alex Chin, Chris Miller, Jared Grauer, Greg Reich, Robert Mitchell, and Pete Flick, “Flying Beyond Flutter with the X-56A Aircraft,” 2023. [Url](#).
- [13] Peter Bauer, Lysandros Anastasopoulos, Franz Michael Sendner, Mirko Hornung, and Balint Vanek, “Identification and Modeling of the Airbrake of an Experimental Unmanned Aircraft,” *Journal of Intelligent and Robotic Systems: Theory and Applications*, Vol. 100, 1 Oct. 2020, pp. 259–287, issn: 15730409. [DOI](#).
- [14] III Perry Boyd, “Methodology for Determining Elevon Deflections to Trim and Maneuver the DAST Vehicle with Negative Static Margin,” NASA Technical Memorandum NASA TM-84499, NASA Langley Research Center, Hampton, Virginia, May 1982.
- [15] Clinton V Eckstrom, David A Seidel, and Maynard C Sandford, “Measurements of Unsteady Pressure and Structural Response for an Elastic Supercritical Wing,” 1994.
- [16] Tamás Luspay, Tamás Péni, and Bálint Vanek, “Control Oriented Reduced Order Modeling of a Flexible Winged Aircraft,” *Proceedings of the IEEE Conference on Control Applications (CCA)*, Sydney, Australia: IEEE, 2015.
- [17] Richard T. Whitcomb, “Review of NASA Supercritical Airfoils,” Technical Report, NTRS ID: 19740058562, NASA Langley Research Center, Hampton, VA, United States, Aug. 1974. [Url](#).
- [18] Charles D. Harris, “NASA Supercritical Airfoils: A Matrix of Family-Related Airfoils,” Technical Paper NASA-TP-2969, NTRS ID: 19900007394, NASA Langley Research Center, Hampton, VA, United States, Mar. 1990. [Url](#).
- [19] Egbert Torenbeek, *Advanced Aircraft Design: Conceptual Design, Analysis and Optimization of Subsonic Civil Airplanes*, 1st ed. Chichester, UK: John Wiley & Sons Ltd, 2013, pp. 397–398, isbn: 978-1-118-56811-8.

- [20] Egbert Torenbeek, *Synthesis of subsonic airplane design: An introduction to the preliminary design of subsonic general aviation and transport aircraft, with emphasis on layout, aerodynamic design, propulsion and performance* Springer Science & Business Media, 2013, isbn: 978-9024727247.
- [21] Parankush Koul, Mahesh Varpe, Pritam Bhat, Aniket Mishra, Chirag Malhotra, and Devang Kalra, "Effects of Leading-edge Tubercles on the Aerodynamic Performance of Rectangular Blades for low-speed Wind Turbine Applications," *International Journal of Scientific Research in Modern Science and Technology*, Vol. 4, Jan. 2025, pp. 01–28. [DOI](#).
- [22] Roelof Vos, "Introduction to Airplane Design," Course material for AE1222-II Aerospace Design and System Engineering Elements. Delft University of Technology, 2023.
- [23] Daniel P. Raymer, *Aircraft Design: A Conceptual Approach*, 2nd ed. Washington, DC: American Institute of Aeronautics and Astronautics, 1992, isbn: 978-1624104909.
- [24] Jan Roskam, *Airplane Design Part VI: Preliminary Calculation of Aerodynamic, Thrust and Power Characteristics* Lawrence, KS: DARcorporation, 2004, pp. 90–98, isbn: 1-884885-52-7.
- [25] John Anderson, *EBOOK: Fundamentals of Aerodynamics (SI units)* McGraw Hill, 2011, isbn: 9781259251344.
- [26] Stanley L. Koutz and Reece V. Hensley, "Loitering and Range Performance of Turbojet-Powered Aircraft Determined by Off-Design Engine Cycle Analysis," Research Memorandum NACA-RM-E51K29, National Advisory Committee for Aeronautics (NACA), Feb. 1952.
- [27] Joel Galos, "Thin-ply composite laminates: A review," *Composite Structures*, Vol. 236, 2020, Article 111920. [DOI](#).
- [28] "Buckling of Thin-Walled Circular Cylinders," NASA SP-8007, National Aeronautics and Space Administration, 1968. [Url](#).
- [29] D. F. Finger, "Comparative Performance and Benefit Assessment of VTOL- and CTOL-UAVs," *Deutscher Luft- und Raumfahrtkongress 2016*, Aachen, Germany, 2016.
- [30] Juntao Xiong, Robert E Bartels, and Nhan Nguyen, "Aerodynamic Optimization of Mach 0.745 Transonic Truss-Braced Wing Aircraft with Variable-Camber Continuous Trailing-Edge Flap," 2021.
- [31] Jan Roskam, *Airplane Design Part IV: Layout Design of Landing Gear and Systems* Lawrence, KS: DARcorporation, 2010, pp. 75–102, isbn: 978-1-884885-53-2.
- [32] James R. Wertz and Wiley J. Larson, *Space Mission Analysis and Design*, 3rd ed. Microcosm Press and Kluwer Academic Publishers, 1999, isbn: 9780792359012.
- [33] Sangeeta Pant, Anuj Kumar, Mangey Ram, Yury Klochkov, and Hitesh Kumar Sharma, "Consistency Indices in Analytic Hierarchy Process: A Review," *Mathematics*, Vol. 10, No. 8, 2022, issn: 2227-7390. [DOI](#).
- [34] Thomas L. Saaty, "How to make a decision: The analytic hierarchy process," *European Journal of Operational Research*, Vol. 48, No. 1, 1990, pp. 9–26, issn: 0377-2217. [DOI](#).
- [35] Enrique Mu and Milagros Pereyra-Rojas, *Understanding the Analytic Hierarchy Process* Springer, 2017, ch. 2, pp. 7–22, isbn: 978-3-319-33861-3. [DOI](#).
- [36] Biplab Chatterjee and Sumit Bhowmik, *Chapter 9 - Evolution of material selection in commercial aviation industry—a review*, Kaushik Kumar, Divya Zindani, and Paulo Davim, Eds. Academic Press, 2019, pp. 199–219, isbn: 978-0-12-816564-5. [DOI](#).
- [37] Steven M. Arnold, David Cebon, and Mike Ashby, "Materials Selection for Aerospace Systems," NASA Technical Memorandum NASA/TM-2012-217411, National Aeronautics and Space Administration (NASA), Cleveland, Ohio, Jan. 2012. [Url](#).
- [38] Pedro Albuquerque, "Structural Loads Handbook," 2011, [Url](#).
- [39] Ding Wang, Yaopeng Niu, and Xiangying Guo, "Dynamic Analysis of Aircraft Engine-Pylon Coupled System," *Journal of Physics: Conference Series*, Vol. 3145, Nov. 2025, p. 012 031. [DOI](#).

- [40] Christopher Miller, Jacob Schaefer, Matthew Boucher, Jeffrey Ouellette, and Scott Howe, "X-56A Flight-test Approach for Envelope Expansion Past Open-loop Flutter Instability," *NATO STO Meeting Proceedings STO-MP-SCI-328*, Edwards, CA, 93523: NASA Armstrong Flight Research Center, 2012, pp. 18-1–18-27, [Url](#).
- [41] C. V. Eckstrom, "Loads Calibrations of Strain Gage Bridges on the DAST Project Aeroelastic Research Wing (ARW-2)," NASA Technical Memorandum NASA-TM-87677, NASA Langley Research Center, Hampton, Virginia, Mar. 1986.
- [42] Jeevanantham V, Vadivelu Ponnusamy, and Manigandan P, "Material Based Structural Analysis of a Typical Landing Gear," *IJISET-International Journal of Innovative Science, Engineering Technology Impact Factor*, Vol. 4, Jan. 2017.
- [43] Hamza Khalid and Ares Gomez, "Substituting Ti-64 with Aa2099 as Material of a Commercial Aircraft Pylon," *Advances in Materials Science*, Vol. 21, Jun. 2021, pp. 77–92. [DOI](#).
- [44] Jens Bachmann, Carme Hidalgo, and Stéphanie Bricout, "Environmental analysis of innovative sustainable composites with potential use in aviation sector—A life cycle assessment review," *Science China Technological Sciences*, Vol. 60, Aug. 2017. [DOI](#).
- [45] Andrzej Katunin, Katarzyna Krukiewicz, Alexander Herega, and G. Catalanotti, "Concept of a Conducting Composite Material for Lightning Strike Protection," *Advances in Materials Science*, Vol. 16, Aug. 2016. [DOI](#).
- [46] Adrian P. Mouritz, *Introduction to Aerospace Materials*, 1st ed., ser. Woodhead Publishing in Materials Cambridge, UK: Woodhead Publishing, 2012, ch. 2,8,9,10,11,13,17,24,25, isbn: 978-1-85573-946-8.
- [47] Patricia Nyamekye, Saeed Rahimpour, Heidi Piili, Pasi Luukka, and Andrzej Kraslawski, "Impact of additive manufacturing on titanium supply chain: Case of titanium alloys in automotive and aerospace industries," *Advances in Industrial and Manufacturing Engineering*, Vol. 6, Jan. 2023, p. 100 112. [DOI](#).
- [48] Shuang–Shuang Li, Xin Yue, Qing–Yuan Li, He–Li Peng, Bai–Xin Dong, Tian–Shu Liu, Hong–Yu Yang, Jun Fan, Shi–Li Shu, Feng Qiu, and Qi–Chuan Jiang, "Development and applications of aluminum alloys for aerospace industry," *Journal of Materials Research and Technology*, Vol. 27, 2023, pp. 944–983, issn: 2238-7854. [DOI](#).
- [49] Stephen D. Antolovich, T.R. Risbeck, Ashok Saxena, and Y. Kawabe, "The effect of microstructure on the fracture toughness of 300 and 350 grade maraging steels," *Engineering Fracture Mechanics*, Vol. 13, No. 4, 1980, pp. 717–739, issn: 0013-7944. [DOI](#).
- [50] Avishreshth Singh, Akhil Charak, Krishna Prapoorna Biligiri, and Venkataraman Pandurangan, "Glass and carbon fiber reinforced polymer composite wastes in pervious concrete: Material characterization and lifecycle assessment," *Resources, Conservation and Recycling*, Vol. 182, 2022, p. 106 304. [DOI](#).
- [51] Madeha Jabbar and Adeela Nasreen, "Composite fabrication and joining," *Composite Solutions for Ballistics*, ser. Woodhead Publishing Series in Composites Science and Engineering, Yasir Nawab, S.M. Sapuan, and Khubab Shaker, Eds., Woodhead Publishing, 2021, pp. 177–197, isbn: 978-0-12-821984-3. [DOI](#).
- [52] Lorraine F. Francis, "Chapter 4 - solid processes," *Materials Processing (Second Edition)*, Second Edition, Academic Press, 2024, pp. 245–341, isbn: 978-0-12-823908-7. [DOI](#).
- [53] Mandy S. Younger and Kenneth Hall Eckelmeyer, "Overcoming Residual Stresses and Machining Distortion in the Production of Aluminum Alloy Satellite Boxes," Sandia Report SAND2007-6811, Sandia National Laboratories, Albuquerque, New Mexico, Nov. 2007. [Url](#).
- [54] Theodor Schmid F., Thomas K., Thomas Körwien, and Matthias Geistbeck, "Bonding of CFRP primary aerospace structures - Discussion of the certification boundary conditions and related technology fields addressing the needs for development," *Composite Interfaces*, Vol. 22, Oct. 2015, pp. 795–808. [DOI](#).

A. Project Planning Diagrams

This Appendix contains the planning diagrams described in chapter 3. Figure A.1 presents the WFD and Figure A.2 shows the N2 diagram. Figure A shows the WBS, and the project design and development logic is presented in Figure A.3. Lastly, the team organogram is shown in Figure A.4.

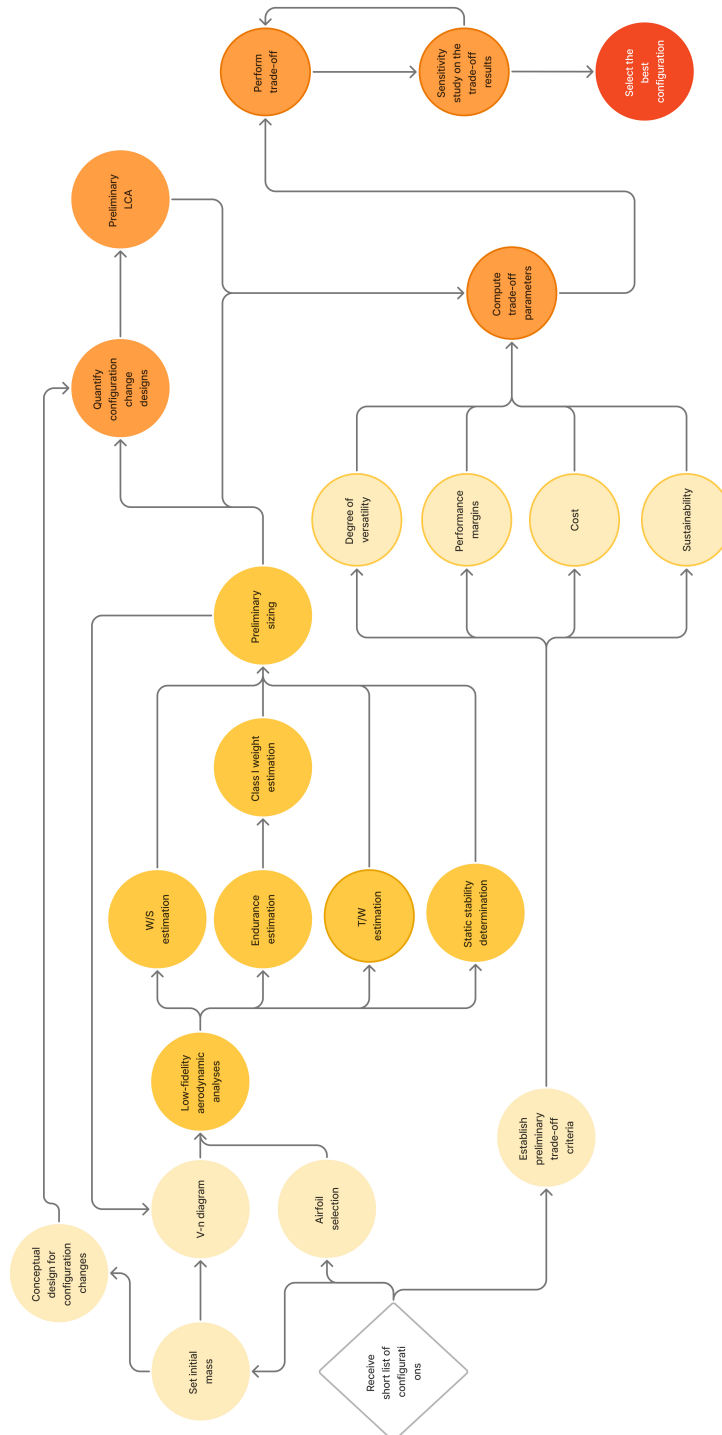
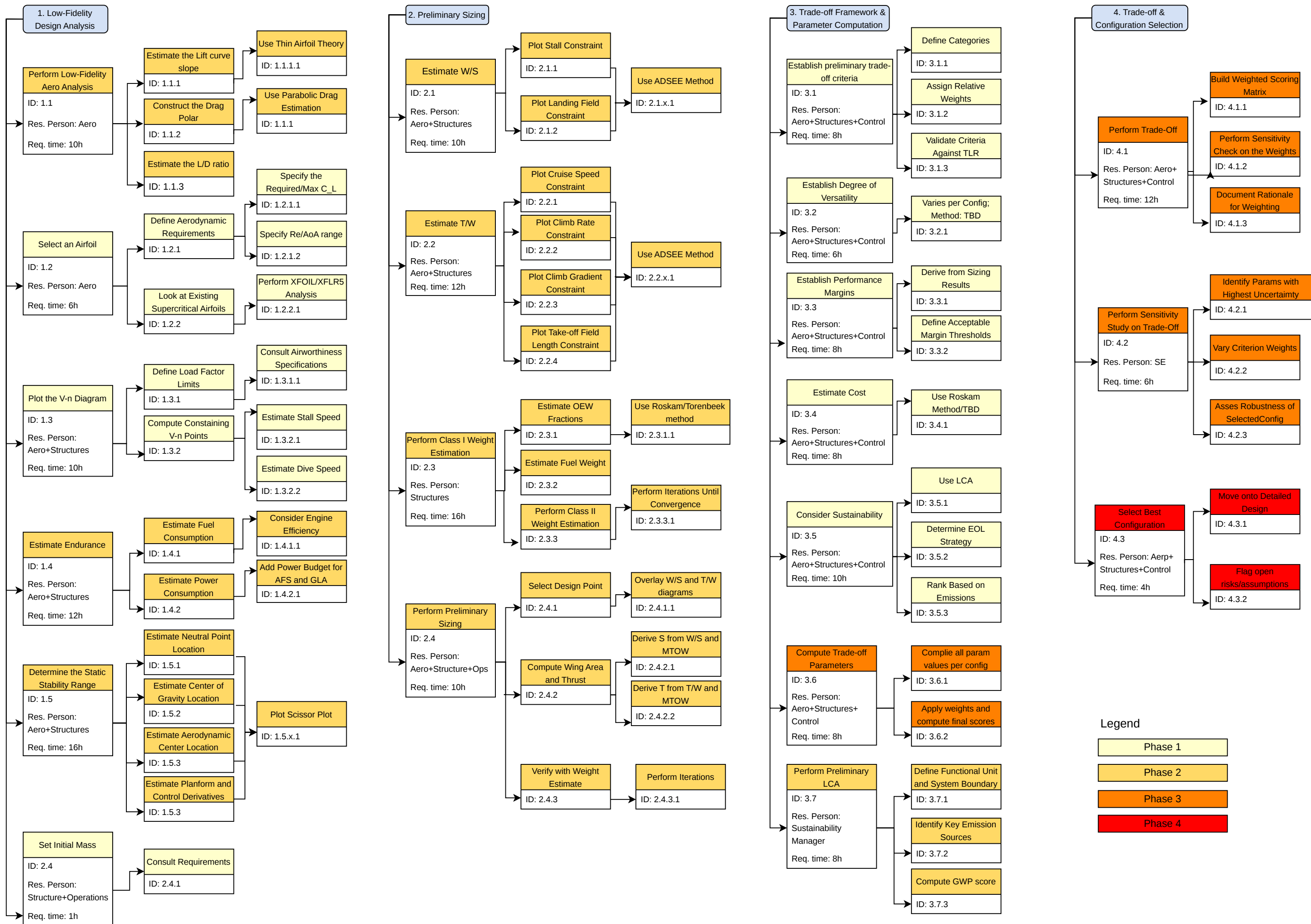


Figure A.1: Midterm Workflow Diagram

Assemble	Integrate the charged battery Integrate the electronics Connect power cables	Connect engine fuel system Fill the fuel tank	Integrate Payload	Fasten structural connections	Power the avionics	Power control	Power the transceiver		Power aircraft sensors	Power payload sensors	Power payload computer	
	Provide electrical power	Start fuel feed system and ignition Arm the aircraft	Initialize the payload		Power the control Retract gear Power high-lift devices							Power the control Extend landing gear Power high-lift
	Generate electric power	Provide thrust		Provide thrust load								
			Carry payload	Contribute to Inertial Load								
		Set Max thrust		Provide load case	Take off							
		Maintain engine & airframe alignment	Secure payload in place	Provide structural integrity								
				Provide load case	Follow mission plan	Provide flight direction		Request velocity				Provide location data
					Feedback attitude information	Control attitude	Output current attitude information					
						Set attitude from commands	Communicate with ground				Set payload settings for testing	Adjust landing plan
		Adjust thrust setting						Set velocity				Set approach speed
	Provide reference thrust				Monitor direction	Provide reference altitude	Record flight data	Provide reference airspeed	Monitor & process flight information (air speed, altitude, temperature, pressure, direction)			Provide attitude
				Provide load case for experimental maneuvers			Transmit measurement data			Record experimental data	Coordinate the experiment	
							Return payload parameters			Set instruments	Communicate with payload	
		Set to idle		Provide load case								Land

Figure A.2: HUGO platform N2 diagram



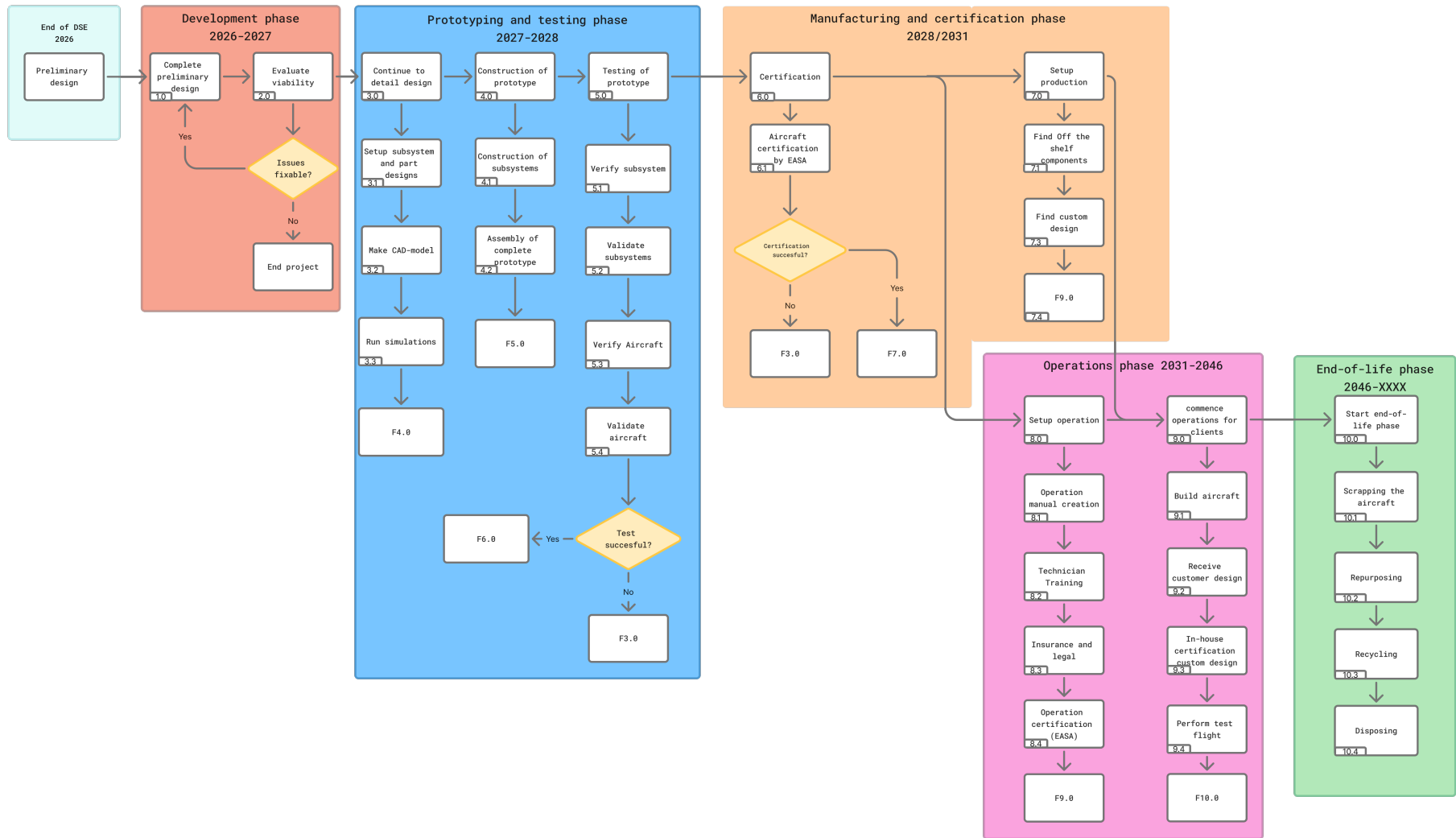


Figure A.3: Project Design & Development Logic

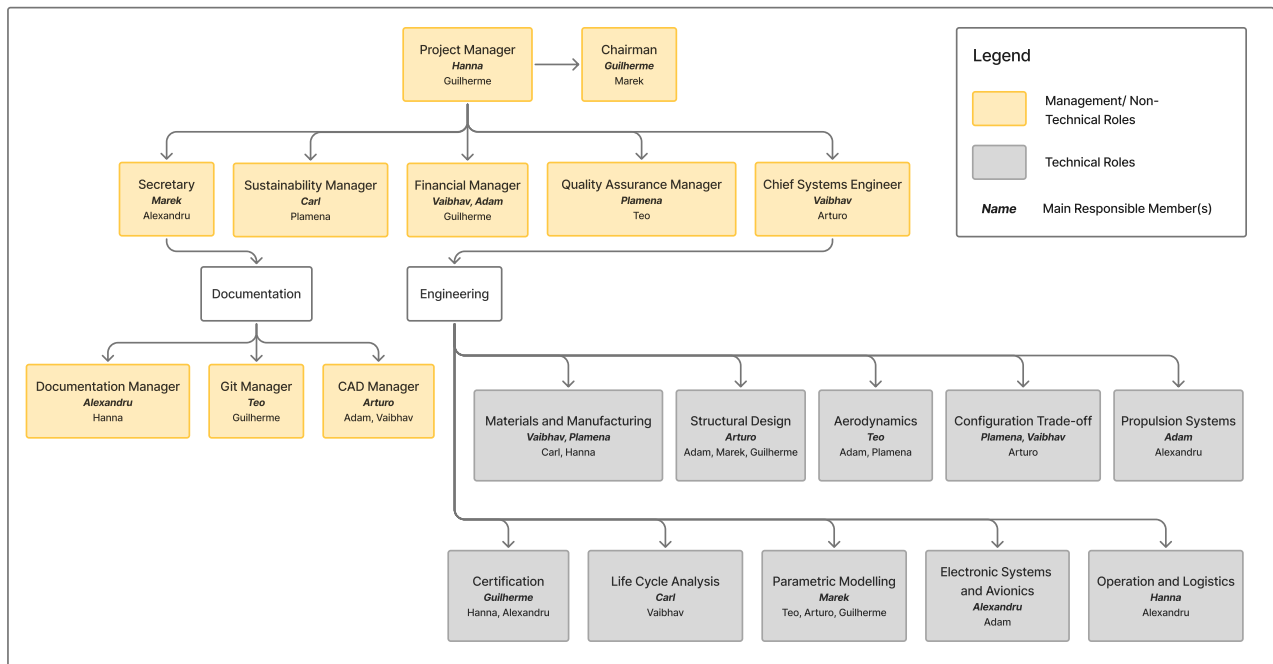


Figure A.4: Team 28 Organogram

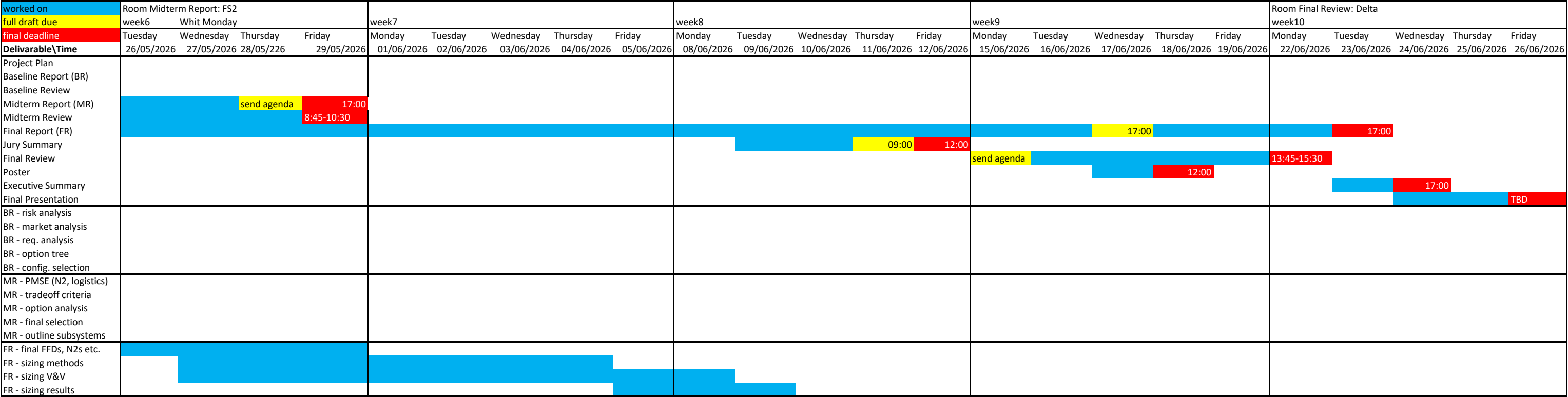
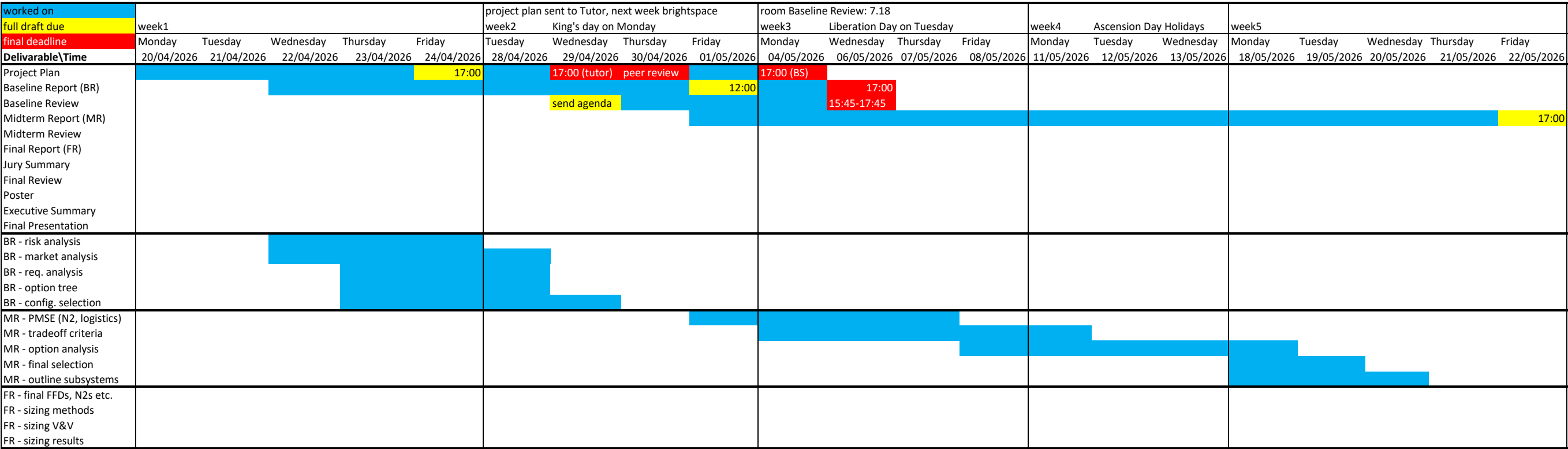


Figure A.5: Project Gantt Chart

B. Trade-Off Rationale

In this Appendix, the reasoning behind numerical values used for the trade-off scores and weight factors is explained.

B.1. Trade-Off Score Values

Table B.1 defines the scores assigned to trade-off criteria in section 7.1. Each criterion was assigned a score from 1 to 5, with 1 being the lowest and 5 the highest in terms of desirability for HUGO.

Table B.1: Trade-Off Score Definition

Criterion	Score				
	1	2	3	4	5
Versatility	One testable configuration (core mission compromised)	Significant limitations	Moderate limitations	Most configurations testable	Full range of configurations testable
Performance	High fuel consumption and weight limiting the achievable Mach at test altitude	Significant fuel consumption and mass penalties, meaningful efficiency reduction	Moderate fuel consumption or mass penalty, noticeable but manageable impact	Slightly elevated fuel consumption or mass, minor impact	Low fuel consumption and weight - clean configuration, no protruding components
Reliability	Catastrophic failure risk: extreme fatigue exposure; accessibility insufficient to prevent escalation	Severe failure risk: moderate fatigue exposure; limited accessibility increases likelihood of undetected degradation	Moderate failure risk: moderate fatigue exposure; good accessibility allows timely intervention	Soft failure risk: minimal fatigue exposure; limited accessibility, preventive maintenance feasible with planning	No anticipated failure: minimal fatigue exposure; good accessibility enabling routine preventive maintenance
Cost	Design violates HUG-FIN-001 (1)	High complexity, significant cost driver	Moderate complexity, high manufacturing and procurement cost	Low complexity, limited manufacturing effort	Minimal complexity
Sustainability	Highest impact: highest fuel consumption and manufacturing impact	High impact: significant manufacturing complexity and elevated fuel burn	Moderate impact: notable manufacturing and operational penalties	Low impact: slightly higher manufacturing or operational impact	Lowest impact: minimal manufacturing complexity, lowest operational fuel burn

B.2. PCM Scores

This Section contains the rationale behind the pairwise comparison scores used in Table 7.6. As an example, element (2, 1) is equal to 6, meaning that versatility is ranked 4 times more important than cost, since a cheap testbed that can't cover the test envelope has no scientific value. Element (1, 2) is the reciprocal, equal to $\frac{1}{4}$.

Below, the remaining pairwise comparisons are justified:

1. **Cost vs Versatility:** The ratio is $\frac{1}{6}$. Coverage of the test envelope is a prerequisite for scientific validity — an inexpensive testbed that cannot meet this requirement offers no value. Versatility is therefore weighted 6× higher than cost.
2. **Cost vs Reliability:** The ratio is $\frac{1}{4}$. Same ratio as cost vs performance, for analogous reasoning — an unreliable aircraft either produces corrupted data or is grounded, making any cost savings irrelevant. Performance and reliability are rated equally against cost because both are non-negotiable prerequisites for mission success.
3. **Cost vs Sustainability:** The ratio is 2. Cost is slightly more important than sustainability because it directly determines whether the aircraft gets built and operated at all. Sustainability matters but has limited influence on mission outcome for a low-flight-hour research aircraft, particularly since SAF is already being used.
4. **Versatility vs Reliability:** The score is 3. Versatility limitations are design-dependent and cannot be corrected post-build, whereas reliability issues can be partially mitigated through redundancy, maintenance procedures, and data post-processing. This makes versatility marginally more important.
5. **Versatility vs Sustainability:** The score is 7. The scientific output of the testbed strongly outweighs environmental considerations. A versatile testbed operating on SAF for limited research hours has a negligible environmental footprint compared to its scientific contribution.
6. **Reliability vs Sustainability:** The score is 4. Unreliable data collection undermines the entire mission, whereas the sustainability impact remains small given the operational profile.

C. Tasks Contributions

This Appendix contains an overview of contributions of AE3200 Team 28 members to the Midterm Report. [Table C.1](#) lists the specific chapters, sections, and/or subsections that have been contributed to by certain team members.

Table C.1: *Student contributions to the report.*

Student	Contributed to the following sections
Adam Janeček	Appendix A , section 6.1 , section 6.6 , section 6.4 , section 9.1 , Configuration sketches: Figure 7.1a , Figure 7.1b , Figure 7.1c , Figure 7.1d , Figure 9.1
Alexandru Gheorghiu	Overleaf Maintenance, section 4.1 , section 4.2 , section 5.3 , section 6.7 , chapter 9 , section 9.1 , section 9.2 , Appendix C .
Arturo Arends Martinez	section 3.5 , section 6.4
Carl Vistisen	section 7.5 , subsection 7.5.2 , subsection 7.5.3 , subsection 6.1.3 , section 5.5 , section 3.4 , chapter 10
Guilherme Castro Lima da Silva	chapter 1 , section 5.2 , section 5.3 , section 6.7 , chapter 4 (feedback implementation)
Hanna Skrobanek	section 4.1 , section 4.2 , section 5.1 , section 5.3 , section 7.5 , section 7.1 , section 7.3 , section 8.1 , Appendix A , Appendix B
Marek Gryszka	section 6.5 , section 6.3 , section 6.8 , chapter 11 , Executive Overview (about engineering design)
Plamena Dimitrova	chapter 2 , section 3.2 , subsection 6.2.1 , section 7.1 , section 7.2 , section 7.3 , section 7.4 section 8.1 , section 8.2 , section 8.3 , Conclusion.
Teo Čokonaj	section 3.1 , section 5.4 , subsection 6.2.2 , section 6.6
Vaibhav Byreddy	chapter 1 , section 3.4 , chapter 7 , section 8.1 , subsection 9.2.2 .